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GPCE500A Programming Guide

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REVISION HISTORY

Revision	Date	By	Remark
V0.1	2024/04/16	Jerry Hsu	Preliminary Programming Guide

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1 IMPORTANT NOTICE BEFORE WRITING YOUR PROGRAM

1.1. Confirmation Sheet

It is highly recommended programmers to obtain a copy of the appropriate confirmation sheet before writing program. The confirmation sheet, a requisite document before placing orders, contains useful information and checklist to help programmers avoiding mistakes during program development. Due to the possible changes may be made in the confirmation sheet occasionally, an up-to-date confirmation sheet can be downloaded from GENERALPLUS web site at http://www.generalplus.com.tw.

1.2. Other Documentations

To obtain the following documentation, please contact your GENERALPLUS sales representatives.

u'nSP Assembly Programming Guide: a manual to introduce *μ'nSP®* assembly instruction set.

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2 INTRODUCTION

2.1. General Description

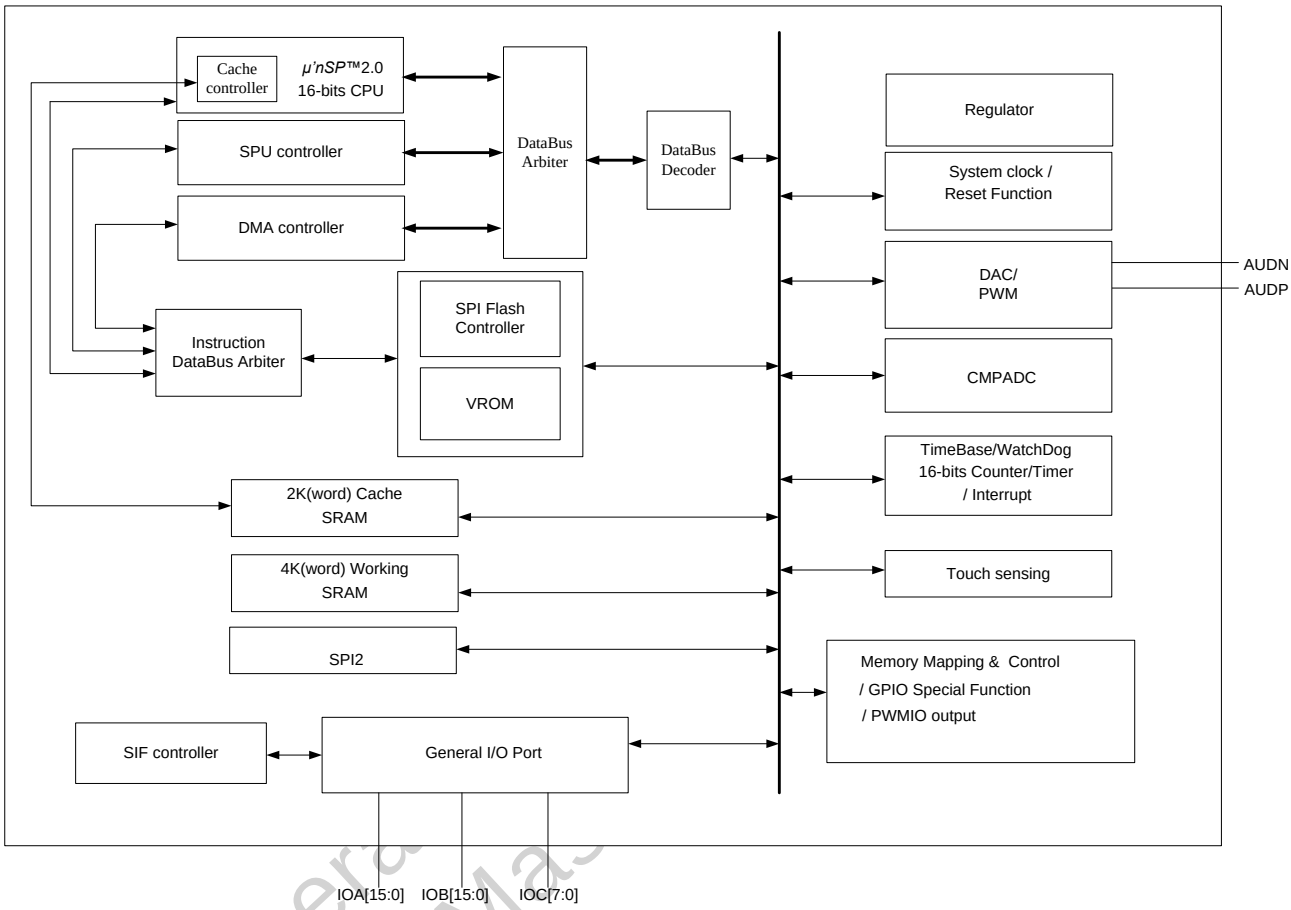
GPCE500A, a 16-bit architecture sound controller, features the newest 16-bit microprocessor, *μ'nSP2.0*™ (pronounced as *micro-n-SP 2.0*). This high processing speed assures the *μ'nSP2.0*™ is capable of handling complex digital signal processes easily and rapidly. Therefore, GPCE500A is applicable for the areas of digital sound process. The operating voltage of 2.0V through 5.5V and speed up to 81.92MHz yield the GPCE500A to be easily used in varieties of applications. The memory capacity includes a 4K-word working SRAM and 2K-word Cache RAM (also can be working RAM). Other features including 40 programmable multi-functional I/Os, five 16-bit timers/counters, 32KHz Real Time Clock, Low Voltage Reset/Detection, eight channels comparator ADC, 24 pads capacitive touch sensor, one SPI FLASH controller, one channel 16-bit voltage DAC and 16-bit PWM out for audio.

2.2. Features

- 16-bit *μ'nSP2.0*™ microprocessor
- CPU Clock: max. 81.92MHz
- Operating Voltage: 2.0V - 5.5V
- Power regulator built-in with input voltage: 2.0~5.5V, output voltage: 2.0~3.0V for LDO30 and 1.5V for LDO15
- IO Port Operating Voltage:
  2.0V - 5.5V for IOA & B and 2.0V – 3.0V for IOC
- Fast speed SPI interface for external FLASH access. Supports run SPI external code and data in auto mode
- Two SPI serial interface I/Os. One is SPI FLASH controller which supports single IO, dual IO, quad IO modes and support auto read/program mode. The other is normal SPI interface.
- 4K-word working SRAM
- 2K-word cache SRAM (also can be released to working-RAM)
- Software-based audio processing
- Two sets of 16-bit software channel with noise filter, mixer and scalar to play high quality sound
- Standby mode for power saving
- Total of five 16-bit timers, including three general-purpose 16-bit timers/counters, two touch-sensing timers (also can released for general-purpose use).
- One 16-bit PWM driver for driving speaker directly
- One channel 16-bit voltage DAC
- 40 general I/Os (including SPI I/F)
- Key wakeup function (IOA[15:0], IOB[15:0] and IOC[7:0])
- Capacitor touch-sensing hardware for max. 24 I/O
- IROSC feature for system clock
- 32KHz Real Time Clock (RTC) with build-in IROSC.
- Eight channels of comparator AD converter
- Low voltage reset and battery voltage detection
- Watchdog Timeout Reset

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2.3. Block Diagram



2.4. Features Table

	GPCE500A	GPCE100A	GPCE200A
Working Voltage	**2.0V~5.5V**	**2.2V~5.5V**	**2.0V~5.5V**
I/O Working Voltage	**2.0V~5.5V for IOA, IOB** **2.0V~3.0V for IOC**	**2.2V~5.5V for IOA, IOB** **2.2V~3.0V for IOC**	**2.0V~5.5V for IOA, IOB** **2.2V~2.85V for IOC**
Max. Speed	81.92MHz	81.92MHz	81.92MHz
CPU	16-bit *μ'nSP(2.0)*®	16-bit *μ'nSP(2.0)*®	16-bit *μ'nSP(2.0)*®
SRAM Size	**Max. 6K-Word (Cache included)**	**Max. 5K-Word (Cache included)**	**Max. 3K-Word (Cache included)**
Cache Size	Max. 2K-Word	Max. 2K-Word	Max. 1K-Word
Memory Size	External SPI FLASH	External SPI FLASH	External SPI FLASH
PORT A	IOA15 ~ 0	IOA15 ~ 0	IOA15 ~ 4
PORT B	IOB15 ~ 0	IOB15 ~ 0	IOB15 ~ 8
PORT C	IOC7 ~ 0	IOC7 ~ 0	IOC6 ~ 3
IOPWM	**IOA8 ~ 0 (x9)**	**IOA15 ~ 8 (x8)**	**IOA11 ~ 8 (x4)**
Audio Output	**16-bit RDAC or 16-bit PWM**	**12-bit DAC or 16-bit PWM**	**12-bit DAC or 16-bit PWM**
Interrupt	Please refer to the Interrupt section	Please refer to the Interrupt section	Please refer to the Interrupt section

	GPCE500A	GPCE100A	GPCE200A
Source			
Wakeup Source	All of Interrupt Source	All of Interrupt Source	All of Interrupt Source
Timer/Counter	Five 16-bit up count Timer/Counter	Five 16-bit up count Timer/Counter	Five 16-bit up count Timer/Counter
SPI x2	Yes	Yes	Yes
12-bit ADC	Yes	Yes	Yes
ADC	Yes, 12-bit @16K S.R.	Yes,12-bit ADC	Yes,8-bit ADC
ADC Channel	Five channel Line-in from IOA11~7	One channel Line-in from IOB9	Ten channel Line-in from IOB9~0
PGA	Yes	No	No
IR (IRTx)	Yes	Yes	Yes
Crystal/ROSC Resonator	32KHz Crystal or build-in 32K ROSC, build-in 81.92M ROSC	built-in 32K ROSC, built-in 81.92M ROSC	32KHz Crystal or build-in 32K ROSC, build-in 81.92M ROSC
Low Voltage Reset	Yes	Yes	Yes
Touch	IOB15~8 + IOA15~0 hardware touch	IOB15~10 + IOA7~0 hardware touch IOB9~0 + IOA15~8 digital touch	IOB15~10 + IOA15~4 hardware touch
SPU	Yes	No	No
DMA	2 channel	No	No
ICE	Yes	Yes	No

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3 MEMORY MAPPING

3.1. Memory Mapping

CPU Address	Memory	Comment
0x0000~0x0FFF	RAM	4K words RAM
0x1000~0x17FF	RAM	2K words Cache RAM
0x3000~0x37FF	Control register	Control register
0x4000~0x43FF	Internal ROM	Generalplus's Reserved area
0x4400~0x47FF	Internal ROM	Not used by users
0x8000~0xFFFFF	External SPI FLASH	SPI Bank 0, Common Bank
0x100000~0x1FFFFF	External SPI FLASH	SPI Bank 1, Common Bank
0x200000~0x2FFFFF	External SPI FLASH	SPI Bank 2, Common Bank
0x300000~0x3FFFFF	External SPI FLASH	SPI Bank 3, Extended to 64MW by Register

Note: User's program allocated in external SPI FLASH memory

3.2. Memory Allocation

The memory allocation is determined at compile time. Generalplus's μ'nSP® IDE automatically allocates the RAM and FLASH space when the project is compiled or built based on the predefined sections such as CODE, NB_DATA, DATA, TEXT, RAM, SRAM, IRAM, ISRAM, ORAM, and OSRAM. For details about the predefined sections, please refer to Generalplus μ'nSP® Assembly Programming Guide and μ'nSP® IDE on-line help.

3.2.1 RAM (0x0000~0x17FF)

The maximum of 6K-word RAM (including 2K-word cache) area is located in 0x0000 ~ 0x17FF with access time of one CPU clock cycle. If cache is not used, the 2K-word cache can be used as ordinary RAM for users. The RAM allocation follows the following priority to allocate the RAM (0x0000~0x17FF). It will allot the user-predefined RAM sections and next declare the RAM block, from the biggest to the smallest. That is, except the designated allocation, the rest of the space will be served by biggest-first rule. The first 64 words in RAM area (0x0000~0x003F) can be used for RAM block allocation only when the RAM block is explicitly declared as SRAM. If not, the RAM block will be allocated in the range from 0x0040 to 0x17FF. The stack will stack up from 0x17FF and move up the stack pointer toward the address 0x0000. The stack size varies at run time as the program thread transits among foreground functions and background interrupts. Consequently, attention must be exercised to avoid the situation that, at run time, the stack size growth conflicts with the allocated RAM blocks.

3.2.2 Internal ROM Memory (0x4000~0x47FF)

GPCE500A has 2K words internal ROM, which includes 1K words ROM area for Test Code and Vector, and 1K words for boot code. User's code cannot be located at internal ROM area. It must be placed at external SPI FLASH.

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3.2.3 External SPI FLASH memory (0x8000~0x3FFFFF)

The SPI FLASH addresses mentioned here are all in word mode.

All user's code are located at this area, where 0x8000~0x80FF is the system boot header. When system is powered on, it reads the system boot header for some basic settings. The 0x8100~0x97FF is the area for user's boot code, which configures the SPI FLASH memory settings. CPU will move the contents of user's boot code to RAM and process it. Next, it executes user's startup and jump to main program. After that, CPU execution will be on external SPI FLASH.

GPCE500A's CPU view 0x8000~0x2FFFFF is the common bank area, always mapping to SPI FLASH's 0x000000~0x2F7FFF. Modifying **P_SPI_Bank**(0x30E9) does not affect reading from common bank.
Example:

- To read FLASH's 0x0F7FFF, CPU can read from either 0x3FFFFF when SPI bank = 0 or 0x0FFFFF, which all have the correct data.

  ```
  R1 = 0x0
  [P_SPI_Bank] = R1;
  R1 = 0x3F;
  DS = R1;
  R2 = 0xFFFF;
  R1 = D:[R2];
  or
  R1 = 0x0F;
  DS = R1;
  R2 = 0xFFFF
  R1 = D:[R2];
  ```

- To read FLASH's 0x4F7FFF, read 0x3FFFFF when SPI bank = 4

  ```
  R1 = 0x4
  [P_SPI_Bank] = R1;
  R1 = 0x3F;
  DS = R1;
  R2 = 0xFFFF;
  R1 = D:[R2];
  ```

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Mapping table of SPI FLASH address and CPU view address:

SPI FLASH Real Address (word)	CPU View Address (word)
0x000000 ~ 0x0F7FFF	**Always mapping to 0x8000~0x0FFFFF** **or** **Bank0: 0x308000~0x3FFFFF**
0x0F8000~0x1F7FFF	**Always mapping to 0x100000~0x1FFFFF** **or** **Bank1: 0x300000~0x3FFFFF**
0x2F8000~0x2F7FFF	**Always mapping to 0x200000~0x2FFFFF** **or** **Bank2: 0x300000~0x3FFFFF**
0x3F8000~0x3F7FFF	Bank3: 0x300000~0x3FFFFF
0x4F8000~0x4F7FFF	Bank4: 0x300000~0x3FFFFF
0x5F8000~0x5F7FFF	Bank5: 0x300000~0x3FFFFF
…	
…	
…	
0x3FF8000~0x3FF7FFF	Bank63 : 0x300000~0x3FFFFF

3.2.4 Engineering issues

Users may opt to manually designate the allocation to the unoccupied location in Generalplus μ'nSP® IDE. To manually allocate the RAM block, please refer to Generalplus μ'nSP® Assembly Programming Guide. The memory allocation examination by user is recommended after completing the program compilation. User can go to the debug or release folder under the project folder and click open the file, project_name.map, for memory allocation details or go to the Generalplus μ'nSP® IDE menu->Tools->Memory Map for graphic presentation.

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4 SYSTEM

4.1. Registers Map

Name	Address	Description
P_System_Clock	0x3030	Controls the system and CPU clocks
P_System_Reset	0x3031	Software reset
P_Reset_BVD_Ctrl	0x3032	LVR and BVD control
P_TimeBase_Clear	0x3033	Timebase reset and accuracy correction.
P_Watchdog_Clear	0x3034	Watchdog reset
P_Wait_Ctrl	0x3035	Set wait state value
P_IOSC32K_Cal_LB	0x3037	IOSC32K calibration low boundary
P_IOSC32K_Cal_HB	0x3038	IOSC32K calibration high boundary
P_Cache_Ctrl	0x3039	Cache control
P_Cache_Hit_Rate	0x303A	Cache hit rate
P_Stack_INT_Level	0x303B	Stack INT level setting
P_IOSC80M_Cal_LB	0x303C	IOSC80M Calibration low Boundary
P_IOSC80M_Cal_HB	0x303D	IOSC80M Calibration high Boundary
P_SPU_Clock_Ctrl	0x303F	SPU clock control

4.2. System Clock

Basically, GPCE500A features two built-in ROSC IOSC81.92MHz and IOSC32KHz(32768Hz). The system clock (F_{SYSTEM}) is provided by IOSC81.92MHz and programmed by the P_System_Clock ($3030H) to determine the frequency of system clock. The default system clock (F_{SYSTEM}) is 5.12MHz. The CPU clock is same as system clock. User can maximal set the system clock (F_{SYSTEM}) up to 81.92MHz.

4.2.1 P_System_Clock (R/W) (0x3030)

The system clock (F_{SYSTEM}) can be configured from 5.12MHz to 81.92MHz through programming b7~b5 of **P_System_Clock** (0x3030H).

P_System_Clock			0x3030		Controls the system and CPU clocks			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	X32K_RDY	-			I8192M_TRIM_SEL	I8192M_CAL_EN	I8192M_RDY	I8192M_CAL_STS
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	SYS_CLK[2:0]			32K_SLP_STS	-	I32K_CAL_EN	I32K_RDY	I32K_CAL_STS
Default	1	1	1	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	X32K_RDY	R	XTAL32K ready status	0 : Not Ready

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Bit	Function	Type	Description	Condition
				1 : Ready
[14:12]	-	-	Reserved	-
[11]	I8192M_TRIM_SEL	R/W	IOSC81.92M trim selection	0: Calibration 1: Fuse
[10]	I8192M_CAL_EN	R/W	IOSC81.92M calibration enable	0 : Disabled 1 : Enabled
[9]	I8192M_RDY	R	IOSC81.92M calibration ready	0 : Not Ready 1 : Ready
[8]	I8192M_CAL_STS	R	IOSC81.92M calibration status	0 : Fail 1 : Pass
[7:5]	SYS_CLK[2:0]	R/W	System clock setting	000: 81.92MHz 001: 81.92MHz/2 010: 81.92MHz/3 011: 81.92MHz/4 100: 81.92MHz/6 101: 81.92MHz/8 110: 81.92MHz/12 111: 81.92MHz/16 (Default)
[4]	32K_SLP_STS	R/W	Determines whether 32.768KHz working or not under sleep mode	1: In sleep mode, 32768Hz is still working. 0: In sleep mode, 32768KHz is OFF (Default).
[3]	-	-	Reserved	-
[2]	I32K_CAL_EN	R/W	IOSC32K calibration enable	0 : Disabled 1 : Enabled
[1]	I32K_RDY	R	IOSC32K calibration ready	0 : Not Ready 1 : Ready
[0]	I32K_CAL_STS	R	IOSC32K calibration status	0 : Fail 1 : Pass

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4.2.2 P_Wait_Ctrl(R/W) (0x3035)

P_Wait_Ctrl			0x3035		Set wait state value			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-					WAIT_STATE[2:0]		
Default	0	0	0	0	0	0	1	0

Bit	Function	Type	Description	Condition
[15:3]	-	-	Reserved	-
[2:0]	WAIT_STATE[2:0]	R/W	Set wait state value	000: 1 cycle 001: 2 cycle 010: 3 cycles(Default) 011: 4 cycles 100: 5 cycles 101: 6 cycles 110: 7 cycles 111: 8 cycles

4.2.3 P_SPU_Clock_Ctrl (R/W) (0x303F)

P_SPU_Clock_Ctrl			0x3035		SPU clock control			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-							SPU_CLK_EN
Default	0	0	0	0	0	0	0	1

Bit	Function	Type	Description	Condition
[15:1]	-	-	Reserved	-
[0]	SPU_CLK_EN	R/W	SPU clock enable	0 : Disabled 1 : Enabled

4.3. IOSC32K Calibration

GPCE500A's IOSC32K can perform the calibration through IOSC81.92M. We must first set the lower limit value (**P_IOSC32K_Cal_LB**(0x3037)) and the upper limit value (**P_IOSC32K_Cal_HB**(0x3038)) and then, write 0x3030[2] to start calibration; wait for 0x3030[1] turning to "1" which means the calibration is completed. Here, we need to read 0x3030[0] again to confirm whether or not the calibration is passed. If it fails, we need to modify the upper and lower limits again, and then re-calibrate it.

Please refer to the following instructions for how to calculate the upper and lower values:

◎ Upper limit: If the I32K intermediate frequency is 3% slower, IOSC81.92MHz will count 2577.3 (A11h) times in a standard I32K cycle.

◎ Lower limit: If the I32K frequency is 3% faster, IOSC81.92MHz will count 2427.2 (97Bh) times in a standard I32K cycle.

Therefore, if +-3% is used as the IOSC32K calibration standard, P_IOSC32K_CAL_HB = 0xA11 and P_IOSC32K_CAL_LB = 0x9C4 should be set.

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Please refer to the following example code:

4.3.1 P_IOSC32K_CAL_LB(R/W) (0x3037)

P_IOSC32K_CAL_LB	0x3037	IOSC32K Calibration Low Boundary						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-				I32K_CAL_LB[11:8]			
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	I32K_CAL_LB[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:12]	-	-	Reserved
[11:0]	I32K_CAL_LB[11:0]	R/W	IOSC32K calibration counter low bond

4.3.2 P_IOSC32K_CAL_HB(R/W) (0x30380)

P_IOSC32K_CAL_HB	0x3037	IOSC32K Calibration Counter High Boundary						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-				I32K_CAL_HB[11:8]			
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	I32K_CAL_HB[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:12]	-	-	Reserved
[11:0]	I32K_CAL_HB[11:0]	R/W	IOSC32K calibration counter high bond

4.4. IOSC81.92M Calibration

GPCE500A is allowed to calibrate IOSC81.92M via an external XTAL32K. First, identify the 32K source is XTAL32K. Read 0x3030[15] to ensure X32K is ready. Assign the upper and lower bound for the frequency calibration. For example, suppose the range of frequency calibration is +-0.2%: Thus, **P_IOSC8192M_CAL_LB** = 2495(0x9BF), **P_IOSC8192M_CAL_HB** = 2505(0x9C9). Next, enable **I8192M_CAL_EN** (0x3030[10]) followed by polling **I8192M_RDY**(0x3030[9]) flag until it becomes "1". Check whether **I8192M_CAL_STS**(0x3030[8]) is pass or not. If failed, it needs to be calibrated again.

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4.4.1 P_IOSC8192M_CAL_LB(R/W) (0x303C)

P_IOSC8092M_CAL_LB	0x303C	IOSC81.92M Calibration low Boundary							
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-				I8192M_CAL_LB[11:8]			
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	I8192M_CAL_LB[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:12]	-	-	Reserved
[11:0]	I8192M_CAL_LB[11:0]	R/W	IOSC81.92M Auto calibration counter low bond

4.4.2 P_IOSC8192M_CAL_HB(R/W) (0x303D)

P_IOSC8192M_CAL_HB	0x303D	IOSC81.92M Auto Calibration Counter HIGH Bond							
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-				I8192M_CAL_HB[11:8]			
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	I8192M_CAL_HB[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:12]	-	-	Reserved
[11:0]	I8192M_CAL_HB[11:0]	R/W	IOSC81.92M Auto calibration counter high bond

4.5. Software Reset

Writing 0x5555H will reset entire system. The hardware reset (pull-low reset pin), except the software reset flag, will set on b14 in **P_Reset_BVD_Ctrl** (R/W) (0x3032).

4.5.1 P_System_Reset (W) (0x3031)

P_System_Reset	0x3031	Software reset						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	SW_RST[15:8]							
Default	-	-	-	-	-	-	-	-

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Bit	7	6	5	4	3	2	1	0
Function	SW_RST [7:0]							
Default	-	-	-	-	-	-	-	-

Bit	Function	Type	Description
[15:0]	SW_RST [15:0]	W	Write 0x5555 to reset the entire system

4.6. LVR/LVD (Low Voltage Reset/Low Voltage Detect)

The system may halt when power drops below LVR level, normally caused by battery-bounce, heavy loading or low-battery condition. With the LVR function, a reset signal is generated to reset the system when the power drops below LVR level for 10 clock cycles.

The LVD reports the circumstance of system voltage. When the system voltage (V_{DD}) drops below the selected LVD level (V_{LVD}), the LVD result-flag (b11 of P_Reset_LVD_Ctrl (R/W) (0x3032)) will be set and will be cleared to "0" when V_{DD} is greater than V_{LVD}. Four LVD levels can be programmed to specific levels by writing P_Reset_LVD_Ctrl (R/W) (0x3032). Suppose, LVD is set to 2.8V when the system power drops below 2.8V, the b11 of P_Reset_LVD_Ctrl (R/W) (0x3032) returns HIGH. As a result, CPU is able to detect the low voltage condition via the programmed voltage level. The default value of LVD function is 3.0V. The LVR and LVD functions are enabled when CPU is running and they will be turned off automatically in sleep mode.

Watchdog reset will not clear the register. On the other hand, all Power on reset, Key reset, LVR and Software reset will clear all registers.

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4.6.1 P_ Reset_LVD_Ctrl (R/W) (0x3032)

P_Reset_LVD_Ctrl		0x3032		Reset flag and LVD control register				
Bit	15	14	13	12	11	10	9	8
Function	WDT_RST	SW_RST	WDT_RST_ERR	-	LVD_STS	-		
Default	0	0	0	0	0	-	-	-

Bit	7	6	5	4	3	2	1	0
Function	-						LVD_CTRL[1:0]	
Default	-	-	-	-	-	-	0	0

Bit	Function	Type	Description	Condition
[15]	WDT_RST	R/W1C	Watchdog reset flag	0: Not occurs 1: Occurs Write '1' to clear the flag
[14]	SW_RST	R/W1C	Software reset flag	0: Not Occurs 1: Occurs Write '1' to clear the flag

[13]	WDT_RST_ERR	R/W1C	Watchdog reset error flag	0: Not occurs 1: Occurs Write '1' to clear the flag
[12]	-	-	Reserved	-
[11]	LVD_STS	R	LVD status flag V_{LVD} is the voltage level set in LVD_CTRL	0: VDD > V_{LVD} 1: VDD < V_{LVD}
[10:2]	-	-	Reserved	-
[1:0]	LVD_CTRL	R/W	LVD voltage level control	00: 3.2V (Default) 01: 2.8V 10: 2.6V 11: 2.4V

4.7. Timebase Clear

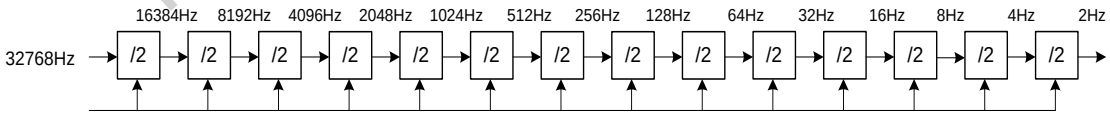
An additional control port, P_Timebase_Clear(W) (0x3033), is for time reset and accuracy correction.

4.7.1 P_Timebase_Clear (W) (0x3033)

P_Timebase_Clear			0x3033		Time reset and accuracy correction			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	TIMEBASE_CLR[15:8]							
Default	-	-	-	-	-	-	-	-

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	TIMEBASE_CLR[7:0]							
Default	-	-	-	-	-	-	-	-

Bit	Function	Type	Description
[15:0]	TIMEBASE_CLR[15:0]	W	Write any data to this port resets all timebase counter stages to zero.



P_Timebase_Clear(W)($3033H)

4.8. Watchdog Timer

The purpose of watchdog is to monitor if the system operates normally. Within 0.875 seconds, watchdog counter must be cleared. If the watchdog is not cleared, CPU assumes the program has been running into an abnormal condition and therefore, CPU will reset the system to the initial state and start running the program all over again. It protects the system from incorrect code execution by launching a system reset when the watchdog timer overflows resulting from the failure of software to clear the watchdog timer within every 0.875 seconds. For GPCE500A, watchdog function can be

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enabled or disabled by option.

In GPCE500A, the watchdog should be cleared within in Twdog period by writing “0x5555” to the P_Watchdog_Clear (W) (0x3034). Thus, the system will not be reset in normal operation. If any inappropriate operation occurred which makes fail to write 0x5555 to the P_Watchdog_Clear (W) (0x3034) in Twdog period, the timer will generate a reset signal to reset the system. If the data written into the P_Watchdog_Clear (W) (0x3034) is not “0x5555”, it will cause a direct system reset. Therefore, user is required to write the designated pattern “0x5555” to P_Watchdog_Clear (W) (0x3034) to clear the watchdog; otherwise, the system may function incorrectly. The reset generates by watchdog function will set a reset flag in P_Reset_LVD_Ctrl (0x3032) bit 15; If watchdog is disabled, those functions are all disabled. (Watchdog time out and write wrong data to reset system)

Example:

```
R1 = 0x5555;
[P_Watchdog_Clear] = R1;       // Reset the watchdog timer
```

4.8.1 P_Watchdog_Clear (W) (0x3034)

P_Watchdog_Clear			0x3034		Write 0x5555 to reset the watchdog timer			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	WDT_CLR[15:8]							
Default	-	-	-	-	-	-	-	-

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	WDT_CLR[7:0]							
Default	-	-	-	-	-	-	-	-

Bit	Function	Type	Description
[15:0]	WDT_CLR[15:0]	W	If the data written into the P_Watchdog_Clear is not “0x5555”, it will cause a direct system reset. Therefore, user is required to write the designated pattern “0x5555” to P_Watchdog_Clear to clear the watchdog;

4.9. Cache Control

Because GPCE500A embeds cache memory, user is allowed to enable cache via configuring P_Cache_Ctrl Register. Cache Clear will clear all data from Cache. If continues to perform this operation, it will cause cache missing and waste resources. When issuing Cache Disable command, 5 additional NOP are required after it to avoid unexpected Pipe line operation.

4.9.1 P_Cache_Ctrl (R/W) (0x3039)

P_Cache_Ctrl			0x3039		Cache control register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	-	-	-	-	-	-	-	-

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Bit	7	6	5	4	3	2	1	0
Function	-						CACHE_CLR	CACHE_EN
Default	-	-	-	-	-	-	0	0

Bit	Function	Type	Description	Condition
[15:2]	-	-	Reserved	-
[1]	CACHE_CLR	R/W	Cache RAM clear	0: None Action 1: Clear Cache RAM
[0]	CACHE_EN	R/W	Cache enable	0: Disabled 1: Enabled

4.9.2 P_Cache_Hit_Rate (R) (0x303A)

P_Cache_Hit_Rate			0x303A		Cache hit rate register			
Bit	15	14	13	12	11	10	9	8
Function	HIT_RATE[15:8]							
Default	-	-	-	-	-	-	-	-

Bit	7	6	5	4	3	2	1	0
Function	HIT_RATE[7:0]							
Default	-	-	-	-	-	-	-	-

Bit	Function	Type	Description
[15:0]	HIT_RATE[15:0]	R	The cache hit rate can be obtained by P_Cache_Hit_Rate Register.

4.10. Stack Point Check

Stack level can be implemented by P_Stack_INT_Level. During program development, it helps to clearify whether or not the stack has used the Data RAM area to avoid abnormal operation.

4.10.1 P_Stack_INT_Level (R/W) (0x303B)

P_Stack_INT_Level			0x303B		Stack level setting			
Bit	15	14	13	12	11	10	9	8
Function	STACK_INT[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	STACK_INT[7:0]							
Default	0	0	0	0	0	0	0	0

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Bit	Function	Type	Description
[15:0]	STACK_INT[15:0]	R/W	When SP(stack pointer) is smaller than P_Stack_INT_Level, b5 of P_INT2_Status(0x3054) will be 1.

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5 SLEEP AND WAKEUP

5.1. General Description

The purpose of sleep mode is to save power while device is not in use. When sleep function acts, the device runs from operating mode to standby mode. On the other hand, wakeup brings system back from sleeping mode to operating mode.

1. Sleep: After power on reset, IC starts working until a sleep command is given. When a sleep signal is accepted, IC will turn off the system clock F_{ISOC}, and then enter sleep mode. Users can change the system to sleep mode by writing '0x5555' into P_System_Sleep (W) ($3036H). After entering sleep mode, program counter will stop at the next line and will start to execute after wakeup interrupt completed and the signal of sleep pin will be high.

2. Wakeup: Waking up from sleep mode (or halt mode) requires a wakeup signal to turn on the system clock F_{ISOC}. At the same time, a wakeup interrupt (IRQ) is also generated. The interrupt signal leads CPU to complete the wakeup process as well as initialization. IRQ4_KEY key-change-wakeup after wakeup interrupt is completed, program counter will continue to execute the next command. In addition, P_IOx_WakeUp_Mask further determines each GPIO's capability for wakeup.

3. All the interrupt happened are wakeup source. Before entering sleep mode, all interrupt flags must be cleared; otherwise, IC is unable to enter sleep mode.

- For key change wakeup source, refer to ***I/O PortA/B/C configuration section*** for details.

Note1: Before entering the sleep mode, users need to disable PWM, DAC and ADC. If PWM, DAC and ADC are not disabled, the GPCE500A will cause power consumption.

Note2: Sleep mode is also called standby mode in this documentation.

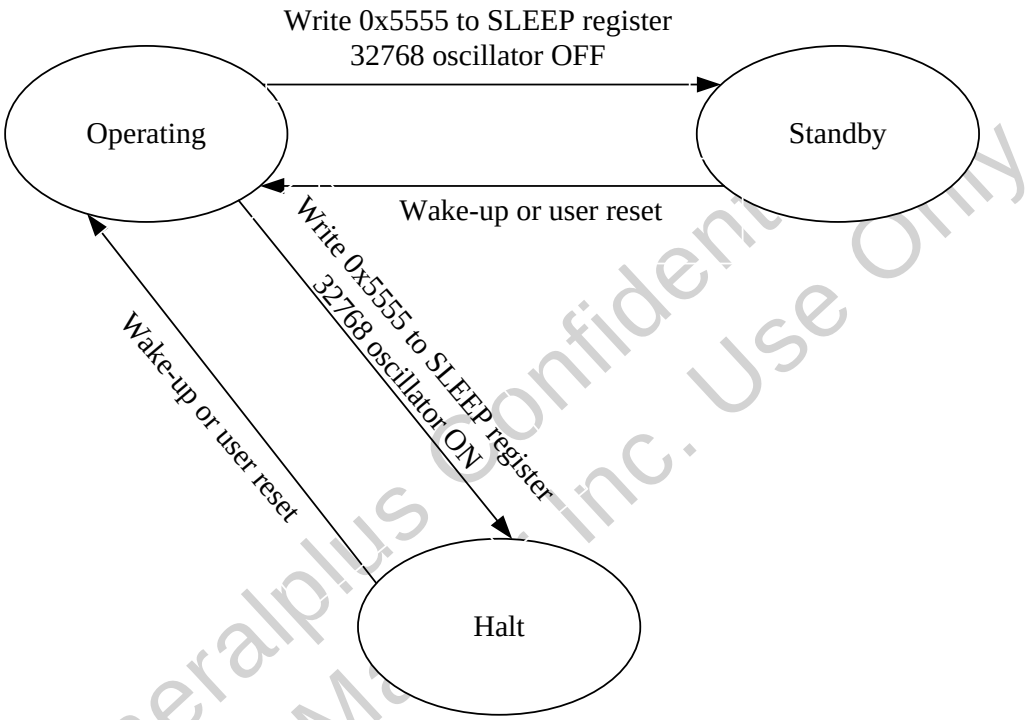
Sleep	Operating	Halt Mode	Standby Mode
CPU	On	Off	Off
32KHz Crystal or build-in IOSC32K	On	On	Off

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P_System_Clock ($3030H) : Bit4

'1' : In sleep mode, 32.768KHz is still working (Default).

'0' : In sleep mode, 32.768KHz is OFF.



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Example:

Enter Sleep mode and wake GPCE500A up with key change.

Before entering into sleep mode, set the IOA[15:0] as input mode and turn on IRQ4_KEY(Key change Wakeup) because Key change Wakeup is one of the interrupt source of IRQ4.

```
// Turn off ADC, DAC, CTS
// Turn off SPI Interface and configure the corresponding setup.  If SPI FLASH is used, refer to AN0176
// application note.  Assure all GPIOs are not in floating and possible current drain leak.

// The following example is about IOA Key Wakeup configuration.  It may be varied based on user's
// demand.

// IOA[15-0] configuration (for other I/O configurations, please refer to the I/O section).
R1 = 0x0000;
[P_IOA_Attrib] = R1;
[P_IOA_Dir] = R1;
[P_IOA_Buffer] = R1;
[P_IOA_WakeUp_Mask] = R1;
```

```
//mask IOB and IOC
R1 = 0xFFFF;
[P_IOB_WakeUp_Mask] = R1;
[P_IOC_WakeUp_Mask] = R1;

//Interrupt configuration (set IRQ4_KEY, Key change interrupt source, for other
//interrupt configurations, please refer to the Interrupt section)
INT OFF;

// Before entering sleep mode, all interrupt flags must be cleared; otherwise, IC is unable to enter sleep
// mode.
R1 = 0xFFFF;
[P_INT_Status] = R1;
[P_INT_Status2] = R1;     // Clear All interrupt Flag

//If IO is a wakeup source, I/O data & attribute ports must be latched before entering sleep mode.
R1 = [P_IOA_Attrib];
R1 = [P_IOA_Data];        // Latch IOA

R1 = C_IRQ4_KEY;
[P_INT_Ctrl] = R1;        // Set Interrupt Source

nop;
nop;
nop;
nop;

// Write “0x5555” to P_System_Sleep (W)$3036H to enter into sleep mode
// (please refer to the System Clock section)
R1 = C_System_Sleep;
[P_System_Sleep] = R1;  // Enter Sleep

nop;
nop;
nop;

IRQ ON;                   //Enable interrupt after wake up
```

…(After PortA key change wakeup source triggered, it will call IRQ4 subroutine)

Sub-routine of IRQ4:

```
_IRQ4:

L_IRQ4_KeyChange_WakeUp:
        :
    (Tasks after wakeup)
```

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```
:
R1 = C_IRQ4_KEY;            //Clear IRQ4(Key Change Wakeup) Event
[P_INT_Status] = R1;
RETI
```

5.2. Registers Map

Name	Address	Description
P_System_Sleep	0x3036	Write 0x5555 to enter sleep mode
P_SRAM_Sleep_Ctrl	0x303E	SRAM sleep control

5.2.1 P_System_Sleep (W) (0x3036)

P_System_Sleep			0x3036		Write 0x5555 to enter sleep mode			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	SystemSleep[15:8]							
Default	-	-	-	-	-	-	-	-

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	SystemSleep[7:0]							
Default	-	-	-	-	-	-	-	-

Bit	Function	Type	Description
[15:0]	SystemSleep[15:0]	W	Users can change the system to sleep mode by writing '0x5555' into P_System_Sleep.

5.2.2 P_SRAM_Sleep_Ctrl (R/W) (0x303E)

P_SRAM_Sleep_Ctrl			0x303E		SRAM sleep control			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	-	-	-	-	-	-	-	-

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-						SRAM_PWR_RDY	SRAM_SLP_EN
Default	-	-	-	-	-	-	0	0

Bit	Function	Type	Description	Condition
[15:2]	-	-	Reserved	-
[1]	SRAM_PWR_RDY	R	SRAM power ready status	0: Not Ready 1: Ready
[0]	SRAM_SLP_EN	R/W	SRAM sleep in sleep mode	0: Disabled 1: Enabled

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6 I/O

6.1. General Description

The purpose of input port and output port is to communicate with other devices. Three programmable I/O ports are available in GPCE500A, Port A IOA[15:0], Port B IOB[15:0] and Port C IOC[7:0]. All of Ports are ordinary I/O with programmable wakeup capability. In addition to regular I/O function, all ports also provide some special functions in certain pins. Please refer to ***Special Function*** for details.

6.2. I/O Configuration

GPCE500A provides a bit-to-bit I/O configuration; every I/O bit can be defined individually. To set up a bit's configuration, three essential setups should be given: Buffer, Attribution, and Direction. The following table is a summary of setting an I/O configuration. The Direction, Attribution and Buffer represent three ports. Each corresponding bit in these ports should be given a value to set one bit's configuration. The setting rules are as follows:

- The direction setting determines whether this pin is an input or output.
- The attribution setting gives a feature to the pin. Float / pull for input, not inverted/ inverted for output.
- The buffer setting affects the initial content of the pin. For inputs, it also decides the pull high or pull low setting.
- IOB[7:0] supporting input with pull high mode
- IOB[15:0] supporting pull low with high resistance, controlled by P_IOB_PL1M
- IOC[1:0] only input mode available; output mode not supported
- IOC[7:0] always enables schmitt trigger.
- IOA、IOB power source from VDD_REGIN, IOC power source from V30_REGO

Please refer to the table below for the functional configuration of I/O pins.

Dir	Attrib	Buffer	PL1M	Function	Wakeup	Description
0	0	0	0	Pull Low*	Yes**	Input with pull low for all IO
0	0	0	1	Pull Low	Yes**	Input with pull low with high resistance for IOB[15:0] Input with pull low for others
0	0	1	x	Pull High Float	Yes**	Input with pull high for IOB[7:0] Input float with schmitt trigger enable for IOB[15:8], IOA[15:0]
0	1	0	x	Float	Yes**	Input with float
0	1	1	x	Float	Yes**	Input with float
1	1	0	x	Output Low	No	Output with buffer (data not inverted)
1	1	1	x	Output High	No	Output with buffer (data not inverted)
1	0	0	x	Output High (inverted)	No	Output with buffer inverted (write "0" to the buffer and will output "1" to the I/O pad)
1	0	1	x	Output Low (inverted)	No	Output with buffer inverted (write "1" to the buffer and will output "0" to the I/O pad)

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*Default: Input floating.
**All of IOA/B/C [15:0] in Input state, user is able to turn off the wakeup function via the corresponding P_IOX_WakeUp_Mask.

The driving and sinking capability on IOC7~IOC2 (SPI1) are identical. However, the driving and sinking capability are not symmetrical on other GPIOs, simply because the SPI1 must achieve 81.92MHz. To transmit signals, user should be alerted to this issue.

6.3. Registers Map

Name	Address	Description
P_IOA_Data	0x3000	IOA data register
P_IOA_Buffer	0x3001	IOA buffer register
P_IOA_Dir	0x3002	IOA direction register
P_IOA_Attrib	0x3003	IOA attribution register
P_IOB_Data	0x3004	IOB data register
P_IOB_Buffer	0x3005	IOB buffer register
P_IOB_Dir	0x3006	IOB direction register
P_IOB_Attrib	0x3007	IOB attribution register
P_IOC_Data	0x3008	IOB data register
P_IOC_Buffer	0x3009	IOB buffer register
P_IOC_Dir	0x300A	IOB direction register
P_IOC_Attrib	0x300B	IOB attribution register
P_IOA_WakeUp_Mask	0x300C	IOA wakeup mask
P_IOB_WakeUp_Mask	0x300D	IOB wakeup mask
P_IOC_WakeUp_Mask	0x300E	IOC wakeup mask
P_IOB_PL1M	0x3029	IOB[15:0] pull low high resistance option
P_IOC_SPI_Swap1	0x302A	IOC SPI swap register 1
P_IOC_SPI_Swap2	0x302B	IOC SPI swap register 2
P_IO_Swap	0x302C	IO swap register
P_IO_First	0x302D	IO first register

6.3.1 P_IOA_Data (R) (0x3000)

P_IOA_Data			0x3000		IOA Pad data			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	IOA_DATA[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	IOA_DATA[7:0]							
Default	0	0	0	0	0	0	0	0

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Bit	Function	Type	Description
[15:0]	IOA_DATA[15:0]	R	Read data from the I/O pad. This port is read only. Read this port to latch data on the I/O Port A for key change wakeup before getting into standby mode.

6.3.2 P_IOA_Buffer (R/W) (0x3001)

P_IOA_Buffer			0x3001		IOA buffer			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	IOA_BUF[15:8]							
Default	1	1	1	1	1	1	1	1

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	IOA_BUF[7:0]							
Default	1	1	1	1	1	1	1	1

Bit	Function	Type	Description
[15:0]	IOA_BUF[15:0]	R/W	Read data from data buffer. Writing data into P_IOA_Buffer(W) will synchronously change the P_IOA_Buffer and P_IOA_Data.

Note: The reading of P_IOA_Data and P_IOA_Buffer is through different physical path. Reading P_IOA_Data is reading data from I/O pad. Reading P_IOA_Buffer is reading data from I/O buffer. Please refer to the I/O block diagram.

6.3.3 P_IOA_Dir (R/W) (0x3002)

P_IOA_Dir			0x3002		IOA direction			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	IOA_DIR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	IOA_DIR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	IOA_DIR[15:0]	R/W	Read/Write direction-vector from/into the Direction Register.

6.3.4 P_IOA_Attrib (R/W) (0x3003)

P_IOA_Attrib			0x3003		IOA attribution			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**

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Function	IOA_ATTRIB[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	IOA_ATTRIB[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	IOA_ATTRIB[15:0]	R/W	Read/Write attribute vector from/into the Attribute Register. Read this port to latch data on the I/O Port A for key change wakeup before get into standby mode.

6.3.5 P_IOB_Data (R) (0x3004)

P_IOB_Data			0x3004		IOB Pad data			
Bit	15	14	13	12	11	10	9	8
Function	IOB_DATA[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	IOB_DATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	IOB_DATA[15:0]	R	Read data from the I/O pad. This port is read only. Read this port to latch data on the I/O Port B for key change wakeup before get into standby mode.

6.3.6 P_IOB_Buffer (R/W) (0x3005)

P_IOB_Buffer			0x3005		IOB buffer			
Bit	15	14	13	12	11	10	9	8
Function	IOB_BUF[15:8]							
Default	1	1	1	1	1	1	1	1

Bit	7	6	5	4	3	2	1	0
Function	IOB_BUF[7:0]							
Default	1	1	1	1	1	1	1	1

Bit	Function	Type	Description

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[15:0]	IOB_BUF[15:0]	R/W	Read data from data buffer. Writing data into P_IOB_Buffer(W) will synchronously change the P_IOB_Buffer and P_IOB_Data.

Note: The reading of P_IOB_Data and P_IOB_Buffer is through different physical path. Reading P_IOB_Data is reading data from I/O pad. Reading P_IOB_Buffer is reading data from I/O buffer. Please refer to the I/O block diagram.

6.3.7 P_IOB_Dir (R/W) (0x3006)

P_IOB_Dir			0x3006		IOB direction			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	IOB_DIR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	IOB_DIR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	IOB_DIR[15:0]	R/W	Read/Write direction vectors from/into the direction register.

6.3.8 P_IOB_Attrib (R/W) (0x3007)

P_IOB_Attrib			0x3007		IOB attribution			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	IOB_ATTRIB[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	IOB_ATTRIB[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	IOB_ATTRIB[15:0]	R/W	Read/Write attribute vector from/into the Attribute Register. Read this port to latch data on the I/O Port B for key change wakeup before entering standby mode.

6.3.9 P_IOC_Data (R) (0x3008)

P_IOC_Data			0x3008		IOC Pad data			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	-	-	-	-	-	-	-	-

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Bit	7	6	5	4	3	2	1	0
Function	IOC_DATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:8]	-	-	Reserved
[7:0]	IOC_DATA[7:0]	R	Read data from the I/O pad. This port is read only. Read this port to latch data on the I/O Port C for key change wakeup before get into standby mode.

6.3.10 P_IOC_Buffer (R/W) (0x3009)

P_IOC_Buffer			0x3009		IOC buffer			
Bit	15	14	13	12	11	10	9	8
Function	-							
Default	-	-	-	-	-	-	-	

Bit	7	6	5	4	3	2	1	0
Function	IOC_BUF[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:8]	-	-	Reserved
[7:0]	IOC_BUF[7:0]	R/W	Read data from data buffer. Writing data into P_IOC_Buffer(W) will synchronously change the P_IOC_Buffer and P_IOC_Data.

Note: The reading of P_IOC_Data and P_IOC_Buffer is through different physical path. Reading P_IOC_Data is reading data from I/O pad. Reading P_IOC_Buffer is reading data from I/O buffer. Please refer to the I/O block diagram.

6.3.11 P_IOC_Dir (R/W) (0x300A)

P_IOC_Dir			0x300A		IOC direction			
Bit	15	14	13	12	11	10	9	8
Function	-							
Default	-	-	-	-	-	-	-	-

Bit	7	6	5	4	3	2	1	0
Function	IOC_DIR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description

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[15:8]	-	-	Reserved
[7:0]	IOC_DIR[7:0]	R/W	Read/Write direction vectors from/into the direction register.

6.3.12 P_IOC_Attrib (R/W) (0x300B)

P_IOC_Attrib			0x300B		IOC attribution			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	-	-	-	-	-	-	-	-

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	IOC_ATTRIB[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:8]	-	-	Reserved
[7:0]	IOC_ATTRIB[7:0]	R/W	Read/Write attribute vector from/into the Attribute Register. Read this port to latch data on the I/O Port C for key change wakeup before entering standby mode.

6.3.13 P_IOA_WakeUp_Mask (R/W) (0x300C)

P_IOA_WakeUp_Mask			0x300C		ADC Setup Register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	WAKEUP_MSK[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	WAKEUP_MSK[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:0]	WAKEUP_MSK[15:0]	R/W	IOA wakeup mask (The IO configuration must be set as input.)	0: UnMask. 1: Mask

6.3.14 P_IOB_WakeUp_Mask (R/W) (0x300D)

P_IOB_WakeUp_Mask			0x300D		IOB wakeup mask			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	WAKEUP_MSK[15:8]							
Default	0	0	0	0	0	0	0	0

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Bit	7	6	5	4	3	2	1	0
Function	WAKEUP_MSK[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:0]	WAKEUP_MSK[15:0]	R/W	IOB wakeup mask (The IO configuration must be set as input.)	0: UnMask. 1: Mask

6.3.15 P_IOC_WakeUp_Mask (R/W) (0x300E)

P_IOC_WakeUp_Mask			0x300D		IOC wakeup mask			
Bit	15	14	13	12	11	10	9	8
Function	WAKEUP_MSK[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	WAKEUP_MSK[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:0]	WAKEUP_MSK[15:0]	R/W	IOC wakeup mask (The IO configuration must be set as input.)	0: UnMask. 1: Mask

6.3.16 P_IOB_PL1M (R/W) (0x3029)

P_IOB_PL1M			0x3029		IOB[15:0] pull low high resistance option			
Bit	15	14	13	12	11	10	9	8
Function	IOB_PL1M[15:8]							
Default	-	-	-	-	-	-	0	0

Bit	7	6	5	4	3	2	1	0
Function	IOB_PL1M[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:0]	IOB_PL1M[15:0]	R/W	Read/Write the PL1M Register to select IO pull low condition.	0: Normal resistance 1: High resistance

6.3.17 P_IOC_SPI_Swap1 (R/W) (0x302A)

P_IOC_SPI_Swap1			0x302A		IOC SPI swap register 1			
Bit	15	14	13	12	11	10	9	8

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Function	-					IOC4_SEL[2:0]		
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-					IOC2_SEL[2:0]		
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:11]	-	-	Reserved	-
[10:8]	IOC4_SEL[2:0]	R/W	IOC4 SPI selection	000 : SIO0 001 : SIO3 010 : SIO2 011 : SIO1 100 : SIO0 101 : CS
[7:3]	-	-	Reserved	-
[2:0]	IOC2_SEL[2:0]	R/W	IOC2 SPI selection	000 : SIO3 001 : SIO3 010 : SIO2 011 : SIO1 100 : SIO0 101 : CS

6.3.18 P_IOC_SPI_Swap2(R/W) (0x302B)

P_IOC_SPI_Swap2	0x302B			IOC SPI swap register 2				
Bit	15	14	13	12	11	10	9	8
Function	-					IOC7_SEL[2:0]		
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-	IOC6_SEL[2:0]			-	IOC5_SEL[2:0]		
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:11]	-	-	Reserved	-
[10:8]	IOC7_SEL[2:0]	R/W	IOC7 SPI selection	000 : SIO2 001 : SIO3 010 : SIO2 011 : SIO1 100 : SIO0 101 : CS

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[7]	-	-	Reserved	-
[6:4]	IOC6_SEL[2:0]	R/W	IOC6 SPI selection	000 : SIO1 001 : SIO3 010 : SIO2 011 : SIO1 100 : SIO0 101 : CS
[3]	-	-	Reserved	-
[2:0]	IOC5_SEL[2:0]	R/W	IOC5 SPI sel	000 : CS 001 : SIO3 010 : SIO2 011 : SIO1 100 : SIO0 101 : CS

6.3.19 P_IO_Swap (R/W) (0x302C)

P_IO_Swap	0x302C	IO swap register						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-				SPI2_IOB_MOSI	SPI2_IOA_MOSI	SP2_IOB_SWAP	SPI2_IOA_SWAP
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-				UART_IOB_INV	UART_IOA_INV	UART_IOB_SWAP	UART_IOA_SWAP
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:12]	-	-	Reserved	-
[11]	SPI2_IOB_MOSI	R/W	SPI2 IOB[15:12] as MOSI	0 : Disabled 1 : Enabled
[10]	SPI2_IOA_MOSI	R/W	SPI2 IOA[3:0] as MOSI	0 : Disabled 1 : Enabled
[9]	SP2_IOB_SWAP	R/W	SPI2 IOB MOSI / MISO swap	0 : Keep 1 : Swap
[8]	SPI2_IOA_SWAP	R/W	SPI2 IOA MOSI / MISO swap	0 : Keep 1 : Swap
[7:4]	-	-	Reserved	-
[3]	UART_IOB_INV	R/W	UART IOB Invert enable	0 : Not inverted 1 : Inverted
[2	UART_IOA_INV	R/W	UART IOA Invert enable	0 : Not inverted 1 : Inverted
[1]	UART_IOB_SWAP	R/W	UART IOB MOSI / MISO swap	0 : Keep 1 : Swap

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[0]	UART_IOA_SWAP	R/W	UART IOA MOSI / MISO swap	0 : Keep 1 : Swap

6.3.20 P_IO_First (R/W) (0x302D)

P_IO_First	0x302D	IO first register						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-				IOB15_FST	IOB14_FST	IOB13_FST	IOB12_FST
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-				IOA3_FST	IOA2_FST	IOA1_FST	IOA0_FST
Default	0	0	0	0	0	0	0	0

Bit	**Function**	**Type**	**Description**	**Condition**
[15:12]	-	-	Reserved	-
[11]	IOB15_FST	R/W	IOB[15] IO function first priority enable	0 : Disabled 1 : Enabled
[10]	IOB14_FST	R/W	IOB[14] IO function first priority enable	0 : Disabled 1 : Enabled
[9]	IOB13_FST	R/W	IOB[13] IO function first priority enable	0 : Disabled 1 : Enabled
[8]	IOB12_FST	R/W	IOB[12] IO function first priority enable	0 : Disabled 1 : Enabled
[7:4]	-	-	Reserved	-
[3]	IOA3_FST	R/W	IOA[3] IO function first priority enable	0 : Disabled 1 : Enabled
[2	IOA2_FST	R/W	IOA[2] IO function first priority enable	0 : Disabled 1 : Enabled
[1]	IOA1_FST	R/W	IOA[1] IO function first priority enable	0 : Disabled 1 : Enabled
[0]	IOA0_FST	R/W	IOA[0] IO function first priority enable	0 : Disabled 1 : Enabled

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7 I/O Special Function

7.1. General Description

Besides serving as an ordinary I/O, Port A/B/C also provides some special functions. In order to execute these functions properly, proper setup should be given. A description is depicted as follows:

All IOs support Key Wakeup Function; thus, we won't list it in the table.

IOA Special Function List

IOA_Func	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IOPWM	-	-	-	-	-	-	-	IOPWM8	IOPWM7	IOPWM6	IOPWM5	IOPWM4	IOPWM3	IOPWM2	IOPWM1	IOPWM0
HighSink	-	-	-	-	-	-	-	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
SPI2	-	-	-	-	-	-	-	-	-	-	SIO2	SIO3	SIO0	SIO1	SCK	CS
UART	-	-	-	-	-	-	-	-	-	-	-	-	TX	RX	-	-
IR	-	-	-	-	-		-	-	IROUT	-	-	IROUT	-	-	-	-
ADC	PGAVM	PGAI	PGAO	INP	INN0	INN1	INN2	INN3	INN4	CMPO	-	-	-	-	-	-
X32K	-	-	-	-	-	-	-	-	-	XTO	XTI	-	-	-	-	-
Touch	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

IOB Special Function List

IOB_Func	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI2	SIO0	SIO1	SCK	CS	SIO3	SIO2	-	-	-	-	-	-	-	-	-	-
UART	TX	RX	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CCP	-	-	-	-	-	-	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ext_INT	-	-	-	-	EXT1	EXT2	-	-	-	-	-		-	-	-	-
FeedBack	-	-	-	-	FI1	FI2	FO1	FO2	-	-	-	-	-		-	-
PL1M	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
PH100K	-	-	-	-	-	-	-	-	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Touch	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	-	-	-	-	-		-	-

IOC Special Function List

IOC_Func	-	-	-	-	-	-	-	-	7	6	5	4	3	2	1	0
SPI1	-	-	-	-	-	-	-	-	SIO2	SIO1	CS	SIO0	SCK	SIO3	-	-
ICE I/F	-	-	-	-	-	-	-	-	-	-	-	-	-	-	SCK	SDA

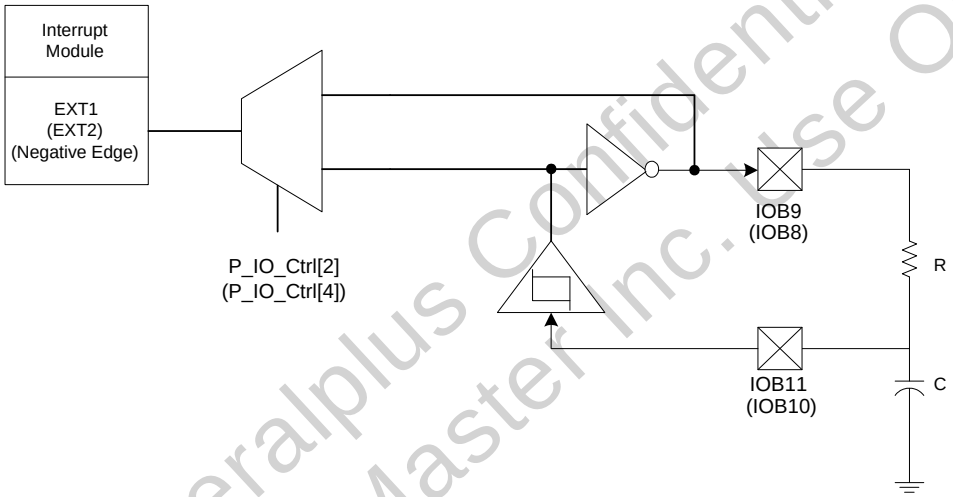
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7.2. IO Feedback Function

The following circuit shows the configuration of IOB8, IOB9, IOB10, and IOB11 with feedback application. With the feedback function, an OSC frequency can be obtained to EXT1 (EXT2) by simply adding a RC circuit between IOB8 (IOB9) and IOB10 (IOB11). To make this feedback function work properly, the IOB8 (IOB9) must be configured as “output inverted” and IOB10 (IOB11) as float input mode.

P_IO_Ctrl [5] 1 Enable the Feedback2 function of IOB10 and IOB8.
0 IOB10 and IOB8 are normal I/Os (Default).

P_IO_Ctrl [3] 1 Enable the Feedback1 function of IOB11 and IOB9.
0 IOB11 and IOB9 are normal I/Os (Default).



IO Feedback Application Circuit

Example:
This program can obtain an OSC frequency (square wave) from IOB9 by connecting an external RC circuit between IOB11 and IOB9. This external clock is able to enable IRQ5_EXT1 with the external interrupt source EXT1 or can be used as an external clock input of a timer.

```
// Set IOB11 as floating input, IOB9 as inverted output
    R1 = 0x0200;
    [P_IOB_Dir] = R1;
    R1 = 0x0800;
    [P_IOB_Attrib] = R1;
// Set feedback oscillator mode with rising edge trigger
    R1 = 0x0008;
    [P_IO_Ctrl] = R1;
```

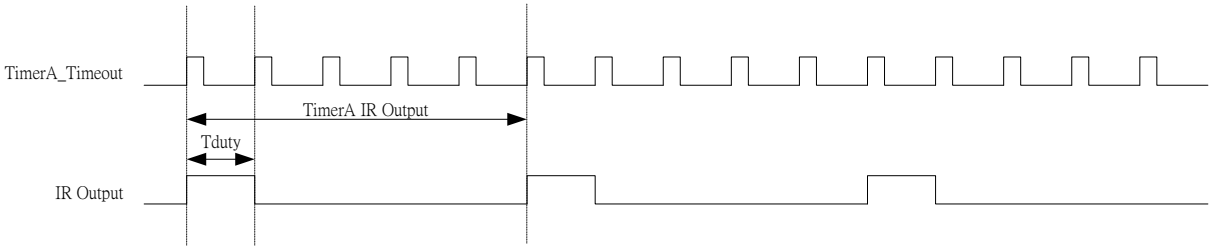
7.3. IR (INFRARED) FUNCTION

The IR allows one device to transmit data to another and vice versa. The GPCE500A IR function is able to transmit IR data through IOA4 or IOA7(IRTx). To make the IR function working properly, IOA4 or IOA7 must be configured as output (for IRTx). To transmit IR signal, you should set the b7~b6 (IRTxEn) of P_IO_Ctrl (R/W) ($300FH) and use the special function in IOA4 or IOA7 (IRTx). To generate the IR transmitting output signal IRTx, the TimerA or TimerB or TimerC must be set properly.

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Example waveform

Condition: Duty = 1/5 with TimerA..



7.4. Register Map

Name	Address	Description
P_IO_Ctrl	0x300F	Special IO function control register

7.4.1 P_IO_Ctrl (R/W) (0x300F)

Special IO function control register

P_IO_Ctrl			0x300F		Special IO function control register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-			IR_SEL	EXT2_SMT_EN	EXT1_SMT_EN	IR_DUTY[1:0]	
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	IR_CTRL[1:0]		EXT2_CTRL[1:0]		EXT1_CTRL[1:0]		UART_SEL	SPI2_SEL
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:13]	-	-	Reserved	-
[12[	IR_SEL	R/W	IR IO select	0 : IOA4 1 : IOA7
[11]	EXT2_SMT_EN	R/W	EXT2 schmitt trigger enable	0 : Disabled 1 : Enabled
[10]	EXT1_SMT_EN	R/W	EXT1 schmitt trigger enable	0 : Disabled 1 : Enabled
[9:8]	IR_DUTY[1:0]	R/W	IR duty cycle select	00 : 1/2 duty 01 : 1/3 duty 10 : 1/4 duty 11 : 1/5 duty
[7:6]	IR_CTRL[1:0]	R/W	IR function enable	00 : Disabled 01 : Timer A as source. 10 : Timer B as source. 11 : Timer C as source.
[5:4]	EXT2_CTRL[1:0]	R/W	EXT2 setting	00 : Normal and rising edge trigger

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Bit	Function	Type	Description	Condition
				01 : Normal and falling edge trigger 10 : Feedback and rising edge trigger 11 : Feedback and falling edge trigger
[3:2]	EXT1_CTRL[1:0]	R/W	EXT1 setting	00 : Normal and rising edge trigger 01 : Normal and falling edge trigger 10 : Feedback and rising edge trigger 11 : Feedback and falling edge trigger
[1]	UART_IOSEL	R/W	UART IO select	0 : IOA[3:2] 1 : IOB[15:14]
[0]	SPI2_IOSEL	R/W	SPI2 IO select	0 : IOA[5:0] 1 : IOB[15:10]

PS：If SPI2 and UART use the same IO port, SPI2 prevails.

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8 TIMER/COUNTER

8.1. General Description

GPCE500A provides three 16-bit timers (counters), Timer A, Timer B and Timer C. The clock source of Timer A/B/C is from clock source 1 and clock source 2 which are ANDED to form varieties of combinations. When timer overflows, a timeout signal (Timeout) is sent to CPU interrupt module to generate a timer interrupt signal. In addition, Timer A/B/C are hardware interrupt events can be used to trigger the latch action of the ADC (auto mode) and the PWM audio output. (Please refer to ***AUDIO OUTPUT*** section).

Writing a N value into P_TimerA_Data(R/W)($3010H) (or P_TimerB_Data(R/W)($3012H)) (or P_TimerC_Data(R/W)($3014H))and selecting an appropriated clock sources, timer will up-count from N, N+1, N+2,... 0xFFFE to 0xFFFF. An INT signal is generated at the moment of timer counting from "0xFFFF" to "0x0000" and the INT signal is processed by INT controller immediately. At the same time, N will be reloaded into timer and the timer starts counting again. (Please refer to ***INTERRUPT*** section).

In Timer A, the clock source 1 is a high frequency source and clock source 2 is a low frequency clock source. The combination of clock source 1 and 2 provides a variety of speeds to Timer A/Counter A. In Timer A block diagram, for the clock source selectors, the "1" represents pass signal, not gating and the "0" indicates deactivating the clock source. Clock source A selector has a "1" and a "0" inputs. Clock source B selector has a "1" input. For instance, if clock source A="1" (that is, P_Timer_Ctrl [b3~b0]="0001"), the clock output is depending on source 2. If clock source 1="0" (that is, P_Timer_Ctrl [b3~b0]="0000"), the Timer A is deactivated. The EXT2 are the external clock sources.

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8.2. Registers Map

Name	Address	Description
P_TimerA_Data	0x3010	Timer A data register
P_TimerA_CNTR	0x3011	Timer A counter register
P_TimerB_Data	0x3012	Timer B data register
P_TimerB_CNTR	0x3013	Timer B counter register
P_TimerC_Data	0x3014	Timer C data register
P_TimerC_CNTR	0x3015	Timer C counter register
P_Timer_Ctrl	0x3016	Timer control register
P_CCP_Ctrl	0x3017	Timer CCP control register
P_CCP_CNTR	0x3018	Timer CCP counter register

8.3. Timer/Counter

8.3.1 P_TimerA_Data (R/W) (0x3010)

P_TimerA_Data	0x3010	Timer A data register

Bit	15	14	13	12	11	10	9	8
Function	TMA_DATA[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	TMA_DATA [7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	TMA_DATA	R/W	Write value into the 16-bit pre-load register or read data from the 16-bit counter value.

Note: If users intend to program Timer A to a specific frequency, they can get the value to write from the following simple formula. Note that the desired timeout signal (timer overflow signal) frequency cannot exceed 200KHz and P_TimerA_Data cannot set value as 0xFFFF. For 200KHz or above, The accuracy of frequency, generated by timer, may be lowered.

Data to Write = 0x10000 – (Clock Source Frequency / Desired Frequency)

"Clock source frequency" is determined by the setting of P_Timer_Ctrl (0x3016).

"Desired Frequency" is the specific frequency as user's expectation.

User can then write the value coming from the formula to the P_TimerA_Data and turn on the specific interrupt for application. For example, under the CPU clock 81.92MHz, if user needs an 8KHz TimerA interrupt, the data to be written will be 0xD800 with the F_{SYSCLK} clock source.

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8.3.2 P_TimerA_CNTR (R/W) (0x3011)

P_TimerA_CNTR			0x3011		Timer A counter register			
Bit	15	14	13	12	11	10	9	8
Function	TMA_CNTR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	TMA_CNTR [7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	TMA_CNTR	R/W	Read this register will get the current Timer/Counter value. Write this register will re-load Timer/Counter value.

8.3.3 P_TimerB_Data (R/W) (0x3012)

P_TimerB_Data			0x3012		Timer B data register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	TMB_DATA[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	TMB_DATA [7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	TMB_DATA	R/W	Write value into the 16-bit pre-load register or read data from the 16-bit counter value.

Note: If users intend to program Timer B to a specific frequency, they can get the value to write from the following simple formula. Note that the desired timeout signal (timer overflow signal) frequency cannot exceed 200KHz and P_TimerB_Data cannot set value as 0xFFFF. For 200KHz or above, The accuracy of frequency, generated by timer, may be lowered.

Data to Write = 0x10000 – (Clock Source Frequency / Desired Frequency)

"Clock source frequency" is determined by the setting of P_Timer_Ctrl (0x3016).

"Desired Frequency" is the specific frequency as user's expectation.

User can then write the value coming from the formula to the P_TimerB_Data and turn on the specific interrupt for application. For example, under the CPU clock 81.92MHz, if user needs an 8KHz TimerB interrupt, the data to be written will be 0xD800 with the F_{SYSCLK} clock source.

8.3.4 P_TimerB_CNTR (R/W) (0x3013)

P_TimerB_CNTR			0x3011		Timer B counter register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	TMB_CNTR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	TMB_CNTR [7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	TMB_CNTR	R/W	Read this register will get the current Timer/Counter value. Write this register will re-load Timer/Counter value.

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8.3.5 P_TimerC_Data (R/W) (0x3014)

P_TimerC_Data			0x3014		Timer C data register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	TMC_DATA[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	TMC_DATA [7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	TMC_DATA	R/W	Write value into the 16-bit pre-load register or read data from the 16-bit counter value.

Note: If users intend to program Timer C to a specific frequency, they can get the value to write from the following simple formula. Note that the desired timeout signal (timer overflow signal) frequency cannot exceed 200KHz and P_TimerC_Data cannot set value as 0xFFFF. For 200KHz or above, The accuracy of frequency, generated by timer, may be lowered.

Data to Write = 0x10000 – (Clock Source Frequency / Desired Frequency)

"Clock source frequency" is determined by the setting of P_Timer_Ctrl (0x3016).

"Desired Frequency" is the specific frequency as user's expectation.

User can then write the value coming from the formula to the P_TimerC_Data and turn on the specific interrupt for application. For example, under the CPU clock 81.92MHz, if user needs an 8KHz TimerC interrupt, the data to be written will be 0xD800 with the F_{SYSCLK} clock source.

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8.3.6 P_TimerC_CNTR (R/W) (0x3015)

P_TimerC_CNTR			0x3015		Timer C counter register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	TMC_CNTR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	TMC_CNTR [7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	TMC_CNTR	R/W	Read this register will get the current Timer/Counter value. Write this register will re-load Timer/Counter value.

8.3.7 P_Timer_Ctrl (R/W) (0x3016)

P_Timer_Ctrl			0x3016	Timer control register				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-			TMA_DMA_EN	TMC_SEL[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	TMB_SEL[3:0]				TMA_SEL[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:13]	-	-	Reserved	-
[12]	TMA_DMA_EN	R/W	TMA DMA enable	0 : Disabled 1 : Enabled
[11:8]	TMC_SEL[3:0]	R/W	determine the clock source of Timer C	Refer to the table below
[7:4]	TMB_SEL[3:0]	R/W	determine the clock source of Timer B	Refer to the table below
[3:0]	TMA_SEL[3:0]	R/W	determine the clock source of Timer A	Refer to the table below

Timer A, B, C input resource= Input 1 AND Input 2 (falling edge counter)

TMXSEL	Input 1	Input 2
0000	'0'	'0'
0001	'1'	'1'
0010	EXT2	'1'
0011	$F_{IOSC32K}$	'1'
0100	F_{SYSCLK}	EXT2
0101	'0'	'0'
0110	F_{AUDCLK}	'1'
0111	$F_{AUDCLK / 8}$	'1'
1000	F_{SYSCLK}	'1'
1001	$F_{SYSCLK / 2}$	'1'
1010	$F_{SYSCLK / 4}$	'1'
1011	$F_{SYSCLK / 8}$	'1'
1100	$F_{SYSCLK / 16}$	'1'
1101	$F_{SYSCLK / 32}$	'1'
1110	$F_{SYSCLK / 64}$	'1'
1111	$F_{SYSCLK / 128}$	'1'

8.4. RTC (Real Time Clock)

Some applications require real time clock for time tracking. The GPCE500A series provides several RTC interrupt sources (from 2Hz to 4096Hz) based on built-in ROSC IOSC32K or Crystal. As an example of IRQ7_2Hz, the system wakes up every 0.5 seconds, which can be used for precise time counting.

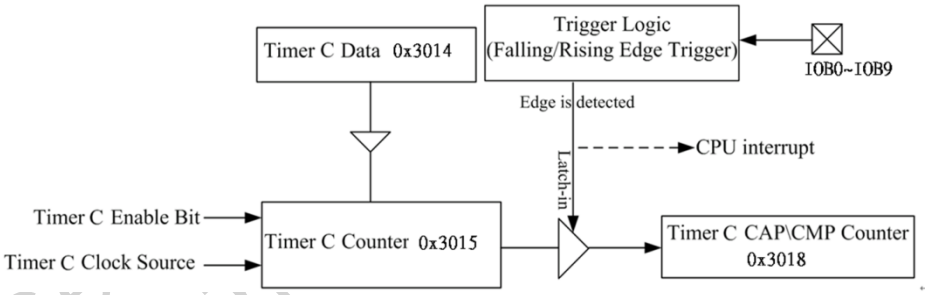
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Note: 32KHz clock source via option to choose IOSC32K or Crystal.

8.5. CCP (Capture/ Compare PWM)

8.5.1 Capture Mode

Timer C not only can be a timer/counter, it can also fetch an input pulse width in capture mode. To use capture mode, the pulse signal must be connected with the designated I/O (IOB0~IOB9), set Timer C to capture mode, select condition for capture (falling edge/rising edge) and set clock source and data for timer. Before Timer A is enabled, hardware will auto-load the timer data into timer counter; when Timer A is enabled, timer counter starts up-counting till it detects a input signal that meets the capture requirement (falling edge or rising edge). It will then, auto-load the timer counter data into CAP/CMP counter and issue an interrupt. The pulse width can be obtained by reading P _CCP_CNTR. Note that when timer counter overflows, interrupt is not issued. It only re-load timer counter data and starts up-counting. The data of CAP/CMP counter will only be changed when the capture requirement is met. Also, when counter is up-counting, write any data into timer counter, it will re-load its data and re-start up-counting again. Finally, before using the capture mode, make sure to set the signal I/O to input mode to avoid abnormal operation.



8.5.2 Comparison Mode

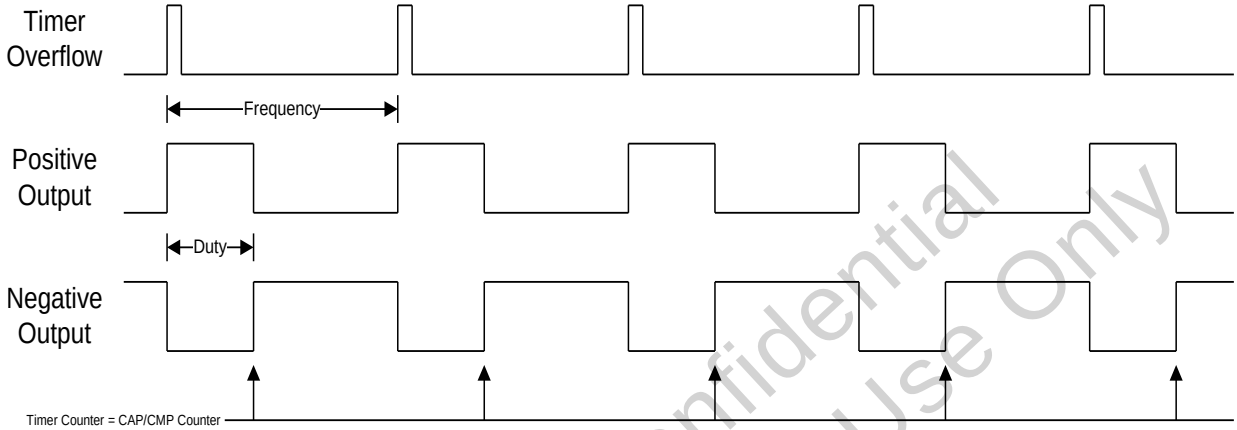
It is allowed to set TimerC to comparison mode for generating PWM pulse output. Also, the PWM pulse frequency and duty cycle are adjustable. Before using this mode, configure the timer to comparison mode, set timer's clock source, data and P _CCP_CNTR. Before the timer is enabled, hardware will auto-load the timer's data into timer counter. The designated I/O will output high or low based on its configuration. When timer is enabled, timer counter starts up-counting until the timer counter's value matches P _CCP_CNTR value. The I/O output will be inversed and issue an interrupt. Finally, when timer counter overflows, it reloads timer's data and starts up counting again. Thus, I/O output will return to the pre-set status (positive output/negative output). Moreover, when counter is up counting, writing any data into timer counter will reload the timer data and re-starts up counting. In conclusion, timer's data determines the PWM frequency and P _CCP_CNTR determines PWM duty cycle.

Note that before using comparison mode, please set the output I/O to output mode or abnormal operation may occur. If I/O is set to output low, the comparison mode is "Positive Output". If it is set to output high, the comparison mode is "Negative Output".

CMP_SEL	Configuration of Selected I/O	Output of Selected I/O
0 (Comparison Output)	Output Low	Positive Output

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CMP_SEL	Configuration of Selected I/O	Output of Selected I/O
0 (Comparison Output)	Output High	Negative Output
1 (Fixed Output)	Output Low	Output Low
1 (Fixed Output)	Output High	Output High



8.5.3 P _CCP_Ctrl (R/W) (0x3017)

P_CCP_Ctrl			0x3017		CCP control register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							CCP_DMA_EN
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	CCP_IO_SEL[3:0]				CMP_SEL	CAP_SEL	CCP_MODE_SEL[1:0]	
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:9]	-	-	Reserved	-
[8]	CCP_DMA_EN	R/W	CCP DMA enable	0 : Disabled 1 : Enabled
[7:4]	CCP_IO_SEL[3:0]	R/W	CCP IO selection	0000 : IOB[0] 0001 : IOB[1] 0010 : IOB[2] 0011 : IOB[3] 0100 : IOB[4] 0101 : IOB[5] 0110 : IOB[6] 0111 : IOB[7] 1000 : IOB[8] 1001 : IOB[9] Others : Reserved

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Bit	Function	Type	Description	Condition
[3]	CMP_SEL	R/W	Compare mode selection	0 : Comparison output 1 : Fixed output
[2]	CAP_SEL	R/W	Capture mode selection	0 : Capture by falling edge 1 : Capture by rising edge
[1:0]	CCP_MODE_SEL	R/W	CCP mode selection	00 : Disabled 01 : Capture 10 : Comparison 11 : Reserve

8.5.4 P _CCP_CNTR (R/W) (0x3018)

P_CCP_CNTR			0x3018	CCP data register				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	CCP_DATA[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	CCP_DATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	CCP_Data[15:0]	R	Latch TimerC data at Capture Mode
		W	Compare TimerC data at Comparison Mode

In compare mode, write this register to compare it with the timer counter value. The trigger signal will be sent to IOB0~9 to generate the PWM waveform. In capture mode, if the signal input by IOB0~9 meets the given edge condition, the capture is triggered and thus, TimerC counter value will be latched up and read by register.

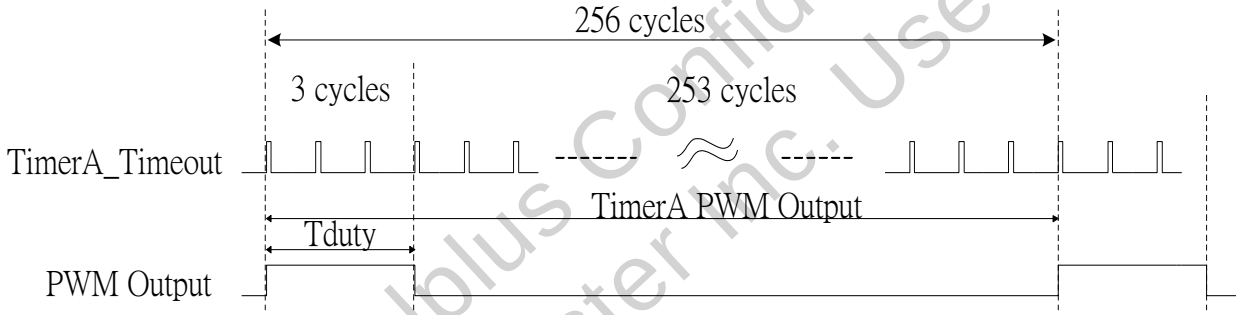
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9 IOPWM

9.1. General Description

GPCE500A provides eight IOPWM functions at IOA[0]~IOA[8], which their nine PWM control registers are configured in the same way. In addition, counter can generate time-out signal for input clock source to a four-bit (256 levels) PWM pulse width counter. A variety of PWM clock duration can be generated and output from IOA[0]~IOA[8]

The following example is a 3/256 duration cycle PWM0 output. The PWM0 waveform is made by selecting a pulse width through P_Timer_Ctrl(W)($3016H) [b3 ~ b0]. As a result, each 256 cycles of TimerA_Timerout will generate a pulse width, Tduty, as we defined in control port, P_PWM0_Ctrl (W)($3020H). These PWM0 signals are able to control the speed of motor or other devices.



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9.2. Registers Map

Name	Address	Description
P_PWM0_Ctrl	0x3020	IO PWM0 control register
P_PWM1_Ctrl	0x3021	IO PWM1 control register
P_PWM2_Ctrl	0x3022	IO PWM2 control register
P_PWM3_Ctrl	0x3023	IO PWM3 control register
P_PWM4_Ctrl	0x3024	IO PWM4 control register
P_PWM5_Ctrl	0x3025	IO PWM5 control register
P_PWM6_Ctrl	0x3026	IO PWM6 control register
P_PWM7_Ctrl	0x3027	IO PWM7 control register
P_PWM8_Ctrl	0x3028	IO PWM8 control register

9.2.1 P_PWMx_Ctrl (R/W) (0x3020~0x3028)

P_PWMx_Ctrl			0x3020		IO PWMx control register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	CNT_CLR	PWM_EN	-	PWM_MSK	PWM_SRC[2:0]			PWM_DUTY[8]
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**

Function	PWM_DUTY[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	CNT_CLR	W	Write only. Clear PWM counter for synchronization, But it can't effect the condition of PWM_Duty = "11111111"	0: None 1: Clear PWM counter
[14]	PWM_EN	R/W	IO PWM enable	0 : Disabled 1 : Enabled
[13]	-	-	Reserved	-
[12]	PWM_MASK*	R/W	IO PWM output mask	0 : UnMask 1 : Mask
[11:9]	PWM_SRC[2:0]	R/W	PWM clock source select	000: F_{32K} / 2 001: Timer A 010: Timer B 011: Timer C 100: F_{SYSTEM} 101: F_{SYSTEM} / 8 110: F_{SYSTEM} / 64 111: F_{SYSTEM} / 1024
[8:0]	PWM_DUTY[8:0]	R/W	PWM duty cycle setting	000h~100h: 0/256~256/256 1FFh: 1/2 duty 102h~1FEh: not available

* Mask IO PWM will disable the IO PWM output, but IO is still in special function, not general purpose IO.

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10 INTERRUPT

10.1. General Description

The *μ'nSP®* has two interrupt (INT) modes: FIQ (Fast Interrupt Request) and IRQ (Interrupt Request). The interrupt controller controls 25 INT sources, and every INT sources can set INT mode as FIQ or IRQ. The FIQ is a high-priority interrupt and IRQ is low-priority interrupt. An IRQ cannot be interrupted by another IRQ even two IRQs are generated simultaneously. If the interrupt requests of the same priority level are received simultaneously, an internal polling priority sequence (IRQ0 > IRQ1 > IRQ2 >…. > IRQ7) determines which request is served. Note that the polling priority in same level takes places only when two INTs occur simultaneously.

The P_INT_Ctrl(R/W)($3050H) and P_INT2_Ctrl(R/W)($3053H) configures the interrupt sources. To enable interrupt(s), write an "1" to the corresponding bit(s) in P_INT_Ctrl(R/W)($3050H) or P_INT2_Ctrl(R/W)($3053H). You can keep track of interrupt enable/disable status by reading this port. Writing '0' to P_INT_Ctrl(R/W)($3050H) or P_INT2_Ctrl(R/W)($3053H) will disable corresponding interrupt(s), but it will not clear the corresponding interrupt status flag existed in P_INT_Status(R/W)($3051) or P_INT2_Status(R/W)($3054). When INT occurs, program should read P_INT_Status(R/W)($3051) or P_INT2_Status(R/W)($3054) to distinguish which interrupt source is activated. CPU will then process the INT according to the IRQ priority. After an interrupt (FIQ or IRQ) is executed completely, its INT vector should be cleared by writing an "1" to P_INT_Status(R/W)($3051) or P_INT2_Status(R/W)($3054) correspondingly and continue to process the rest of low priority IRQs until all INTs are executed. Next, CPU will leave from interrupt service routine and continue to execute other instructions. After each interrupt is completed, writing data to the correspondent bits in P_INT_Status(R/W)($3051) or P_INT2_Status(R/W)($3054) to clear the interrupt will allow next interrupt to come in.

For example, suppose the b10 of P_INT2_Ctrl(R/W)($3050H) is enabled, the IRQ6_4KHz is activated. When the INT occurs, the b10 of P_INT2_Status(R/W)($3051) turns HIGH. CPU should process the IRQ6_4KHz in INT service routine. After the INT is completed, write '0x0400' to P_INT2_Status(R/W)($3051) to clear the IRQ6_4KHz. This method will directly clear the processed IRQ without influencing other enabled interrupt sources.

In order to activate the interrupt service, the IRQ, FIQ or INT instructions need to be programmed correspondingly as well. User needs to explicitly activate the desired interrupt set after the P_INT_Ctrl(R/W)($3050H) or P_INT2_Ctrl(R/W)($3053H) is correctly set such as "FIQ ON" or "IRQ ON". In case that user uses INT to activate the interrupt set and only one of the parameters.

To access the interrupts, *μ'nSP®* reserves the following words for the interrupt entry points:
_FIQ, _IRQ0, _IRQ1, _IRQ2, _IRQ3, _IRQ4, _IRQ5, _IRQ6, _IRQ7
It also reserves _BREAK (INT Vector: 0x4015) for "Software Break" interrupt entry point.

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Interrupt Source	Interrupt	Priority	Reserved Word
FIQ (selected by P_FIQ_SEL)	FIQ	HIGH	_FIQ
Timer A	IRQ0_TMA	LOW	_IRQ0
Timer B	IRQ1_TMB	LOW	_IRQ1
Timer C	IRQ2_TMC	LOW	_IRQ2
DMA1	IRQ3_DMA1	LOW	_IRQ3
DMA2	IRQ3_DMA2	LOW	_IRQ3
SPI1	IRQ3_SPI1	LOW	_IRQ3
SPI2	IRQ3_SPI2	LOW	_IRQ3
ADC	IRQ3_ADC	LOW	_IRQ3
KEY	IRQ4_KEY	LOW	_IRQ4
EXT1	IRQ5_EXT1	LOW	_IRQ5
EXT2	IRQ5_EXT2	LOW	_IRQ5
SPU	IRQ5_SPU	LOW	_IRQ5
4KHz	IRQ6_4KHz	LOW	_IRQ6
2KHz	IRQ6_2KHz	LOW	_IRQ6
512Hz	IRQ6_512Hz	LOW	_IRQ6
CTSTMA	IRQ6_CTSTMA	LOW	_IRQ6
CTSTMB	IRQ6_CTSTMB	LOW	_IRQ6
64Hz	IRQ7_64Hz	LOW	_IRQ7
16Hz	IRQ7_16Hz	LOW	_IRQ7
2Hz	IRQ7_2Hz	LOW	_IRQ7
UART	IRQ7_UART	LOW	_IRQ7
Stack	IRQ7_STACK	LOW	_IRQ7
CPURX	IRQ7_CPURX	LOW	_IRQ7

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10.2. Registers Map

Name	Address	Description
P_INT_Ctrl	0x3050	Interrupt control register
P_INT_Status	0x3051	Interrupt status register
P_FIQ_Sel	0x3052	FIQ select register
P_INT2_Ctrl	0x3053	Interrupt control register 2
P_INT2_Status	0x3054	Interrupt status register 2
P_FIQ2_Sel	0x3055	FIQ select register 2

10.2.1 P_INT_Ctrl (R/W) (0x3050)

P_INT_Ctrl			0x3050		Interrupt control register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	IRQ0_TMA	IRQ1_TMB	IRQ2_TMC	IRQ3_DMA1	IRQ3_ADC	-	IRQ3_DMA2	IRQ3_SPI2
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	IRQ3_SPI1	-	-	IRQ4_KEY	-	IRQ5_SPU	IRQ5_EXT1	IRQ5_EXT2
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	IRQ0_TMA	R/W	Timer A interrupt selection	0 : Disabled 1 : Enabled
[14]	IRQ1_TMB	R/W	Timer B interrupt selection	0 : Disabled 1 : Enabled
[13]	IRQ2_TMC	R/W	Timer C interrupt selection	0 : Disabled 1 : Enabled
[12]	IRQ3_DMA1	R/W	DMA1 interrupt selection	0 : Disabled 1 : Enabled
[11]	IRQ3_ADC	R/W	ADC interrupt selection	0: Disabled 1: Enabled
[10]	-	-	Reserved. Must be '0'	-
[9]	IRQ3_DMA2	R/W	DMA2 interrupt selection	0 : Disabled 1 : Enabled
[8]	IRQ3_SPI2	R/W	SPI2 interrupt selection	0 : Disabled 1 : Enabled
[7]	IRQ3_SPI1	R/W	SPI1 interrupt selection	0 : Disabled 1 : Enabled
[6:5]	-	-	Reserved	-
[4]	IRQ4_KEY	R/W	Key change Wakeup interrupt selection	0 : Disabled 1 : Enabled
[3]	-	-	Reserved	-
[2]	IRQ5_SPU	R/W	SPU interrupt selection	0 : Disabled 1 : Enabled
[1]	IRQ5_EXT1	R/W	EXT1 interrupt selection	0 : Disabled 1 : Enabled
[0]	IRQ5_EXT2	R/W	EXT2 interrupt selection	0 : Disabled 1 : Enabled

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10.2.2 P_INT_Status (R/W) (0x3051)

P_INT_Status			0x3051		Interrupt status register			
Bit	15	14	13	12	11	10	9	8
Function	IRQ0_TMA	IRQ1_TMB	IRQ2_TMC	IRQ3_DMA1	IRQ3_ADC	-	IRQ3_DMA2	IRQ3_SPI2
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	IRQ3_SPI1	-	-	IRQ4_KEY	-	IRQ5_SPU	IRQ5_EXT1	IRQ5_EXT2

Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	IRQ0_TMA	R/W1C	Timer A interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[14]	IRQ1_TMB	R/W1C	Timer B interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[13]	IRQ2_TMC	R/W1C	Timer C interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[12]	IRQ3_DMA1	R/W1C	Timer C interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[11]	IRQ3_ADC	R/W1C	ADC interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[10]	-	-	Reserved. Must be ‘0’	-
[9]	IRQ3_DMA2	R/W1C	DMA2 interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[8]	IRQ3_SPI2	R/W1C	SPI2 interrupt status	Write: 0: No effect 1: Clear status Read:

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Bit	Function	Type	Description	Condition
				0: No interrupt 1: Interrupt event is issued
[7]	IRQ3_SPI1	R/W1C	SPI1 interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[6:5]	-	-	Reserved	-
[4]	IRQ4_KEY	R/W1C	Key change Wakeup interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[3]	-	-	-	-
[2]	IRQ5_SPU	R/W1C	SPU interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[1]	IRQ5_EXT1	R/W1C	EXT1 interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[0]	IRQ5_EXT2	R/W1C	EXT2 interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued

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10.2.3 P_FIQ_Sel (R/W) (0x3052)

P_FIQ_Sel			0x3052		FIQ select register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	IRQ0_TMA	IRQ1_TMB	IRQ2_TMC	IRQ3_DMA1	IRQ3_ADC	-	IRQ3_DMA2	IRQ3_SPI2
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	IRQ3_SPI1	-	-	IRQ4_KEY	-	IRQ5_SPU	IRQ5_EXT1	IRQ5_EXT2
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	IRQ0_TMA	R/W	Timer A interrupt selection	0: IRQ 1: FIQ
[14]	IRQ1_TMB	R/W	Timer B interrupt selection	0: IRQ 1: FIQ
[13]	IRQ2_TMC	R/W	Timer C interrupt selection	0: IRQ 1: FIQ
[12]	IRQ3_DMA1	R/W	DMA1 interrupt selection	0: IRQ 1: FIQ
[11]	IRQ3_ADC	R/W	ADC interrupt selection	0: IRQ 1: FIQ
[10]	-	-	Reserved. Must be '0'	-
[9]	IRQ3_DMA2	R/W	DMA2 interrupt selection	0: IRQ 1: FIQ
[8]	IRQ3_SPI2	R/W	SPI2 interrupt selection	0: IRQ 1: FIQ
[7]	IRQ3_SPI1	R/W	SPI1 interrupt selection	0: IRQ 1: FIQ
[6:5]	-	-	Reserved	-
[4]	IRQ4_KEY	R/W	Key change Wakeup interrupt selection	0: IRQ 1: FIQ
[3]	-	-	Reserved	-
[2]	IRQ5_SPU	R/W	SPU interrupt selection	0: IRQ 1: FIQ
[1]	IRQ5_EXT1	R/W	EXT1 interrupt selection	0: IRQ 1: FIQ
[0]	IRQ5_EXT2	R/W	EXT2 interrupt selection	0: IRQ 1: FIQ

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10.2.4 P_INT2_Ctrl (R/W) (0x3053)

P_INT2_Ctrl			0x3053		Interrupt control register 2			
Bit	15	14	13	12	11	10	9	8
Function	-			IRQ6_CTSTMA	IRQ6_CTSTMB	IRQ6_4KHz	IRQ6_2KHz	IRQ6_512Hz
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-		IRQ7_STACK	IRQ7_CPURX	IRQ7_UART	IRQ7_64Hz	IRQ7_16Hz	IRQ7_2Hz

Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:13]	-	-	Reserved	-
[12]	IRQ6_CTSTMA	R/W	CTS TimerA interrupt selection	0: Disabled 1: Enabled
[11]	IRQ6_CTSTMB	R/W	CTS TimerB interrupt selection	0: Disabled 1: Enabled
[10]	IRQ6_4KHz	R/W	4096Hz interrupt selection	0: Disabled 1: Enabled
[9]	IRQ6_2KHz	R/W	2048Hz interrupt selection	0: Disabled 1: Enabled
[8]	IRQ6_512Hz	R/W	512Hz interrupt selection	0: Disabled 1: Enabled
[7:6]	-	-	Reserved	-
[5]	IRQ7_STACK	R/W	Stack underflow interrupt selection	0: Disabled 1: Enabled
[4]	IRQ7_CPURX	R/W	-	0: Disabled 1: Enabled
[3]	IRQ7_UART	R/W	UART interrupt selection	0: Disabled 1: Enabled
[2]	IRQ7_64Hz	R/W	64Hz interrupt selection	0: Disabled 1: Enabled
[1]	IRQ7_16Hz	R/W	16Hz interrupt selection	0: Disabled 1: Enabled
[0]	IRQ7_2Hz	R/W	2Hz interrupt selection	0: Disabled 1: Enabled

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10.2.5 P_INT2_Status (R/W) (0x3054)

P_INT2_Status	0x3054	Interrupt status register 2

Bit	15	14	13	12	11	10	9	8
Function	-			IRQ6_CTSTMA	IRQ6_CTSTMB	IRQ6_4KHz	IRQ6_2KHz	IRQ6_512Hz
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-		IRQ7_STACK	IRQ7_CPURX	IRQ7_UART	IRQ7_64Hz	IRQ7_16Hz	IRQ7_2Hz
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:13]	-	-	Reserved	-
[12]	IRQ6_CTSTMA	R/W1C	CTS TimerA interrupt status	Write:

Bit	Function	Type	Description	Condition
				0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[11]	IRQ6_CTSTMB	R/W1C	CTS TimerB interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[10]	IRQ6_4KHz	R/W1C	4096Hz interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[9]	IRQ6_2KHz	R/W1C	2048Hz interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[8]	IRQ6_512Hz	R/W1C	512Hz interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[7:6]	-	-	Reserved	-
[5]	IRQ7_STACK	R/W1C	Stack underflow interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[4]	IRQ7_CPURX	R/W1C	-	Write:

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Bit	Function	Type	Description	Condition
				0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[3]	IRQ7_UART	R/W1C	UART interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[2]	IRQ7_64Hz	R/W1C	64Hz interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[1]	IRQ7_16Hz	R/W1C	16Hz interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[0]	IRQ7_2Hz	R/W1C	2Hz interrupt status	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued

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10.2.6 P_FIQ2_Sel (R/W) (0x3055)

P_FIQ2_Sel			0x3055	FIQ select register 2				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-			IRQ6_CTSTMA	IRQ6_CTSTMB	IRQ6_4KHz	IRQ6_2KHz	IRQ6_512Hz
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-		IRQ7_STACK	IRQ7_CPURX	IRQ7_UART	IRQ7_64Hz	IRQ7_16Hz	IRQ7_2Hz
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:13]	-	-	Reserved	-
[12]	IRQ6_CTSTMA	R/W	CTS TimerA interrupt selection	0: IRQ 1: FIQ
[11]	IRQ6_CTSTMB	R/W	CTS TimerB interrupt selection	0: IRQ 1: FIQ
[10]	IRQ6_4KHz	R/W	4096Hz interrupt selection	0: IRQ 1: FIQ
[9]	IRQ6_2KHz	R/W	2048Hz interrupt selection	0: IRQ 1: FIQ
[8]	IRQ6_512Hz	R/W	512Hz interrupt selection	0: IRQ 1: FIQ
[7:6]	-	-	Reserved	-
[5]	IRQ7_STACK	R/W	Stack underflow interrupt selection	0: IRQ 1: FIQ
[4]	IRQ7_CPURX	R/W	-	0: IRQ 1: FIQ
[3]	IRQ7_UART	R/W	UART interrupt selection	0: IRQ 1: FIQ
[2]	IRQ7_64Hz	R/W	64Hz interrupt selection	0: IRQ 1: FIQ
[1]	IRQ7_16Hz	R/W	16Hz interrupt selection	0: IRQ 1: FIQ
[0]	IRQ7_2Hz	R/W	2Hz interrupt selection	0: IRQ 1: FIQ

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10.3. Summary of Interrupt Control/Status/FIQ_SEL Ports

P_INTx_Ctrl (W) (1)	Enable INT/wakeup function.
P_INTx_Ctrl (W) (0)	Disable INT/wakeup function, but it will not clear the existed interrupt event in P_INT_Status (R).
P_INTx_Ctrl (R) (1)	Interrupt Enable Status
P_INTx_Ctrl (R) (0)	Interrupt Disable Status
P_INTx_Status (W) (1)	Clear the INT event
P_INTx_Status (W) (0)	Non-changed INT status
P_INTx_Status (R) (1)	INT event occurred
P_INTx_Status (R) (0)	INT event not occurred
P_FIQx_SEL (W) (1)	Configure as FIQ
P_FIQx_SEL (W) (0)	Configure as IRQ (default)

P_FIQx_SEL (R) (1)	INT status as FIQ
P_FIQx_SEL (R) (0)	INT status as IRQ

Ps. INTx include INT and INT2.

Example:

Demonstrate the difference of using FIQ, IRQ and INT to activate the interrupt service.

```
// Use IRQ to turn on/off IRQ
IRQ ON;              // IRQ is on
IRQ OFF;             // IRQ is off

// Use FIQ to turn on/off FIQ
FIQ ON;              // FIQ is on
FIQ OFF;             // FIQ is off

// Use INT to turn on/off FIQ, IRQ
INT FIQ IRQ;         // FIQ and IRQ are on
INT FIQ;             // FIQ is on and IRQ is off
INT IRQ;             // FIQ is off and IRQ is on
INT OFF;             // FIQ and IRQ are off
```

Example:

Use Timer A to enable FIQ_TMA(Fast interrupt source), and output a toggle signal from IOB.

```
//Set IOB as output low
    R1 = 0xFFFF;
    [P_IOB_Attrib] = R1;
    [P_IOB_Dir] = R1;
    R1 = 0x0000;
    [P_IOB_Buffer] = R1;
//Set TimerA = 8KHz at FSYSCLK Speed = 81.92MHz
    R1 = 0xD800;
    [P_TimerA_Data] = R1;
    [P_TimerA_CNTR] = R1;
    R1 = 0x000F;
    [P_TimerA_Ctrl] = R1;
//Set FIQ_TMA source
    R1 = 0x8000;
    [P_INT_Ctrl] = R1;
    [P_FIQ_SEL] = R1;
    FIQ ON;              // Turn on FIQ
// When TimerA (Interrupt source) is set, GPCE500A will enter FIQ sub-routine according
// to user-define frequency.

……
//FIQ sub-routine
```

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```
_FIQ:
        PUSH R1, R5 to [SP];
        R1 = 0x8000;
        TEST R1, [P_INT_Ctrl];
        JNE L_Quit_FIQ;

L_FIQ_TMA:                      // FIQ TMA entrance
        [P_INT_Status] = R1;    //Clear FIQ TMA Event
        R1 = [P_IOB_Buffer];
        R1 ^= 0xFFFF;           // Toggle the IOB signal when FIQ_TMA occurs
        [P_IOB_Buffer] = R1;

L_Quit_FIQ:
        POP R1, R5 FROM [SP];
        RETI;                   //Return to main program

_IRQ0:
RETI;

_IRQ1:
RETI;

_IRQ2:
RETI;

_IRQ3:
RETI;

_IRQ4:
RETI;

_IRQ5:
RETI;

_IRQ6:
RETI;

_IRQ7:
RETI;
```

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11 SPU

11.1. General Description

A SPU (Sound Processing Unit) is a hardware 8-channel wave-table synthesizer which has many features like A3400proE 4-bit/ A3400proE 5-bit/PCM 8-bit/PCM 16-bit, loop/concatenation, envelope volume control, main volume control, and enable/disable interrupts control, etc. The SPU sequencer and the envelope updater are simulated by software. The synthesizer generates interrupts at the end of a speech or tone color. The A3400proE 4-bit/A3400proE 5-bit also provide interrupt event tags, programmers can edit event tag to trigger customized events.

11.1.1 PCM/ A3400pro 4-bit Format for Wave-Table

SPU synthesizer totally supports two kinds of speech and instrument formats: A3400proE 4-bit/ A3400proE 5-bit provided by Generalplus) format, and PCM format. By configuring P_SPU_ChX_DPTR_H[15:14] (P_SPU_ChX_DPTR_H = Data Pointer High word), synthesizer will recognize the speech and instrument formats exactly, A3400pro 4-bit (P_SPU_ChX_DPTR_H[15:14] = 00), A3400pro 5-bit (P_SPU_ChX_DPTR_H[15:14] = 01), PCM 8-bit (P_SPU_ChX_DPTR_H[15:14] = 10) or PCM 16-bit (P_SPU_ChX_DPTR_H[15:14] = 11).

11.1.2 Tone Color Relative/Absolute/Concatenate Jump Mode

After reaching the end of a tone color, there are three modes (relative, absolute and concatenated) for the synthesizer to jump to the next tone color. Read the following subsections carefully to understand these three modes.

11.1.3 Relative Jump Mode

In relative jump mode, if the tone color is an A3400proE 4-bit/ A3400proE 5-bit format, the synthesizer will directly fetch the loop length from encoded tone-color data. If tone color is a PCM format, the synthesizer will fetch the LPLEN after high byte and low byte of P_SPU_ChX_LoopLength are set by user. In relative jump mode of PCM, the P_Ch0LpLen cannot be set “1” by user, or the noise will be presented. On the other hand, in relative jump mode of A3400proE 4-bit/ A3400proE 5-bit, the P_SPU_ChX_LoopLength is used directly by SPU and user cannot use it. The following diagram shows an example of PCM tone colors, A and B.

A	A+1	A+2	End Code	B	B+1	B+2	End Code

(1). P_SPU_ChX_LoopLength = 0, SPU will play A→A+1→A+2→B→B+1…(Play next tone color of the memory)

(2). P_SPU_ChX_LoopLength = 1, not allowed

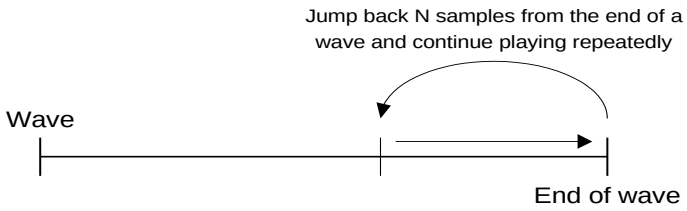
(3). P_SPU_ChX_LoopLength = 2, SPU will play A→A+1→A+2→A+2→A+2…(Used to stop tone color)

(4). P_SPU_ChX_LoopLength = 3, SPU will play A→A+1→A+2→A+1→A+2→A+1→A+2…(Jump back to last 2 samples)

(5). P_SPU_ChX_LoopLength = N (N>1), SPU will repeat the last (N-1) sample

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The diagram of relative jump mode:



11.2. Registers Map

Name	Address	Description
P_SPU_Volume_Ctrl	0x3060	Master volume control register
P_SPU_Mixer_Ctrl	0x3061	Channel mixer control register
P_SPU_CH_Ctrl	0x3062	SPU channel control register
P_SPU_INT_Ctrl	0x3063	SPU channel interrupt control register
P_SPU_INT_Status	0x3069	SPU channel interrupt status flag
P_SPU_CLK_Ctrl	0x306F	SPU base clock

Name	Address	Description
P_SPU_ChX_PhaseAdder	0x28x0	Phase adder counter register
P_SPU_ChX_SampleFreq	0x28x1	Sample frequency register
P_SPU_ChX_DPTR_L	0x28x2	Data pointer low word register
P_SPU_ChX_DPTR_H	0x28x3	Data pointer high word register
P_SPU_ChX_LoopLength	0x28x4	Loop length register
P_SPU_ChX_Filter_S2	0x28x5	Hardware filter register
P_SPU_ChX_Filter_S1	0x28x6	Hardware filter register
P_SPU_ChX_Volume_Ctrl	0x28x7	Channel volume & Next channel volume
P_SPU_ChX_NextDPTR_L	0x28x8	Next data pointer low word register
P_SPU_ChX_NextDPTR_H	0x28x9	Next data pointer high word register
P_SPU_ChX_EventTag	0x28xA	Updated when Event Tag is detected by SPU hardware

Note: 'X' is channel number and the value would be 0~7

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	Address	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
P_SPU_ChX_PhaseAdder	0x28x0	PhaseCnt[15:0]															
P_SPU_ChX_SampleFreq	0x28x1	FS[15:0]															
P_SPU_ChX_DPTR_L	0x28x2	DPTR[15:0]															
P_SPU_ChX_DPTR_H	0x28x3	FORMAT		ABS	Reserved		DPTR[16:16]										
P_SPU_ChX_LoopLength	0x28x4	LELEN[15:8]								LELEN[7:0]							
P_SPU_ChX_Filter_S2	0x28x5	S2															
P_SPU_ChX_Filter_S1	0x28x6	S1															
P_SPU_ChX_Volume_Ctrl	0x28x7	VOLN								VOL							
P_SPU_ChX_NextDPTR_L	0x28x8	DPTR_N[15:0]															
P_SPU_ChX_NextDPTR_H	0x28x9	FORMATN		ABSN	Reserved		DPTR_N[26:16]										
P_SPU_ChX_EventTag	0x28xA	Ext_Tag								Int_Tag							

Note: 'x' is channel number and the value would be 0~7

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11.2.1 P_SPU_Volume_Ctrl (R/W) (0x3060)

P_SPU_Volume_Ctrl		0x3060		Master volume control register				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	SWING	MVC[6:0]						
Default	0	1	1	1	1	1	1	1

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	SWING	R/W	Set the maximum swing range of sound for each channel Write	0: 1/2 swing 1: Full swing
[14:8]	MVC[6:0]	R/W	Master volume control	0x00: Mute 0x7F: Maximum output
[7:0]	-	-	Reserved	-

11.2.2 P_SPU_Mixer_Ctrl (R/W) (0x3061)

P_SPU_Mixer_Ctrl		0x3061		Channel mixer control register				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	MIX7	MIX6	MIX5	MIX4	MIX3	MIX2	MIX1	MIX0
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	MIX7	R/W	Enable/Disable the channel 7 to be mixed	0: Disabled 1: Enabled
[14]	MIX6	R/W	Enable/Disable the channel 6 to be mixed	0: Disabled 1: Enabled
[13]	MIX5	R/W	Enable/Disable the channel 5 to be mixed	0: Disabled 1: Enabled
[12]	MIX4	R/W	Enable/Disable the channel 4 to be mixed	0: Disabled 1: Enabled
[11]	MIX3	R/W	Enable/Disable the channel 3 to be mixed	0: Disabled 1: Enabled
[10]	MIX2	R/W	Enable/Disable the channel 2 to be mixed	0: Disabled 1: Enabled

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Bit	Function	Type	Description	Condition
[9]	MIX1	R/W	Enable/Disable the channel 1 to be mixed	0: Disabled 1: Enabled
[8]	MIX0	R/W	Enable/Disable the channel 0 to be mixed	0: Disabled 1: Enabled
[7:0]	-	-	Reserved	-

11.2.3 P_SPU_CH_Ctrl (R/W) (0x3062)

P_SPU_CH_Ctrl			0x3062		SPU channel control register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	CONC7	CONC6	CONC5	CONC4	CONC3	CONC2	CONC1	CONC0
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	CH7_EN	CH6_EN	CH5_EN	CH4_EN	CH3_EN	CH2_EN	CH1_EN	CH0_EN
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	CONC7	R/W	Channel 7 concatenated jump mode	0: Disabled 1: Enabled
[14]	CONC6	R/W	Channel 6 concatenated jump mode	0: Disabled 1: Enabled
[13]	CONC5	R/W	Channel 5 concatenated jump mode	0: Disabled 1: Enabled
[12]	CONC4	R/W	Channel 4 concatenated jump mode	0: Disabled 1: Enabled
[11]	CONC3	R/W	Channel 3 concatenated jump mode	0: Disabled 1: Enabled
[10]	CONC2	R/W	Channel 2 concatenated jump mode	0: Disabled 1: Enabled
[9]	CONC1	R/W	Channel 1 concatenated jump mode	0: Disabled 1: Enabled
[8]	CONC0	R/W	Channel 0 concatenated jump mode	0: Disabled 1: Enabled
[7]	CH7_EN	R/W	Enable/Disable the channel 7	0: Disabled 1: Enabled
[6]	CH6_EN	R/W	Enable/Disable the channel 6	0: Disabled 1: Enabled
[5]	CH5_EN	R/W	Enable/Disable the channel 5	0: Disabled 1: Enabled
[4]	CH4_EN	R/W	Enable/Disable the channel 4	0: Disabled 1: Enabled
[3]	CH3_EN	R/W	Enable/Disable the channel 3	0: Disabled 1: Enabled

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Bit	Function	Type	Description	Condition
[2]	CH2_EN	R/W	Enable/Disable the channel 2	0: Disabled 1: Enabled
[1]	CH1_EN	R/W	Enable/Disable the channel 1	0: Disabled 1: Enabled
[0]	CH0_EN	R/W	Enable/Disable the channel 0	0: Disabled 1: Enabled

11.2.4 P_SPU_INT_Ctrl (R/W) (0x3063)

P_SPU_INT_Ctrl			0x3063		SPU channel interrupt control register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	CH7_INTE	CH6_INTE	CH5_INTE	CH4_INTE	CH3_INTE	CH2_INTE	CH1_INTE	CH0_INTE
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:8]	-	-	Reserved	-
[7]	CH7_INTE	R/W	Enable/Disable the channel 7 interrupt	0: Disabled 1: Enabled
[6]	CH6_INTE	R/W	Enable/Disable the channel 6 interrupt	0: Disabled 1: Enabled
[5]	CH5_INTE	R/W	Enable/Disable the channel 5 interrupt	0: Disabled 1: Enabled
[4]	CH4_INTE	R/W	Enable/Disable the channel 4 interrupt	0: Disabled 1: Enabled
[3]	CH3_INTE	R/W	Enable/Disable the channel 3 interrupt	0: Disabled 1: Enabled
[2]	CH2_INTE	R/W	Enable/Disable the channel 2 interrupt	0: Disabled 1: Enabled
[1]	CH1_INTE	R/W	Enable/Disable the channel 1 interrupt	0: Disabled 1: Enabled
[0]	CH0_INTE	R/W	Enable/Disable the channel 0 interrupt	0: Disabled 1: Enabled

11.2.5 P_SPU_INT_Status (R/W) (0x3069)

P_SPU_INT_Status			0x3069		SPU channel interrupt status			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	0	0	0	0	0	0	0	0

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Bit	7	6	5	4	3	2	1	0
Function	CH7_STS	CH6_STS	CH5_STS	CH4_STS	CH3_STS	CH2_STS	CH1_STS	CH0_STS
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:8]	-	-	-	-
[7]	CH7_STS	R/W	The interrupt status of channel 7	Write : 0: No effect 1: Clear this bit Read : 0: CH7 interrupt not occurred 1: CH7 interrupt occurred
[6]	CH6_STS	R/W	The interrupt status of channel 6	Write : 0: No effect 1: Clear this bit Read : 0: CH6 interrupt not occurred 1: CH6 interrupt occurred
[5]	CH5_STS	R/W	The interrupt status of channel 5	Write : 0: No effect 1: Clear this bit Read : 0: CH5 interrupt not occurred 1: CH5 interrupt occurred
[4]	CH4_STS	R/W	The interrupt status of channel 4	Write : 0: No effect 1: Clear this bit Read : 0: CH4 interrupt not occurred 1: CH4 interrupt occurred
[3]	CH3_STS	R/W	The interrupt status of channel 3	Write : 0: No effect 1: Clear this bit Read : 0: CH3 interrupt not occurred 1: CH3 interrupt occurred
[2]	CH2_STS	R/W	The interrupt status of channel 2	Write : 0: No effect 1: Clear this bit Read : 0: CH2 interrupt not occurred 1: CH2 interrupt occurred
[1]	CH1_STS	R/W	The interrupt status of channel 1	Write : 0: No effect 1: Clear this bit Read :

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Bit	Function	Type	Description	Condition
				0: CH1 interrupt not occurred 1: CH1 interrupt occurred
[0]	CH0_STS	R/W	The interrupt status of channel 0	Write : 0: No effect 1: Clear this bit Read : 0: CH0 interrupt not occurred 1: CH0 interrupt occurred

11.2.6 P_SPU_CLK_Ctrl (R/W) (0x306F)

P_SPU_CLK_Ctrl	0x306F	SPU base clock

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-					SPU_BASE_CLK[2:0]		
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:3]	-	-	-	-
[2:0]	SPU_BASE_CLK[2:0]	R/W	SPU clock control	000: FSpu = Fsystem/2 001: FSpu = Fsystem/2 010: FSpu = Fsystem/3 011: FSpu = Fsystem/4 100: FSpu = Fsystem/5 101: FSpu = Fsystem/6 110: FSpu = Fsystem/7 111: FSpu = Fsystem/8

11.2.7 P_SPU_Chx_PhaseAdder (R/W) (0x28x0)

P_SPU_Chx_PhaseAdder	0x28x0	Phase adder counter

Bit	15	14	13	12	11	10	9	8
Function	PHASE_CNT[15:8]							
Default	-	-	-	-	-	-	-	-

Bit	7	6	5	4	3	2	1	0
Function	PHASE_CNT[7:0]							
Default	-	-	-	-	-	-	-	-

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Bit	Function	Type	Description
[15:0]	PHASE_CNT[15:0]	R	Reserved for hardware, users don't care.

Note:

1. 'X' represents the specific channel, from 0 ~ 7.

11.2.8 P_SPU_Chx_SampleFreq (R/W) (0x28x1)

P_SPU_Chx_SampleFreq	0x28x1	Sample frequency register						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	FS[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	FS[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	FS[15:0]	R/W	Set sampling rate of tone color or speech The following equation is used to find the FS value according to the sample frequency (F_S). **$FS = (F_S/F_{SPU}) * 2^{24}$**

Note:

1. 'x' means the specific channel, from 0 to 7.
2. For example, when F_{SPU} = 12MHz and F_S is 32KHz, then the FS value is (32K/12M) * 2^{24} = 44739 = 0xAEC3.
3. The maximum sample frequency is determined by $F_{SPU}/256$. For example, when **F_{SPU}** = 12MHz => maximum sample frequency is 46.875KHz, i.e. fill FS with 0xFFFF will get this sample frequency.

11.2.9 P_SPU_Chx_DPTR_L (R/W) (0x28x2)

P_SPU_Chx_DPTR_L	0x28x2	Data pointer low word register						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	DPTR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DPTR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:0]	DPTR[15:0]	R/W	The current address low word of tone color or speech	Write: Write the address low word

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Bit	Function	Type	Description	Condition
				Read: Represent the address low word

Note:

1. 'x' the specific channel, from 0 to 7.

11.2.10 P_SPU_Chx_DPTR_H (R/W) (0x28x3)

P_SPU_Chx_DPTR_H	0x28x3	Data pointer high word register

Bit	15	14	13	12	11	10	9	8
Function	FORMAT[1:0]		ABS	-		DPTR[26:24]		
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DPTR[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:14]	FORMAT[1:0]	R/W	Set the format of current tone color or speech	00: A3400proE 4-bit 01: A3400proE 5-bit 10: PCM 8-bit 11: PCM 16-bit
[13]	ABS	R/W	Set the jump mode of current tone color or speech	0 : Relative jump mode 1 : Absolute jump mode
[12:11]	-	-	Reserved	-
[10:0]	DPTR[10:0]	R/W	The current address high word of tone color or speech	Write: Write the address high word Read: Represent the address high word

Note:

1. 'x' the specific channel, from 0 to 7.

11.2.11 P_SPU_Chx_LoopLength (R/W) (0x28x4)

P_SPU_Chx_LoopLength	0x28x4	Loop length register

Bit	15	14	13	12	11	10	9	8
Function	LEN[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	LEN[7:0]							

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Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:0]	LEN[15:0]	R/W	Loop length of a tone color or a speech	Write: Write the loop length Read: Represent the loop length

Note:

1. 'x' the specific channel, from 0 to 7.
2. In A3400pro 4-bit/ A3400pro 5-bit format, it is reserved for SPU.
3. In PCM format, the value of the loop length must be increased by 1 by software.

11.2.12 P_SPU_Chx_Filter_S2 (R/W) (0x28x5)

P_SPU_Chx_Filter_S2	0x28x5	Hardware filter register						
Bit	15	14	13	12	11	10	9	8
Function	S2[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	S2[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	S2[15:0]	R/W	This RAM must be cleared to initialize SPU by user before SPU is enabled. This is for SPU usage

Note:

1. 'x' the specific channel, from 0 to 7.

11.2.13 P_SPU_Chx_Filter_S1 (R/W) (0x28x6)

P_SPU_Chx_Filter_S1	0x28x6	Hardware filter register						
Bit	15	14	13	12	11	10	9	8
Function	S1[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	S1[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	S1[15:0]	R/W	This RAM must be cleared to initialize SPU by user before SPU is

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			enabled. This is for SPU usage

Note:

1. 'x' the specific channel, from 0 to 7.

11.2.14 P_SPU_Chx_Volume_Ctrl (R/W) (0x28x7)

P_SPU_Chx_Volume_Ctrl			0x28x7		Channel volume & Next channel volume			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	VOL_N[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	VOL[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:8]	VOL_N[7:0]	R/W	Set channel volume of next tone color or speech	Write: Set the channel volume Read: Represent the channel volume
[7:0]	VOL[7:0]	R/W	Set channel volume of current tone color or speech.	Write: Set the channel volume Read: Represent the channel volume

Note:

1. 'x' the specific channel, from 0 to 7.

11.2.15 P_SPU_Chx_NextDPTR_L (R/W) (0x28x8)

P_SPU_Chx_NextDPTR_L			0x28x8		Next data pointer low word register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	DPTR_N[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	DPTR_N[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:0]	DPTR_N[15:0]	R/W	The start address low byte of next tone color or speech.	Write: Write the address low word Read:

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Bit	Function	Type	Description	Condition
				Represent the address low word

Note:

1. ‘x’ the specific channel, from 0 to 7.
2. It will be reloaded to the P_SPU_ChX_DPTR_L (0x28x2) automatically for SPU to play next tone color.
3. The low word address must be decreased by 1 by software.

11.2.16 P_SPU_Chx_NextDPTR_H (R/W) (0x28x9)

P_SPU_Chx_NextDPTR_H	0x28x9	Next data pointer high word register						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	FORMAT_N[1:0]		ABSN	-		DPTR_N[26:24]		
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DPTR_N[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:14]	FORMATN[1:0]	R/W	Set the format of next tone color or speech	00: A3400proE 4-bit 01: A3400proE 5-bit 10: PCM 8-bit 11: PCM 16-bit
[13]	ABSN	R/W	Set the jump mode of next tone color or speech	0 : Relative jump mode 1 : Absolute jump mode
[12:11]	-	-	Reserved	-
[10:0]	DPTR_N[26:16]	R/W	The start address high word of next tone color or speech	Write: Write the address high word Read: Represent the address high word

Note:

1. ‘x’ the specific channel, from 0 to 7.
2. It will be reloaded to the P_SPU_Chx_DPTR_H (0x28x3) automatically for SPU to play next tone color.

11.2.17 P_SPU_Chx_EventTag (R/W) (0x28xA)

P_SPU_Chx_EventTag	0x28xA	Updated when Event Tag is detected by SPU hardware						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	Ext_Tag[7:0]							
Default	0	0	0	0	0	0	0	0

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Bit	7	6	5	4	3	2	1	0
Function	Int_Tag[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:8]	Ext_Tag[7:0]	R/W	Be updated when Group 2 Event Tag is detected by SPU hardware
[7:0]	Int_Tag[7:0]	R/W	Be updated when Group 1 Event Tag is detected by SPU hardware

Note:

1. ‘x’ the specific channel, from 0 to 7.

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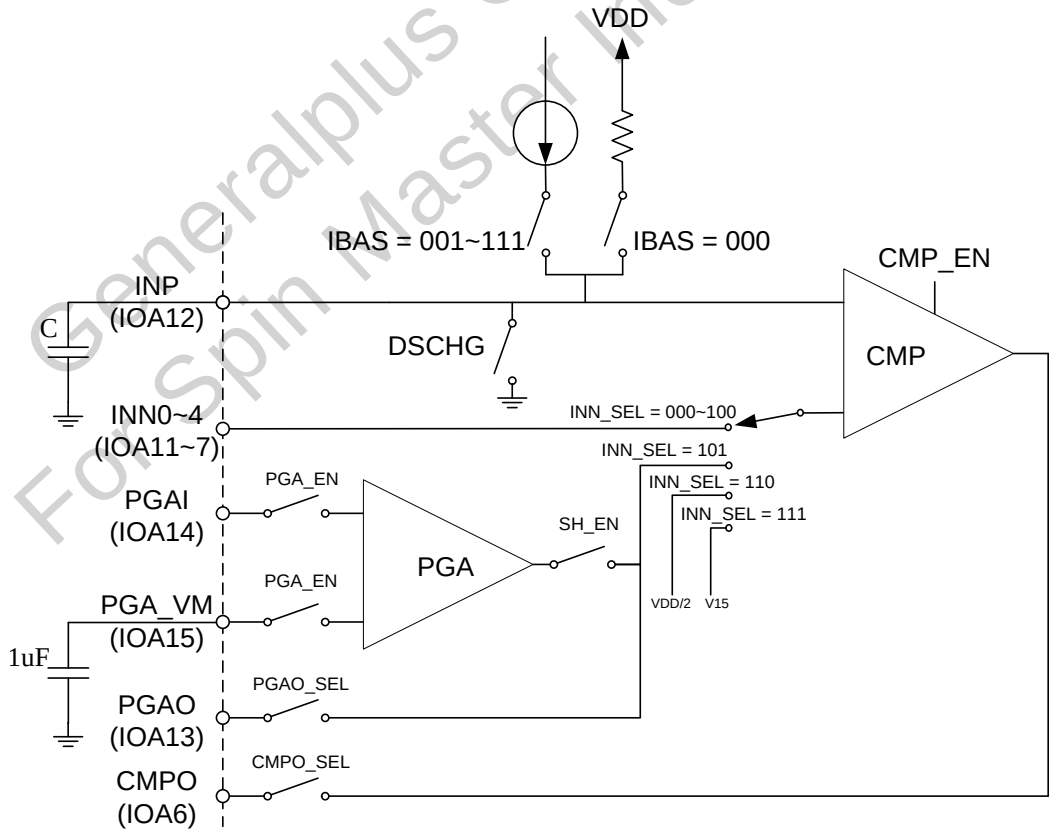
12 COMPARATOR ADC

12.1. GENERAL Description

GPCE500A features a Comparator ADC or a.k.a. slope ADC, which is a constant current source intended to charge an external capacitor (C) connected to INP, and conduct counting as well. When the voltage of capacitor (C) is greater than the input voltage, we can latch the counter value to obtain the outcome of ADC. It contains one PGA circuit provided eight channels of input to be selected: 5 channel Line-In, VDD/2, V15, and PGAO, in which PGAO signal including Sample and Hold function.

How to pick a proper capacitor (C):
Use the recording resolution of 12-bit as an example. Suppose maximum Vin = 3.3V, IBAS = 70uA, and SystemClock = 81.92MHz:
The time to charge the capacitor is 4096 / 81.92M = 50us, sample rate must be less than 20KHz.
C = (I*T) / V, C = (70u * 50u) / 3.3 ≅ 1000pF

12.2. Block Diagram



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12.3. Register Map

Name	Address	Description
P_CMPADC_Ctrl0	0x3070	Compare ADC control register 0
P_CMPADC_Ctrl1	0x3071	Compare ADC control register 1
P_CMPADC_Status	0x3072	Compare ADC status register
P_CMPADC_Data	0x3073	Compare ADC data register
P_PGA_Ctrl	0x3075	PGA control register

12.3.1 P_CMPADC_Ctrl0(R/W) (0x3070)

P_CMPADC_Ctrl0	0x3070	Compare ADC control register 0

Bit	15	14	13	12	11	10	9	8
Function	-			INN_SEL[2:0]			SH_SEL[1:0]	
Default	0	0	0	0	0	0	0	1

Bit	7	6	5	4	3	2	1	0
Function	DSCHG	HYS_EN	-		IBAS_SEL[2:0]			CMP_EN
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:13]	-	-	Reserved	-
[12:10]	INN_SEL[2:0]	R/W	INN selection	000 : INN0 (IOA11) 001 : INN1 (IOA10) 010 : INN2 (IOA9) 011 : INN3 (IOA8) 100 : INN4 (IOA7) 101 : PGAO (IOA13) 110 : VDD / 2 111 : 1.5V
[9:8]	SH_SEL[1:0]	R/W	Sample & hold selection **Note:** Only effective in auto mode	00 : 1us @ SysClk = 81.92MHz 01 : 2us @ SysClk = 81.92MHz 10 : 3us @ SysClk = 81.92MHz 11 : 4us @ SysClk = 81.92MHz
[7]	DSCHG	R/W	Discharge control **Note:** Only effective in manual mode	0 : Disabled 1 : Enabled
[6]	HYS_EN	R/W	Hysteresis enable	0 : Disabled 1 : Enabled
[5:4]	-	-	Reserved	-
[3:1]	IBAS_SEL[2:0]	R/W	IBAS selection	000 : RC 001 : 10uA 010 : 20uA 011 : 30uA 100 : 40uA 101 : 50uA 110 : 60uA

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Bit	Function	Type	Description	Condition
				111 : 70uA
[0]	CMP_EN	R/W	Compare enable	0 : Disabled 1 : Enabled

12.3.2 P_CMPADC_Ctrl1(R/W) (0x3071)

P_CMPADC_Ctrl1	0x3071	Compare ADC control register 1						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							ADC_DMA_EN
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	AUTO	-			CMPO_SEL	TMR_SEL[1:0]		STR
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:9]	-	-	Reserved	-
[8]	ADC_DMA_EN	R/W	Compare ADC DMA enable	0 : Disabled 1 : Enabled
[7]	AUTO	R/W	AUTO mode enable	0 : Disabled 1 : Enabled
[6:4]	-	-	Reserved	-
[3]	CMPO_SEL	R/W	CMPO output selection	0 : None 1 : IOA[6]
[2:1]	TMR_SEL[1:0]	R/W	Auto mode timer selection	00 : Timer A 01 : Timer B 10 : Timer C 11 : Manual
[0]	STR	R/W	Compare ADC start count	0 : Disabled 1 : Enabled

12.3.3 P_CMPADC_Status(R/W) (0x3072)

P_CMPADC_Status	0x3072	Compare ADC status register						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	FLAG	TIMEOUT	-					STS
Default	0	0	0	0	0	0	0	0

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Bit	Function	Type	Description	Condition
[15:8]	-	-	Reserved	-
[7]	FLAG	R/W1C	Compare ADC interrupt flag	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[6]	TIMEOUT	R/W1C	Compare ADC timeout flag	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[5:1]	-	-	Reserved	-
[0]	STS	R	Compare ADC status	0 : INP < INN 1 : INP ≧ INN

12.3.4 P_CMPADC_Data(R/W) (0x3073)

P_CMPADC_Data			0x3073		Compare ADC data register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	DATA[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	DATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:0]	DATA[15:0]	R	Compare ADC data	

12.3.5 P _PGA_Ctrl(R/W) (0x3075)

P_PGA_Ctrl			0x3075		PGA control register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	PGA_EN	-	PGAO_SEL	-	SH_EN	GAIN[2:0]		

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Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:8]	-	-	Reserved	-
[7]	PGA_EN	R/W	PGA enable	0 : Disabled 1 : Enabled
[6]	-	-	Reserved	-
[5]	PGAO_SEL	R/W	PGAO output selection	0 : None 1 : IOA[13]
[4]	-	-	Reserved	-
[3]	SH_EN	R/W	Sample & hold enable	0 : Disabled 1 : Enabled
[2:0]	GAIN[2:0]	R/W	PGA gain selection	000 : 0 dB 001 : 9 dB 010 : 17 dB 011 : 23 dB 100 : 29 dB 101 : 34 dB 110 : 39 dB 111 : 42 dB

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13 AUDIO OUTPUT

13.1. General Description

A voltage DAC(16-bit) and PWM(16-bit) are built in GPCE500A. User can select output as voltage DAC or PWM mode via P_AUDIO_Ctrl1 (R/W)($3043) and P_AUDIO_Ctrl3 (R/W)(#3044). Using voltage DAC requires an external amplifier, which is not required when using PWM. The data of audio should be delivered to P_AUDIO_CH1_Data (W) ($3041) and P_AUDIO_CH1_Data (W) ($3042). GPCE500A supports two channels data port. User is allowed to send the various data into different channels and thus, IC will perform the mix operation before DAC/PWM conversion.

13.2. Registers Map

Name	Address	Description
P_AUDIO_Ctrl2	0x3040	Audio output control register 2
P_AUDIO_CH1_Data	0x3041	Audio data of CH1
P_AUDIO_CH2_Data	0x3042	Audio data of CH2
P_AUDIO_Ctrl1	0x3043	Audio output control register 1
P_AUDIO_Ctrl3	0x3044	Audio output control register 3
P_AUDIO_Ctrl4	0x3045	Audio output control register 4

13.2.1 P_AUDIO_Ctrl2 (R/W) (0x3040)

P_AUDIO_Ctrl2		0x3040		Audio output control register 2				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-			SCALER_EN	CH2_HALF_VOL	CH2_UPSAMPLE	CH2_TMR_SEL[1:0]	
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-				CH1_HALF_VOL	CH1_UPSAMPLE	CH1_TMR_SEL[1:0]	
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:13]	-	-	Reserved	-
[12]	SCALER_EN	R/W	Scaler enable	0: Disabled 1: Enabled
[11]	CH2_HALF_VOL	R/W	CH2 half volume enable	0: Disabled 1: Enabled
[10]	CH2_UPSAMPLE	R/W	CH2 up sample enable	0: Disabled 1: Enabled
[9:8]	CH2_TMR_SEL[1:0]	R/W	CH2 latch data timer selection	00: Manual 01: Latch CH2 data by TimerA 10: Latch CH2 data by TimerB

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Bit	Function	Type	Description	Condition
				11: Latch CH2 data by TimerC
[7:4]	-	-	Reserved	-
[3]	CH1_HALF_VOL	R/W	CH1 half volume enable	0: Disabled 1: Enabled
[2]	CH1_UPSAMPLE	R/W	CH1 up sample enable	0: Disabled 1: Enabled
[1:0]	CH1_TMR_SEL[1:0]	R/W	CH1 latch data timer selection	00: Manual 01: Latch CH1 data by TimerA 10: Latch CH1 data by TimerB 11: Latch CH1 data by TimerC

13.2.2 P_AUDIO_CH1_Data (R/W) (0x3041)

P_AUDIO_CH1_Data	0x3041			Audio data of CH1				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	CH1_DATA[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	CH1_DATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	CH1 _DATA[15:0]	R/W	Audio data of CH1

13.2.3 P_AUDIO_CH2_Data (R/W) (0x3042)

P_AUDIO_CH2_Data	0x3042			Audio data of CH2				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	CH2_ DATA[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	CH2_ DATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	CH2 _DATA[15:0]	R/W	Audio data of CH2

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13.2.4 P_AUDIO_Ctrl1 (R/W) (0x3043)

P_AUDIO_Ctrl1	0x3043			Audio output control register 1				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-		CH2_DMA_EN	CH1_DMA_EN	DS_SEL	-	PWMIP_EN	PWM_EN
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	MUTE_CTRL[1:0]		AUD_GAIN[5:0]					
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:14]	-	-	Reserved	-
[13]	CH2_DMA_EN	R/W	DMA enable for CH2	0: Disabled 1: Enabled
[12]	CH1_DMA_EN	R/W	DMA enable for CH1	0: Disabled 1: Enabled
[11]	DS_SEL	R/W	DS selection	0 : DS3 1 : DS6
[10]	-	-	Reserved	-
[9]	PWMIP_EN	R/W	PWM IP enable	0: Disabled 1: Enabled
[8]	PWM_EN	R/W	PWM enable	0: Disabled 1: Enabled
[7:6]	MUTE_CTRL[1:0]	R/W	PWM auto mute control	00: Auto mute off 01: entry auto mute by MUTE_TIME_IN setting , exit mute by 1 data exchange 10: entry auto mute by MUTE_TIME_IN setting , exit mute by MUTE_TIME_OUT setting 11: auto mute off
[5:0]	AUD_GAIN[5:0]	R/W	Gain control of volume	000000: Lelve 0 000001: Level 1 …… 111111 = Level 63 (Max Volume)

13.2.5 P_AUDIO_Ctrl3 (R/W) (0x3044)

P_AUDIO_Ctrl3	0x3044			Audio output control register 3				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							SIGN
Default	0	0	0	0	0	0	0	0

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Bit	7	6	5	4	3	2	1	0
Function	-	MUTE_TYPE	-			RDAC_EN	AUDP_O	AUDN_O
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:13]	-	-	Reserved	-
[8]	SIGN	R/W	Audio PWM input data signed/unsigned selection	0: Signed 1: Unsigned
[7]	-	-	Reserved	-
[6]	MUTE_TYPE	R/W	AUDP/AUDN output in auto mute mode	0 : OFF (floating) 1 : 50% duty
[5:3]	-	-	Reserved	-
[2]	RDAC_EN	R/W	RDAC enable	0 : Disabled 1 : Enabled
[1]	AUDP_O	R/W	AUDP Output control	Refer to the table below
[0]	AUDN_O	R/W	AUDN Output control	Refer to the table below

RDAC_EN	PWMIP_EN	PWM_EN	AUDP	AUDN
0	0	0	Floating	AUDN_O (=1, pull high) (=0, pull low)
0	0	1	Floating	AUDN_O (=1, pull high) (=0, pull low)
0	1	0	AUDP_O	AUDN_O
0	1	1	AUDPWM	AUDPWM
1	0	0	Floating	Output Low
1	0	1	Floating	RDAC
1	1	0	AUDP_O	Output Low
1	1	1	AUDP_O	RDAC

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13.2.6 P_AUDIO_Ctrl4 (R/W) (0x3045)

P_AUDIO_Ctrl4	0x3045			Audio output control register 4				
Bit	15	14	13	12	11	10	9	8
Function	MUTE_OUT_SEL[6:0]							MUTE_IN_SEL[6]
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	MUTE_IN_SEL[5:0]						MUTE_OUT_CLR[1:0]	
Default	0	0	0	0	0	0	0	

Bit	Function	Type	Description	Condition
[15:9]	MUTE_OUT_SEL [6:0]	R/W	Exit auto mute timing setting	7'h7FH: 62.5us 7'h7EH: 125us 7'h7DH: 187.5us …… 7'h00H: 8ms
[8:2]	MUTE_IN_SEL [6:0]	R/W	Entry auto mute timing setting	7'h7FH: 125us 7'h7EH: 250us 7'h7DH: 375us …… 7'h00H: 16ms
[1:0]	MUTE_OUT_CLR[1:0]	R/W	Recount MUTE_TIME_OUT control setting, when exit auto mute, detect data no change, Recount MUTE_TIME_OUT setting.	00: 15.625us 01: 31.25us 10: 62.5us 11: 125us

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14 DMA

14.1. General Description

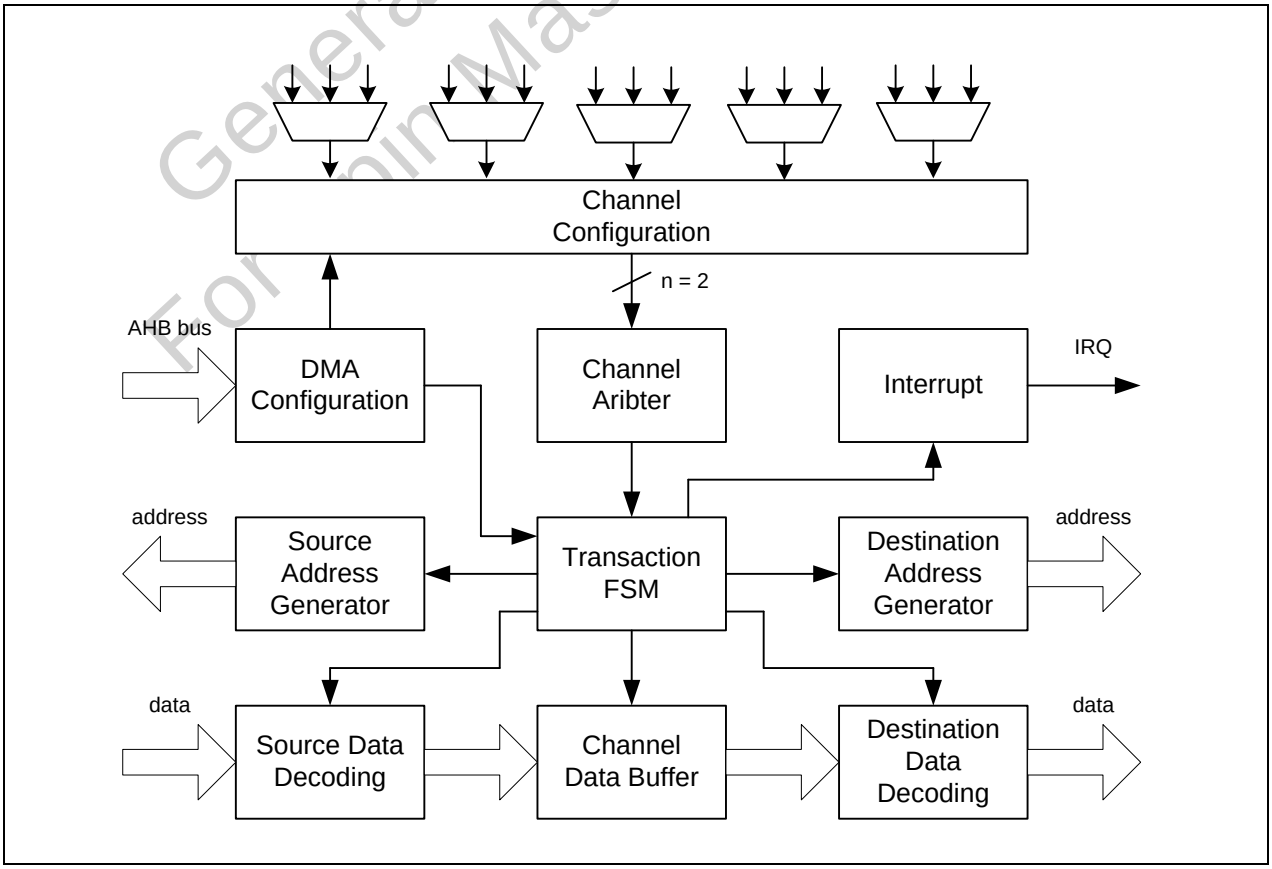
Direct memory access (DMA) is used in order to provide high-speed data transfer between peripherals and memory as well as memory to memory. Data can be quickly moved by DMA without any CPU actions. This keeps CPU resources free for other operations.

The DMA controllers have 2 channels, each dedicated to managing memory access requests from one or more peripherals. It has an arbiter for handling the priority between DMA requests.

14.2. Features

- Two independently configurable channels (requests)
- Each of the channels is connected to dedicated hardware DMA request, software trigger is also supported on each channel.
- Priorities between requests were based on channel number, such as request in channel 1 has higher priority than request channel 2.
- Support for burst operation.
- DMA Transfer complete send a interrupt request for each channel
- Support Memory-to-memory transfer, peripheral-to-memory, and memory-to-peripheral transfers with the programmable source and destination addresses
- Programmable number of data to be transferred: up to 65536

14.3. Block Diagram



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14.4. Function Description

14.4.1 DMA Transaction

After an event, the peripheral sends a request signal to the DMA Controller. The DMA controller serves the request depending on the channel priorities. As soon as the DMA Controller accesses the peripheral, an Acknowledge is sent to the peripheral by the DMA Controller. The peripheral releases its request as soon as it gets the Acknowledge from the DMA Controller.

14.4.2 DMA Request Arbiter

The arbiter manages the channel requests based on their priority and launches the peripheral/memory access sequences.

If 2 requests have the same software priority level, the channel with the lowest number will get priority versus the channel with the highest number. For example, channel 0 gets priority over channel 1.

14.4.3 DMA Channels

Each channel can handle DMA transfer between a source address and a destination address. The amount of data to be transferred (up to 65535) is programmable. The register which contains the amount of data items to be transferred is decremented after each transaction.

Address Increment

Source and destination address can optionally be automatically post-incremented after each transaction depending on the **DMA_SRCINC** and **DMA_DSTINC** bits in the DMA_CTRLx register.

If incremented mode is enabled, the address of the next transfer will be the address of the previous one incremented by 1. The first transfer address is the one programmed in the **DMA_SRCADR** / **DMA_DSTADR** registers. During transfer operations, these registers keep the initially programmed value. The current transfer addresses (in the current internal source/destination address register) are not accessible by software.

Burst mode

In DMA burst mode, the first request must be sent from software or peripheral device. After first transfer completed, DMA controller will automatically execute next transfer and decrease the internal transfer counter, reloaded from **DMA_DTN.** When internal transfer counter is not equivalent to 0**,** DMA controller will execute transfer and decrease the internal transfer counter until total transfer number completed.

14.4.4 Error management

When DMA read or write access cycle be occupied by another master such CPU over 256 system clocks, the transfer error flag (TEIF) will be asserted.

14.4.5 Interrupts

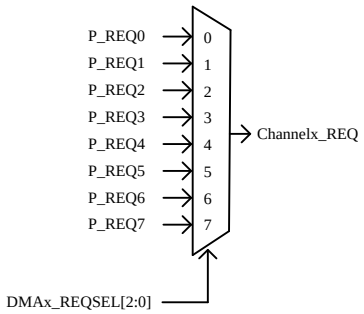
An interrupt can be produced on a Transfer complete or Transfer error for each DMA channel. Separate interrupt enable bits are available for flexibility.

14.4.6 Channel selection

Each channel is associated with a DMA request that can be selected out of 8 possible peripheral requests. The selection is controlled by the DMA_REQSEL[3:0] bits in the DMA_CTRLx register.

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Channel selection diagram



DMA request mapping

Peripherals	Channel 0/1
ADC	P-REQ0
DAC1	P-REQ1
DAC2	P-REQ2
SPI2	P-REQ3
UART	P-REQ4
CCP	P-REQ5
TIMERA	P-REQ6
MEMORY	P-REQ7

14.5. Register Map

Name	Address	Description
P_DMA_INTSTS	0x30A0	DMA interrupt status register
P_DMA1_CTRL0	0x30A1	DMA1 control register 0
P_DMA1_CTRL1	0x30A2	DMA1 control register 1
P_DMA1_DTN	0x30A3	DMA1 data transfer number register
P_DMA1_SRCADR	0x30A4	DMA1 source address register
P_DMA1_DSTADR	0x30A6	DMA1 destination address register
P_DMA2_CTRL0	0x30B1	DMA2 control register 0
P_DMA2_CTRL1	0x30B2	DMA2 control register 1
P_DMA2_DTN	0x30B3	DMA2 data transfer number register
P_DMA2_SRCADR	0x30B4	DMA2 source address register
P_DMA2_DSTADR	0x30B6	DMA2 destination address register

14.5.1 P_DMA_INTSTS (R/W) (0x30A0)

P_DMA_INTSTS		0x30A0		DMA interrupt status register				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-						DMA2_TEIF	DMA1_TEIF
Default	0	0	0	0	0	0	0	0

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Bit	7	6	5	4	3	2	1	0
Function	-						DMA2_TCIF	DMA1_TCIF
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:10]	-	-	Reserved	-
[9]	DMA2_TEIF	R/W1C	DMA2 transfer error flag	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[8]	DMA1_TEIF	R/W1C	DMA1 transfer error flag	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[7:2]	-	-	Reserved	-
[1]	DMA2_TCIF	R/W1C	DMA2 transfer complete flag	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued
[0]	DMA1_TCIF	R/W1C	DMA1 transfer complete flag	Write: 0: No effect 1: Clear status Read: 0: No interrupt 1: Interrupt event is issued

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14.5.2 P_DMA1_CTRL0 (R/W) (0x30A1)

P_DMA1_CTRL0	0x30A1			DMA1 control register 0				
Bit	15	14	13	12	11	10	9	8
Function	DMA_REQ[3:0]				-			
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_CONT	DMA_BURST	DMA_SRCINC	DMA_DSTINC	DMA_CIRC	DMA_TEIE	DMA_TCIE	DMA_EN
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:12]	DMA_REQ[3:0]	R/W	Peripheral request selection **Note:** DMA_REQ only can be changed when DMA_BUSY = 0	0000 = ADC 0001 = DAC1 0010 = DAC2 0011 = SPI2 0100 = UART 0101 = CCP 0110 = TMA 0111 = MEMORY others = Reserved
[11:8]	-	-	Reserved	-
[7]	DMA_CONT	R/W	DMA continuous mode Transfer done signal will not be passed to peripheral.	0 : Disabled 1 : Enabled
[6]	DMA_BURST	R/W	DMA burst transfer mode **Note:** When DMA receives a request, it will automatically transfer all the data at once according to the transfer number set by user. **Note:** After burst mode is enabled, DMA will continue transferring the data, and other requests will be ignore during the time of transferring.	0 : Disabled 1 : Enabled
[5]	DMA_SRCINC	R/W	DMA source address increment mode	0 : Disabled 1 : Enabled
[4]	DMA_DSTINC	R/W	DMA destination address increment mode	0 : Disabled 1 : Enabled
[3]	DMA_CIRC	R/W	DMA circular mode **Note:** If circular mode is enabled, DMA can perform data moving in infinite loops until user set DMA_STOP bit.	0 : Disabled 1 : Enabled
[2]	DMA_TEIE	R/W	DMA transfer error interrupt enable	0 : Disabled 1 : Enabled
[1]	DMA_TCIE	R/W	DMA transfer complete interrupt enable	0 : Disabled 1 : Enabled
[0]	DMA_EN	R/W	DMA enable	0 : Disabled 1 : Enabled

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14.5.3 P_DMA1_CTRL1 (R/W) (0x30A2)

P_DMA1_CTRL1	0x30A2			DMA1 control register 1				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-						DMA_DBF	DMA_BUSY
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-				DMA_DBM	DMA_STPS	DMA_STP	DMA_STR
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:10]	-	-	Reserved	-
[9]	DMA_DBF	R	DMA double buffer full flag	0 : Not Full 1 : Full
[8]	DMA_BUSY	R	DMA operation status This flag will keep high during data transfer.	0 : Idle 1 : Busy
[7:4]	-	-	Reserved	-
[3]	DMA_DBM	R/W	DMA double buffer mode **Note:** DMA will move the data twice and thus trigger two INTs. After moving the data length set by the user for the first time, the first DMA INT will be triggered, and DMA will keep moving data at this time. The second DMA INT will be triggered after DMA finishes the second data moving.	0 : Single buffer 1 : Double buffer
[2]	DMA_STPS	R/W	DMA software stop request selection	0 : Stop immediately 1 : Stop when data transfer is completed.
[1]	DMA_STP	W	DMA software stop request **Note:** When this bit = 1, DMA_STPS will determine to stop DMA immediately or to stop DMA after the transmission is completed.	0 : No function 1 : Stop
[0]	DMA_STR	W	DMA software trigger request **Note:** If no trigger source is in use, for example in the case if memory to memory, DMA data moving will be triggered by software trigger function.	0 : No function 1 : Start

14.5.4 P_DMA1_DTN (R/W) (0x30A3)

P_DMA1_DTN	0x30A3	DMA1 data transfer number register

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Bit	15	14	13	12	11	10	9	8
Function	DMA_DTN[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_DTN[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:0]	DMA_DTN[15:0]	R/W	DMA data transfer number counter The number of data to be transferred. If this register is set as 0, there won't be any transfer whether the DMA_EN is enabled or not. When the DMA_EN is enabled from disable state, DMA_DTN will be reloaded to internal counter, and the counter will be decreased by one every time a transfer is completed. Even if DMA_DTN is changed by the software, the internal counter register will not be updated when DMA_EN is enabled.	-

14.5.5 P_DMA1_SRCADR (R/W) (0x30A4)

P_DMA1_SRCADR	0x30A4		DMA1 source address register				
Bit	15	14	13	12	11	10	9	8
Function	DMA_SRCADR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_SRCADR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:0]	DMA_SRCADR[15:0]	R/W	DMA source data transfer address Base address of the source area from which the data will be read.	-

14.5.6 P_DMA1_DSTADR (R/W) (0x30A6)

P_DMA1_DSTADR	0x30A6	DMA1 destination address register

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Bit	15	14	13	12	11	10	9	8
Function	DMA_DSTADR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_DSTADR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:0]	DMA_DSTADR[15:0]	R/W	DMA destination data transfer address Base address of the destination area to which the data will be written.	-

14.5.7 P_DMA2_CTRL0 (R/W) (0x30B1)

P_DMA2_CTRL0	0x30B1	DMA2 control register 0						
Bit	15	14	13	12	11	10	9	8
Function	DMA_REQ[3:0]				-			
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_CONT	DMA_BURST	DMA_SRCINC	DMA_DSTINC	DMA_CIRC	DMA_TEIE	DMA_TCIE	DMA_EN
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:12]	DMA_REQ[3:0]	R/W	Peripheral request selection **Note:** DMA_REQ only can be changed when DMA_BUSY = 0	0000 = ADC 0001 = DAC1 0010 = DAC2 0011 = SPI2 0100 = UART 0101 = CCP 0110 = TMA 0111 = MEMORY others = Reserved
[11:8]	-	-	Reserved	-
[7]	DMA_CONT	R/W	DMA continuous mode Transfer done signal will not be passed to peripheral.	0 : Disabled 1 : Enabled
[6]	DMA_BURST	R/W	DMA burst transfer mode **Note:** When DMA receives a request, it will automatically transfer all the data at once according to the transfer number set by user. **Note:** After burst mode is enabled,	0 : Disabled 1 : Enabled

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Bit	Function	Type	Description	Condition
			DMA will continue transferring the data, and other requests will be ignore during the time of transferring.	
[5]	DMA_SRCINC	R/W	DMA source address increment mode	0 : Disabled 1 : Enabled
[4]	DMA_DSTINC	R/W	DMA destination address increment mode	0 : Disabled 1 : Enabled
[3]	DMA_CIRC	R/W	DMA circular mode **Note:** If circular mode is enabled, DMA can perform data moving in infinite loops until user set DMA_STOP bit.	0 : Disabled 1 : Enabled
[2]	DMA_TEIE	R/W	DMA transfer error interrupt enable	0 : Disabled 1 : Enabled
[1]	DMA_TCIE	R/W	DMA transfer complete interrupt enable	0 : Disabled 1 : Enabled
[0]	DMA_EN	R/W	DMA enable	0 : Disabled 1 : Enabled

14.5.8 P_DMA2_CTRL1 (R/W) (0x30B2)

P_DMA2_CTRL1	0x30B2			DMA2 control register 1				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-						DMA_DBF	DMA_BUSY
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-				DMA_DBM	DMA_STPS	DMA_STP	DMA_STR
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:10]	-	-	Reserved	-
[9]	DMA_DBF	R	DMA double buffer full flag	0 : Not Full 1 : Full
[8]	DMA_BUSY	R	DMA operation status This flag will keep high during data transfer.	0 : Idle 1 : Busy
[7:4]	-	-	Reserved	-
[3]	DMA_DBM	R/W	DMA double buffer mode **Note:** DMA will move the data twice and thus trigger two INTs. After moving the data length set by the user for the first time, the first DMA INT will be triggered, and DMA will keep moving	0 : Single buffer 1 : Double buffer

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Bit	Function	Type	Description	Condition
			data at this time. The second DMA INT will be triggered after DMA finishes the second data moving.	
[2]	DMA_STPS	R/W	DMA software stop request selection	0 : Stop immediately 1 : Stop when data transfer is completed.
[1]	DMA_STP	W	DMA software stop request **Note:** When this bit = 1, DMA_STPS will determine to stop DMA immediately or to stop DMA after the transmission is completed.	0 : No function 1 : Stop
[0]	DMA_STR	W	DMA software trigger request **Note:** If no trigger source is in use, for example in the case if memory to memory, DMA data moving will be triggered by software trigger function.	0 : No function 1 : Start

14.5.9 P_DMA2_DTN (R/W) (0x30B3)

P_DMA2_DTN	0x30B3	DMA2 data transfer number register						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	DMA_DTN[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	DMA_DTN[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:0]	DMA_DTN[15:0]	R/W	DMA data transfer number counter The number of data to be transferred. If this register is set as 0, there won't be any transfer whether the DMA_EN is enabled or not. When the DMA_EN is enabled from disable state, DMA_DTN will be reloaded to internal counter, and the counter will be decreased by one every time a transfer is completed. Even if DMA_DTN is changed by the software, the internal counter register will not be updated when DMA_EN is enabled.	-

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14.5.10 P_DMA2_SRCADR (R/W) (0x30B4)

P_DMA2_SRCADR		0x30B4		DMA2 source address register				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	DMA_SRCADR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	DMA_SRCADR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	**Function**	**Type**	**Description**	**Condition**
[15:0]	DMA_SRCADR[15:0]	R/W	DMA source data transfer address Base address of the source area from which the data will be read.	-

14.5.11 P_DMA2_DSTADR (R/W) (0x30B6)

P_DMA2_DSTADR		0x30B6		DMA2 destination address register				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	DMA_DSTADR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	DMA_DSTADR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	**Function**	**Type**	**Description**	**Condition**
[15:0]	DMA_DSTADR[15:0]	R/W	DMA destination data transfer address Base address of the destination area to which the data will be written.	-

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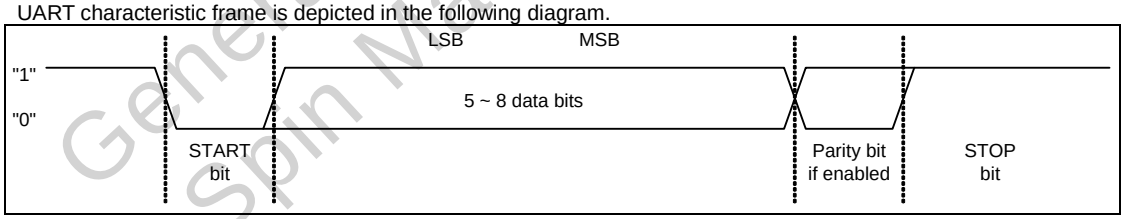
15 UART

15.1. General Description

The UART module built in GPCE500A performs serial-to-parallel conversion on data received from an external device and it also performs parallel-to-serial conversion on data transmitted to the external device. The transmission and reception paths are individually buffered with internal FIFO memories, allowing up to 16 bytes. This module provides the following features.

- Programmable baud rate
- Data width: 5, 6, 7, or 8 bits
- Parity: even, odd or disabled
- Stop bit width: 1 or 2 bits
- Individual 8-bit transmit and 11-bit (8-bit data + 3-bit status) receive FIFOs (16 First-in & first-out buffer)
- 16 byte-deep transmitting and receiving FIFO, respectively
- Programmable UART SIR Input /Output (IOA2 and IOA3 or IOB14 and IOB15)
- Programmable transmitting and receiving FIFO interrupt
- Supports receiving timeout interrupt
- Frame, Parity, and break error detection during reception (error information also saved in FIFO)
- Overrun detection
- Supports high data rate (a division of the system clock)
- Supports Loop Back Testing for system diagnosis or mass production testing

UART characteristic frame is depicted in the following diagram.



Note:

"1" and "0" mean logical level, representing VCC and GND in GPCE500A; on the other hand, "1" and "0" mean –12V and +12V in RS232 compatible line.

15.2. UART Control Pin Configuration

GPCE500A can select UART IO options via P_IO_Ctrl[1] ($300F).

Name	Description
P_IO_Ctrl[1] = 0	UART TX : IOA[3] UART RX : IOA[2]
P_IO_Ctrl[1] = 1	UART TX : IOB[15] UART RX : IOB[14]

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15.3. Registers Map

Name	Address	Description
P_UART_Data	0x30C0	UART data register
P_UART_RX_Status	0x30C1	UART received error status register
P_UART_Ctrl	0x30C2	UART control register
P_UART_BaudRate	0x30C3	UART baud rate register
P_UART_Status	0x30C4	UART interrupt status register

15.3.1 P_UART_Data (R/W) (0x30C0)

P_UART_Data			0x30C0		UART data register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-				OE	BE	PE	FE
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	UARTDATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:12]	-	-	Reserved	-
[11]	OE	R	**Overrun Error** Same as P_UART_RX_Status[3] This bit would be set to “1” if data is received and the FIFO is full.	0: Not Occurred 1: Occurred
[10]	BE	R	**Break Error** Same as P_UART_RX_Status[2] This bit would be set to “1” if a break condition is detected. Indicate that the received data input is held LOW for more than a full-word transmission time (defined as start, data, parity and stop bits).	0: Not Occurred 1: Occurred
[9]	PE	R	**Parity Error** Same as P_UART_RX_Status[1] This bit would be set to “1” if the parity of the received data character does not match the parity selected in PSEL Control bit.	0: Not Occurred 1: Occurred
[8]	FE	R	**Frame Error** Same as P_UART_RX_Status[0] This bit is set to “1” if the received character does not have a valid stop bit (a valid stop bit is 1 bit).	0: Not Occurred 1: Occurred
[7:0]	UARTDATA[7:0]	R/W	**UART Data Read/Write Register** If the FIFO is enabled, the data byte is extracted	-

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Bit	Function	Type	Description	Condition
			and a 3-bit status (break, frame and parity) is pushed into the 11-bit wide receiving FIFO. This control register represents the entry to transmitting / receiving FIFO. If the FIFO is disabled, this control register represents the transmitting/receiving hold register.	

15.3.2 P_UART_RX_Status (R/W) (0x30C1)

P_UART_RX_Status	0x30C1	UART received error status register						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-				OE	BE	PE	FE
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:4]	-	-	-	-
[3]	OE	R/W1C	**Overrun Error** This bit would be set to “1” if data is received and the FIFO is full.	Read: 0: Not Occurred 1: Occurred Write: 0: No effect 1: Clear flag
[2]	BE	R/W1C	**Break Error** This bit would be set to “1” if a break condition is detected. Indicate that the received data input is held LOW for more than a full-word transmission time (defined as start, data, parity and stop bits).	Read: 0: Not Occurred 1: Occurred Write: 0: No effect 1: Clear flag
[1]	PE	R/W1C	**Parity Error** This bit would be set to “1” if the parity of the received data character does not match the parity selected in PSEL Control bit.	Read: 0: Not Occurred 1: Occurred Write: 0: No effect 1: Clear flag
[0]	FE	R/W1C	**Frame Error** This bit is set to “1” if the received character does not have a valid stop bit (a valid stop bit is 1 bit).	Read: 0: Not Occurred 1: Occurred Write:

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Bit	Function	Type	Description	Condition
				0: No effect 1: Clear flag

15.3.3 P_UART_Ctrl (R/W) (0x30C2)

P_UART_Ctrl			0x30C2		UART control register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	RXIE	TXIE	RTIE	UART_EN	RST	-	RX_DMA_EN	TX_DMA_EN
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-	DATA_LEN[1:0]		FIFO_EN	STP_SEL	PARITY_SEL	PARITY_EN	SEND_BREAK
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	RXIE	R/W	**Receiving Interrupt Enabling selection** **(a) FIFO is enabled (16-byte deep, FEN=1):** If this bit is set to “1”, and the data number in receiving FIFO is more or equal to 8, hardware will issue an IRQ7 or FIQ to CPU. **(b) FIFO is disabled (1-byte deep, FEN=0):** If this bit is set to “1”, and the receiving buffer is just held one new-coming data, hardware will issue an IRQ7 or FIQ to CPU. If this bit is cleared to “0”, this receiving interrupt will be masked. To select between IRQ7 and FIQ, please refer to Chapter Interrupt.	0: Disabled 1: Enabled
[14]	TXIE	R/W	**Transmitting Interrupt Enabling selection** **(a) FIFO is enabled (16-byte deep, FEN=1):** If this bit is set to “1”, and the data number in transmitting FIFO is less or equal to 8, hardware will issue an IRQ7 or FIQ to CPU. **(b) FIFO is disable (1-byte deep, FEN=0):** If this bit is set to “1”, and the transmit buffer is empty, hardware will issue an IRQ7 or FIQ to CPU. If this bit is cleared to “0”, this receiving interrupt will be masked. To select between IRQ7 and FIQ, please refer to Chapter Interrupt.	0: Disabled 1: Enabled
[13]	RTIE	R/W	**Receiving Timeout Interrupt Enable**	0: Disabled

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Bit	Function	Type	Description	Condition
			If this bit is set to “1”, and when the receiving FIFO is not empty and no further data is received over a 32-bit period, hardware will issue an IRQ7 or FIQ to CPU. If this bit is cleared to “0”, this receiving interrupt will be masked. To select between IRQ7 and FIQ, please refer to Chapter Interrupt.	1: Enabled
[12]	UART_EN	R/W	**UART Enable** If this bit is set to “1”, the UART Interface is enabled.	0: Disabled 1: Enabled
[11]	RST	W	**Reset the UART module** Write “1” to generate a single internal reset pulse to clear the UART FIFO data. It does not reset UART control register setting. **Note:** after reset UART FIFO data, programmer needs dummy read this control register once to ensure UART function work correctly.	0: No effect 1: Reset
[10]	-	-	Reserved	-
[9]	RX_DMA_EN	R/W	RX DMA enable	0: Disabled 1: Enabled
[8]	TX_DMA_EN	R/W	TX DMA enable	0: Disabled 1: Enabled
[7]	-	-	Reserved	-
[6:5]	DATA_LEN[1:0]	R/W	**Data Length Definition** Indicate number of data bits transmitted or received in a frame.	00: 5 bits 01: 6 bits 10: 7 bits 11: 8 bits
[4]	FIFO_EN	R/W	**FIFO Buffer Enable/Disable** Setting this bit to “1” will enable the 16-byte deep FIFO buffer on both transmitting and receiving operation. When clearing this bit to “0”, the FIFO becomes 1-byte deep hold registers.	0: Disabled 1: Enabled
[3]	STP_SEL	R/W	**Stop Bit Size Selection** When this bit is set to “1”, two stop bits are transmitted at the end of the frame. It cannot recognize for data with two received stop bits in receiving module.	0: 1 Stop bit 1: 2 Stop bits
[2]	PARITY_SEL	R/W	**Parity Selection** If this bit is set to “1”, even parity generation and	0: Odd Parity 1: Even Parity

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Bit	Function	Type	Description	Condition
			checking are performed during transmission and reception, which checks for an even number of 1s in data and parity bits. When cleared to “0” the odd parity is performed to check for an odd number of 1s. This bit has no effect until parity is enabled by setting PEN Control bit to “1”.	
[1]	PARITY_EN	R/W	**Parity Enable** If this bit is set to “1”, parity checking and generation is enabled; else parity is disabled and no parity bit is added to the data frame.	0: Disabled 1: Enabled
[0]	SEND_BREAK	R/W	**Send Break** If this bit is set to “1”, a low level is continuously output on the TX output pin after completing the current character transmission. This bit must be asserted for at least one complete frame period in order to generate a break condition. The transmitting FIFO contents remain ineffective during a break condition. For normal usage, this bit must be cleared to “0”.	0: No effect 1: Send break

Note:

```
R1 = [ P_UART_Ctrl]
R1 |= 0x0800                ; set reset FIFO register
[P_UART_Ctrl] = R1
R1 = [ P_UART_Ctrl]         ; after reset UART FIFO data, need a dummy read.
```

15.3.4 P_UART_BaudRate (R/W) (0x30C3)

P_UART_BaudRate			0x30C3		UART baud rate register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	UART_BUD[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	UART_BUD[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	UART_BUD[15:0]	R/W	UART Baud Rate Divisor.

The Baud Rate Divisor value is calculated as follows

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UARTBAUD = [SYSCLK / (16 X Baud Rate)] –1

For example, if the system clock is 81.92MHz, to obtain 115200bps baud rate, the divisor must be set to UARTBAUD = [81920000/ (16 X 115200)] –1 = 43 = 0x002B

Baud Rate Setting Table for System Clock =81.92 MHz

Programmable divisor (value on UARTBUD)	Available UART Baud Rate (Bit Per Second)
0x000A	460,800
0x0015	230,400
0x002B	115,200
0x0041	76,800
0x0057	57,600
0x0084	38,400
0x00B0	28,800
0x0109	19,200
0x0162	14,400
0x0214	9,600
0x0854	2,400
0x10A9	1,200

15.3.5 P_UART_Status (R) (0x30C4)

P_UART_Status			0x30C4		UART interrupt status register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	RXIF	TXIF	RTIF	-				
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	TXEF	RXEF	TXFF	RXEF	BUSYF	-		
Default	1	0	0	1	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	RXIF	R	**Receive Interrupt Flag** **FIFO is enabled (16-byte deep,** FEN=1**)**: This bit is set to “1” by hardware if the receiving interrupt enable bit is set to “1”, and the data number in receiving FIFO is more or equal to 8. This bit is cleared to “0” by hardware if data number in receive FIFO is less than 8. In other word, receive interrupt is cleared by reading data from the receive FIFO until it becomes less than	0: data number in RX FIFO < 8 1: data number in RX FIFO >= 8

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Bit	Function	Type	Description	Condition
			8 data.	
			(b) FIFO is disabled (1-byte deep, FEN =0**):** This bit is set to “1” by hardware if the receiving interrupt enable bit is set to “1”, and the data is received completely. The receive interrupt is cleared by performing a single read of receive data register (P_UART_Data). 0 = RX Data is not Ready 1 = RX Data is Ready	0: RX Data is not Ready 1: RX Data is Ready
[14]	TXIF	R	**Transmit Interrupt Flag** **(a) FIFO is enabled (16-byte deep,** FEN=1): This bit is set to “1” by hardware if the transmitting interrupt enable bit is set to “1”, and the data number in transmitting FIFO is less or equal to 8. This bit is cleared to “0” by hardware if data number in transmitting FIFO is more than 8. In other word, transmit interrupt is cleared by filling data to transmit FIFO until it becomes more than 8 data.	0: data number in TX FIFO < 8 1: data number in TX FIFO >= 8
			(b) FIFO is disabled (1-byte deep, FEN=0**):** This bit is set to “1” by hardware if the transmitting interrupt enable bit is set to “1”, and there is no data in the transmitter. This flag is cleared to “0”, by performing a single write to the transmit data register (P_UART_Data).	0: TX Buffer is not Ready 1: TX Buffer is Ready
[13]	RTIF	R	**Receive Timeout Interrupt Flag** This bit is set to “1” by hardware if the receiving timeout interrupt is asserted. (When receive FIFO is not empty and no further data is received over a 32-bit period). This interrupt flag is cleared to “0” by hardware when the FIFO becomes empty through reading all the data (or by reading the holding register).	0: Not occurred 1: Occurred
[12:8]	-	-	Reserved	-
[7]	TXEF	R	**Transmit FIFO Empty Flag** **(a) FIFO is enabled (16-byte deep,** FEN=1**)**: This bit is set to “1” when the transmitting FIFO is empty.	0: data number in TX FIFO > 0 1: data number in TX FIFO = 0
			(b) FIFO is disabled (1-byte deep, FEN=0**):**	0: TX buffer is not

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Bit	Function	Type	Description	Condition
			This bit is set to “1” when the transmitting buffer is empty. Note that this flag is read-only; hardware will set or clear this flag automatically.	empty 1: TX buffer is empty
[6]	RXEF	R	**Receive FIFO Full Flag** **(a) FIFO is enabled (16-byte deep,** FEN=1**)**: This bit is set to “1” when the receiving FIFO is full.	0: data number in RX FIFO < 16 1: data number in RX FIFO = 16
			(b) FIFO is disabled (1-byte deep, FEN=0**):** This bit is set to “1” when the receiving buffer is full. Note that this flag is read-only; hardware will set or clear this flag automatically.	0: RX buffer is not full 1: RX buffer is full
[5]	TXFF	R	**Transmit FIFO Full Flag** **(a) FIFO is enabled (16-byte deep,** FEN=1**)**: This bit is set to “1” when the transmitting FIFO is full.	0: data number in TX FIFO < 16 1: data number in TX FIFO = 16
			(b) FIFO is disabled (1-byte deep, FEN=0**):** This bit is set to “1” when the transmitting buffer is full. Note that this flag is read-only; hardware will set or clear this flag automatically.	0: TX buffer is not full 1: TX buffer is full
[4]	RXEF	R	**Receive FIFO Empty Flag** **(a) FIFO is enabled (16-byte deep,** FEN=1**)**: This bit is set to “1” when the receive FIFO is empty. 0 = no. in RX FIFO > 0 1 = no. in RX FIFO = 0	0: data number in RX FIFO > 0 1: data number in RX FIFO = 0
			(b)FIFO is disabled (1-byte deep, FEN=0**):** This bit is set to “1” when the receiving buffer is empty. Note that this flag is read-only; hardware will set or clear this flag automatically.	0: RX buffer is not empty 1: RX buffer is empty
[3]	BUSYF	R	**BUSY Flag** If this bit is set to “1”, the UART module is busy in transmitting data. This bit remains set until the complete byte, including all the stop bits, has been sent from the shift register.	0: Ready 1: Busy

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Bit	Function	Type	Description	Condition
			Note that this flag is read-only; hardware will set or clear this flag automatically.	
[2:0]	-	-	Reserved	-

Note:

*The above three interrupt flags are combined into a single interrupt flag on P_INT_Status1[11] by an OR logic. Programmers should determine which interrupt occurs from this control register (P_UART_Status [15:13]). This bit is effect only when receive interrupt enable bit (P_UART_Ctrl[15]) is set.

** Also note that TX interrupt or RX interrupt flag is “clear” by the write and read operation on the corresponding FIFOs, they are not cleared by any specified control bit.

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16 SERIAL PERIPHERAL INTERFACE (SPI)

16.1. General Description

There are two Serial Peripheral Interface (SPI) controller built in GPCE500A to facilitate communicating with other devices and components. The SPI controller built in GPCE500A include four master mode and one slave mode. GPCE500A always run SPI external code by SPI1.

Users can decide the SPICS configuration in the master mode, if P_SPIx_Ctrl[12] set '0', the CS_pin as GPIO function, if P_SPIx_Ctrl[12] set '1', the CS pin as SPICS hardware function. While SPI module is enabled by corresponding control bit, these four/six pins cannot be GPIOs. In other words, any setting on corresponding GPIO control register will have no effect.

While SPI Interface is not in transmitting data, The I/O configuration to DO and DI is Input Floating. Therefore, before entering sleep mode, SPI interface must be turned off first and configure corresponding setup to prevent standby current issue. If SPI FLASH is used, please refer to AN0176 application note.

For SPI1, the driving and sinking capability on IOC2~IOC7 (SPI1) are identical. However, the driving and sinking capability are not symmetrical on other GPIOs, simply because the SPI1 must reach speed of 81.92MHz. To transmit signals, user should be alerted to this issue.

The SPI provides following features

- Supports single byte and consecutive bytes transferring
- Supports overrun error indication
- Supports transmitting / receiving interrupt request
- Programmable phase and polarity of master clock
- Selectable data sampling time (end or middle of clock period)
- For SPI1, Programmable master SPICK clock frequency: System Clock /1, /2, /16
- For SPI2, Programmable master SPICK clock frequency: System Clock /2, / 4, /8, /16, /32, /64, /128 and TimerA / 2
- Built-in 8-depth 8-bit FIFOs in both transmit and receive direction, the interrupt level of these two FIFOs are both programmable
- Supports SPI 1, 2, 4 I/O mode

SPI1 has the following features:

- Supports run SPI external code and data in auto mode

16.1.1 Master Mode

In master mode, the shifting clock (SPICK) is generated by GPCE500A. There are two control bits to control the clock phase and polarity. The transmission starts immediately from a register writes to the SPIBUF control register. As long as there is data in the FIFO, the transmitter will start automatically.

The SPI shifts the data from MSB to LSB through the SPIDO pin. The 8-bit data is shifted out after 8 clock cycles. At the same time, the data is also shifted in through slave device SPIDI pin. When the

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transmit FIFO level is lower than the interrupt trigger level, the SPITXIF flag bit will be set; besides, a SPI interrupt will be generated if the SPITXIRQEN bit is set. When the receive FIFO level is higher than the interrupt trigger level, the SPIRXIF flag bit will be set; moreover, a SPI interrupt will be generated if the SPIRXIRQEN bit is set. Programmer can read SPI data from SPIRXD control register.

The following diagram depicts the timing scheme on SPI master mode for different operation types (polarity control bit equals “1” or “0”, phase control bit equals “1” or “0”, and sample strobe control bit equals “1” or “0”).

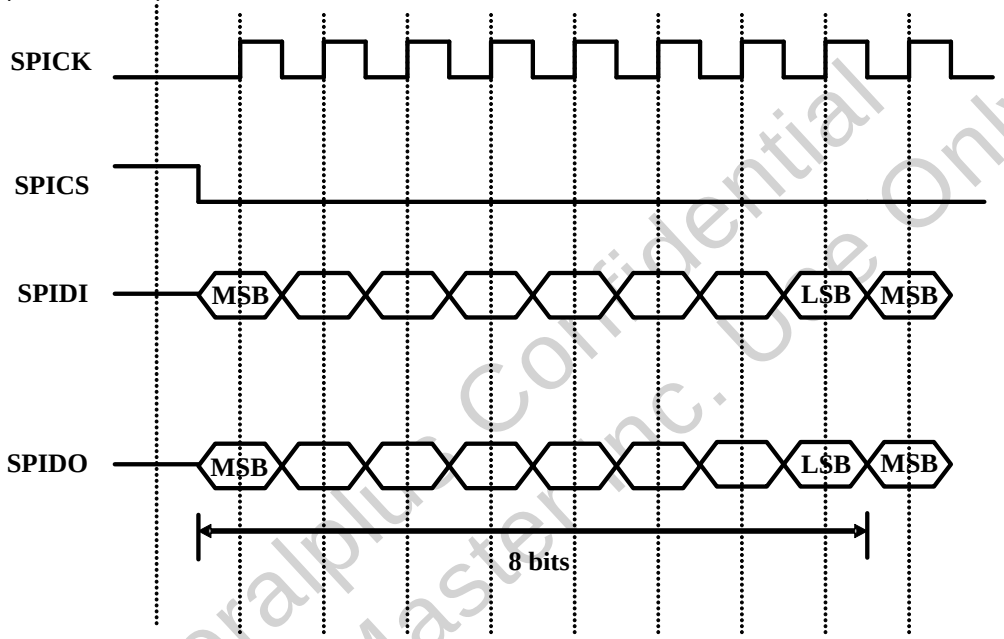


Fig1. Master Mode, POLARITY=0, PHASE=0.
At this setting, GPCE500A will sample data at rising edge of SPICK.

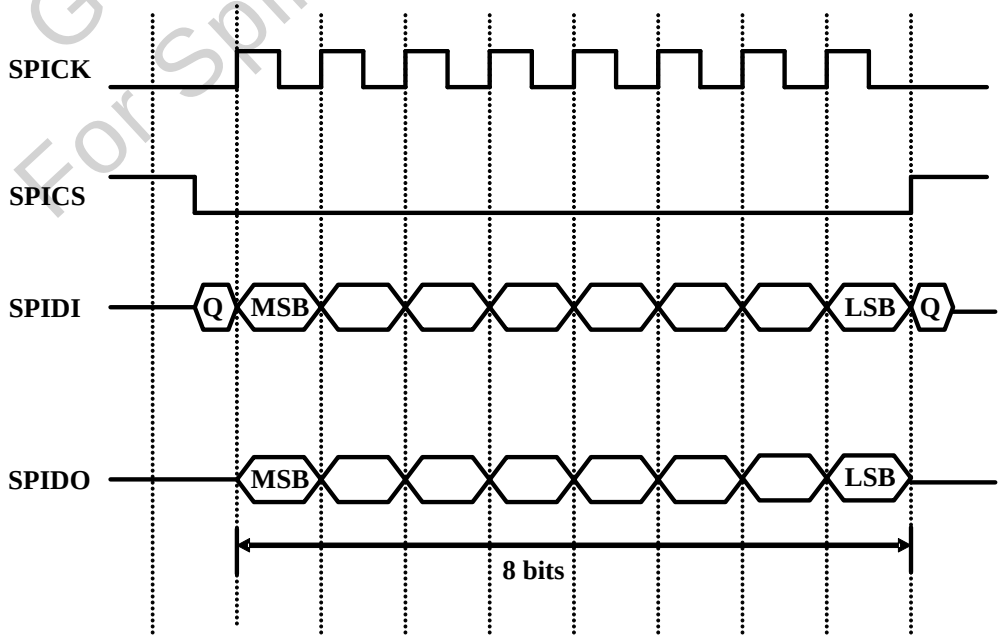


Fig2. Master Mode, POLARITY=0, PHASE=1
At this setting, GPCE500A will sample data at falling edge of SPICK.

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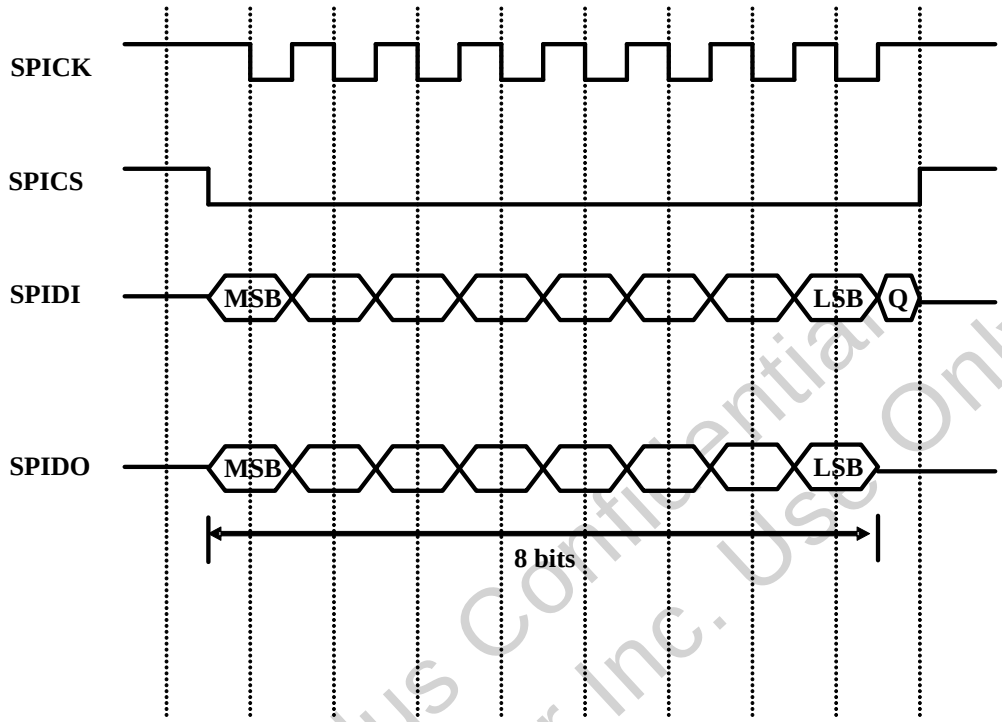


Fig3. Master Mode, POLARITY=1, PHASE=0

At this setting, GPCE500A will sample data at falling edge of SPICK.

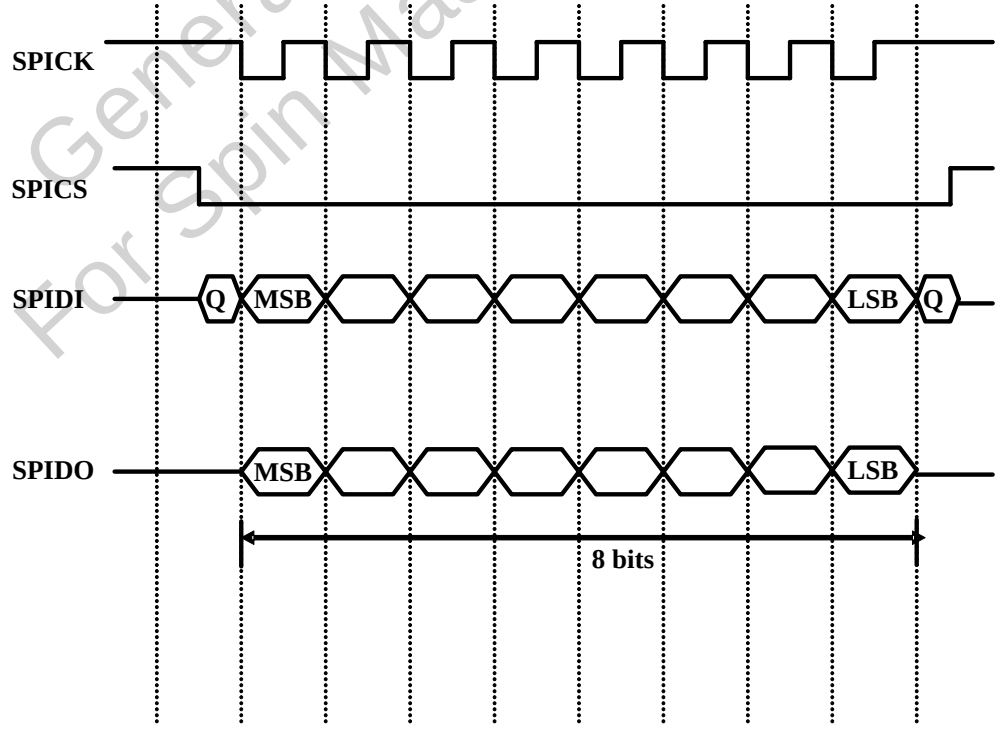


Fig4. Master Mode, POLARITY=1, PHASE=1

At this setting, GPCE500A will sample data at rising edge of SPICK.

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16.1.2 Slave Mode

In slave mode, the SPICK becomes an input pin that receives external clock. And all clock and data are valid when SPICS at low state. In GPCE500A, SPI slave mode only supports polarity=0, phase=0 mode. Each time transmitting data to GPCE500A slave, the SPICS needs to switch low for GPCE500A SPI clock and data synchronous and switch SPICS to high when data transmission is finished.

16.1.3 Consecutive Bytes Transfer

Consecutive bytes transfer is available in master modes. In transmission, software is able to send the data consecutively as long as the SPITXIF flag is not set. In reception, software will check for overrun error to monitor if there is any missing data due to the polling rate is too low.

16.2. SPI Control Pin Configuration

SPI1:

Name	I/O	Description
SPICS	O	Serial Peripheral Interface, Chip Select, Low Active (Shared with IOC[5])
SPICK	O	Serial Peripheral Interface, Clock Pin (Shared with IOC[3])
SPIDI (SIO1)	I/O	Serial Peripheral Interface, Data In Pin (Shared with IOC[6])
SPIDO (SIO0)	I/O	Serial Peripheral Interface, Data Out Pin (Shared with IOC[4])
SPIDATA (SIO2)	I/O	Serial Peripheral Interface, Data Pin (Shared with IOC[7])
SPIDATA (SIO3)	I/O	Serial Peripheral Interface, Data Pin (Shared with IOC[2])

Ps. SIO pins are for two-I/O and four-I/O mode.

SPI2 :

Name	I/O	Description
SPI2CS	O	Serial Peripheral Interface, Chip Select, Low Active (Shared with IOB[12] / IOA[0])
SPI2CK	O	Serial Peripheral Interface, Clock Pin (Shared with IOB[13] / IOA[1])
SPI2DI (SIO1)	I/O	Serial Peripheral Interface, Data In Pin (Shared with IOB[14] / IOA[2])
SPI2DO (SIO0)	I/O	Serial Peripheral Interface, Data Out Pin (Shared with IOB[15] / IOA[3])
SPIDATA (SIO2)	I/O	Serial Peripheral Interface, Data Pin (Shared with IOB[10] / IOA[5])
SPIDATA (SIO3)	I/O	Serial Peripheral Interface, Data Pin (Shared with IOB[11] / IOA[4])

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16.3. Register Map

Name	Address	Description
P_SPI_Ctrl(R/W)	0x30E0	SPI control register
P_SPI_TX_Status(R/W)	0x30E1	SPI transmit status register
P_SPI_TX_Data(R/W)	0x30E2	SPI transmit FIFO register
P_SPI_RX_Status(R/W)	0x30E3	SPI receive status register
P_SPI_RX_Data(R/W)	0x30E4	SPI receive FIFO register
P_SPI_Misc(R/W)	0x30E5	SPI miscellaneous control register
P_SPI_Man_Ctrl(R/W)	0x30E6	SPI manual control register
P_SPI_Auto_Ctrl(R/W)	0x30E7	SPI auto control register
P_SPI_Enhance_Data(R/W)	0x30E8	SPI enhance data register
P_SPI_Bank(R/W)	0x30E9	SPI bank control register
P_SPI_Auto_Settting(R/W)	0x30EA	SPI auto mode setting register
P_SPI_Scramble_Ctrl(R/W)	0x30EB	SPI manual mode scramble control register
P_SPI2_Ctrl(R/W)	0x3090	SPI2 control register
P_SPI2_TXStatus(R/W)	0x3091	SPI2 transmit status register
P_SPI2_TXData(R/W)	0x3092	SPI2 transmit FIFO register
P_SPI2_RXStatus(R/W)	0x3093	SPI2 receive status register
P_SPI2_RXData(R/W)	0x3094	SPI2 receive FIFO register
P_SPI2_Misc(R/W)	0x3095	SPI2 miscellaneous control register
P_SPI2_Man_Ctrl(R/W)	0x3096	SPI2 manual control register

16.3.1 P_SPI_Ctrl (R/W) (0x30E0)

P_SPI_Ctrl			0x30E0		SPI control register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	SPI_EN	AUTO_EN	-	CS_SEL	RST	MIO[1:0]		MOD
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-	DTR	SCK_PHA	SCK_POL	BUSY	SCK_SEL[2:0]		
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	SPI_EN	R/W	SPI enabled If SPI is enabled, any further corresponding setting on the selected GPIOs will be no effect.	0 : Disabled 1 : Enabled
[14]	AUTO_EN	R/W	Auto SPI enabled	0 : Disabled 1 : Enabled
[13]	-	-	Reserved	-
[12]	CS_SEL	R/W	SPI CS Pin Mode Control	0 : SPI_CS controlled by IO

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Bit	Function	Type	Description	Condition
				1 : SPI_CS auto controlled
[11]	SPIRST	W	SPI Soft Reset If this bit is written by “1”, the state machine of SPI controller and FIFO pointer will return to the original value.	0 : No effect 1 : Reset SPI Controller
[10:9]	MIO[1:0]	R/W	Multi-IO mode	00 : 1 I/O 01 : 2 I/O 10 : 4 I/O 11 : Reserved
[8]	MOD	R/W	SPI Mode Selection register In slave mode only supports SCKPHA = 0, SCKPOL = 0 timing. And SPICS become SPI chip select pin.	0 : Master mode 1 : Slave mode
[7]	-	-	Reserved	-
[6]	DTR	R/W	Double Transfer Rate mode	0 : Normal Transfer Rate 1 : Double Transfer Rate
[5]	SCK_PHA	R/W	SPI clock phase This bit should not be changed after **SPI_EN** is 1.	Refer to timing scheme on previous section
[4]	SCK_POL	R/W	SPI clock polarity This bit should not be changed after **SPI_EN** is 1.	Refer to timing scheme on previous section
[3]	BUSY	R	SPI module busy	0 : not busy 1 : busy
[2:0]	SCK_SEL[2:0]	R/W	Master mode clock selection	000 : SYSCLK 001 : SYSCLK / 2 010~110 : Reserved 111 : SYSCLK / 16

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16.3.2 P_SPI_TX_Status (R/W) (0x30E1)

P_SPI_TX_Status			0x30E1		SPI Transmit Status Register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	TXIF	TXIEN	-					
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	TXFLEV[3:0]				TXFFLAG[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	TXIF	R/W1C	SPI Transmit Interrupt flag.	Read:

Bit	Function	Type	Description	Condition
			This bit is set to “1” by hardware when the transmit FIFO level is lower than the value set by the user. When SMART is set in **P_SPI_Misc** register, the bit will be cleared as long as the transmit FIFO level is higher than interrupt level, else you should write “1” to this control bit to clear this flag.	0: Not Occurred 1: Occurred Write: 0: No effect 1: Clear flag
[14]	TXIEN	R/W	SPI Transmit Interrupt Enable If this bit is set to “1”, and SPI interrupt (when 8-bit TX FIFO level in lower than interrupt level) occurs, hardware will issue an IRQ3 or FIQ to CPU. If this bit is cleared to “0”, this interrupt will be masked. To select between IRQ3 and FIQ, please refer to ***Chapter Interrupt***.	0 : Disabled 1 : Enabled **Note:** If SPI TX interrupt is enabled, we suggest programmer enable smart interrupt clear function. (**P_SPI_Misc[8]**)
[13:8]	-	-	Reserved	-
[7:4]	TXFLEV[3:0]	R/W	Transmit FIFO interrupt level register. This register is used to indicate how many empty slots are required when issuing interrupt. The smaller value is set, the lower interrupt penalty you have because you can write more data in one interrupt.	0000 : Data no. in FIFO < 1, 8 write is allowed. 0001 : Data no. in FIFO < 2, 7 write is allowed. 0010 : Data no. in FIFO < 3, 6 write is allowed. 0011 : Data no. in FIFO < 4, 5 write is allowed. 0100 : Data no. in FIFO < 5, 4 write is allowed. 0101 : Data no. in FIFO < 6, 3 write is allowed. 0110 : Data no. in FIFO < 7, 2 write is allowed. 0111 : Data no. in FIFO < 8, 1 write is allowed. 1000~1111 : Not valid
[3:0]	TXFFLAG[3:0]	R	Transmit FIFO Data Level The register is used to indicate how much data is still in the FIFO.	0000 : No data in FIFO or 8 bytes in FIFO. 0001 : 1 byte in FIFO. 0010 : 2 bytes in FIFO. 0011 : 3 bytes in FIFO. 0100 : 4 bytes in FIFO. 0101 : 5 bytes in FIFO.

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Bit	Function	Type	Description	Condition
				0110 : 6 bytes in FIFO. 0111 : 7 bytes in FIFO.

16.3.3 P_SPI_TX_Data (W) (0x30E2)

P_SPI_TX_Data			0x30E2		SPI transmit FIFO register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	TXDATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:8]	-	-	Reserved
[7:0]	TXDATA[7:0]	W	Write data to SPI Transmit FIFO.

16.3.4 P_SPI_RX_Status (R/W) (0x30E3)

P_SPI_RX_Status			0x30E3		SPI receive status register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	RXIF	RXIEN	-				RXFULL	RXFOV
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	RXFLEV[3:0]				RXFFLAG[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	RXIF	R/W1C	SPI Transmit Interrupt flag. This bit is set to “1” by hardware when the received FIFO level is higher than the value set by the user. When SMART is set in **P_SPI_Misc** register, the bit will be cleared as long as the received FIFO level is lower than interrupt level, else you should write “1” to this control bit to clear this flag.	Read: 0: Not Occurred 1: Occurred Write: 0: No effect 1: Clear flag
[14]	RXIEN	R/W	SPI Receive Interrupt Enable	0 : Disabled

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Bit	Function	Type	Description	Condition
			If this bit is set to “1”, and SPI interrupt (when 8-bit RX FIFO level in higher than interrupt level) occurs, hardware will issue an IRQ3 or FIQ to CPU. If this bit is cleared to “0”, this interrupt will be masked. To select between IRQ3 and FIQ, please refer to Chapter Interrupt.	1 : Enabled **Note:** If SPI RX interrupt is enabled, we suggest programmer enable smart interrupt clear function. (**P_SPI_Misc[8]**)
[13:10]	-	-	Reserved	-
[9]	RXFULL	R	Receive FIFO full register This bit will be set by hardware when the received FIFO is full. And it will be cleared by hardware when the received FIFO is not full.	0 = Not Full 1 = Full
[8]	RXFOV	R/W1C	Received FIFO over run register If the RX FIFO is full, and there is a next data coming in, this flag will be set. The new coming data will overwrite the last input data or be skipped, please refer **P_SPI_Misc**	Read: 0: Not Occurred 1: Occurred Write: 0: No effect 1: Clear flag
[7:4]	RXFLEV[3:0]	R/W	Receive FIFO interrupt level register. This register is used to indicate number of bytes are stored in received FIFO when issuing interrupt. The larger value is set, the lower interrupt penalty you have since you can read more data in one interrupt.	0000 : Data no. in FIFO >= 1, 1 read is allowed. 0001 : Data no. in FIFO >= 2, 2 read is allowed. 0010 : Data no. in FIFO >= 3, 3 read is allowed. 0011 : Data no. in FIFO >= 4, 4 read is allowed. 0100 : Data no. in FIFO >= 5, 5 read is allowed. 0101 : Data no. in FIFO >= 6, 6 read is allowed. 0110 : Data no. in FIFO >= 7, 7 read is allowed. 0111 : Data no. in FIFO >= 8, 8 read is allowed. 1000~1111 : Not valid
[3:0]	RXFFLAG[3:0]	R	Receive FIFO Data Level The register is used to indicate how much data is still in the FIFO.	0000 : No data in FIFO or 8 bytes in FIFO 0001 : 1 byte in FIFO. 0010 : 2 bytes in FIFO. 0011 : 3 bytes in FIFO.

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Bit	Function	Type	Description	Condition
				0100 : 4 bytes in FIFO. 0101 : 5 bytes in FIFO. 0110 : 6 bytes in FIFO. 0111 : 7 bytes in FIFO.

16.3.5 P_SPI_RX_Data (R) (0x30E4)

P_SPI_RX_Data			0x30E4		SPI receive FIFO register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	RXDATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:8]	-	-	Reserved
[7:0]	RXDATA[7:0]	R	Read data from SPI Transmit FIFO

16.3.6 P_SPI_Misc (R/W) (0x30E5)

P_SPI_Misc			0x30E5		SPI miscellaneous control register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-						OVER	SMART
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-			BSY	RFF	RNE	TNF	TFE
Default	0	0	0	0	0	0	1	1

Bit	Function	Type	Description	Condition
[15:10]	-	-	Reserved	-
[9]	OVER	R/W	SPI FIFO Over Write Mode This register is used to control the data will be overwrite or skipped when TX/RX FIFO is full.	0 : The further write to the full FIFO will be skipped. 1 : The further write to the full FIFO will overwrite the last written data in the FIFO.
[8]	SMART	R/W	SPI FIFO SMART Mode Register When this bit is set to “1”, programmer don’t need to clear the transmit/receive	0 : Normal Interrupt Clear 1 : Smart Interrupt Clear **Note:** In Smart Mode, if

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Bit	Function	Type	Description	Condition
			interrupt flag when the FIFO status is reach, programmer only needs write-to/read-from transmit/receive FIFO and let the FIFO level lower/higher than the interrupt level, and the interrupt flag will clear automatically.	IRQ/FIQ is enabled , when interrupt occurs, user must Write/Read FIFO; otherwise, interrupt flag will not be cleared and thus, program will keep entering interrupt subroutine.
[7:5]	-	-	Reserved	-
[4]	BSY	R	SPI Controller Busy Flag This bit is used to indicate if the SPI controller is busy or not. For SPI1, When send data to **P_SPI_TX_Data** port, please wait for another one SPI1 clock before probing SPI busy flag. For SPI2, When sending data to **P_SPI2_TX_Data** port, user doesn't need to wait clock before probing SPI busy flag.	0 : Idle 1 : Busy
[3]	RFF	R	Received FIFO Full Flag This bit is used to indicate if the SPI controller is full or not. If the received FIFO is full, that means any data read from SPI bus can't be written into the FIFO, and the RFOV bit will be set in this situation.	0 : Receive FIFO not full 1 : Receive FIFO full, not empty slot.
[2]	RNE	R	Received FIFO Not Empty Flag This bit is used to indicate if there is any data currently in the received FIFO.	0 : Receive FIFO is empty 1 : Receive FIFO is not empty
[1]	TNF	R	Transmit FIFO Not Full Flag. This bit is used to indicate if there is any empty slots in the transmit FIFO.	0 : Transmit FIFO is full, no more data can be written into it. 1 : Transmit FIFO is not full.
[0]	TFE	R	Transmit FIFO Empty Flag This bit is used to indicate if the transmit FIFO is empty or not.	0 : Transmit FIFO is not empty. 1 : Transmit FIFO is empty.

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16.3.7 P_SPI_Man_Ctrl (R/W) (0x30E6)

P_SPI_Man_Ctrl			0x30E6		SPI manual control register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**

Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-						MAN_CSB	MAN_OE
Default	0	0	0	0	0	0	1	0

Bit	Function	Type	Description	Condition
[15:2]	-	-	Reserved	-
[1]	MAN_CSB	R/W	SPI manual mode CSB	0 : CSB = Low 1 : CSB = High
[0]	MAN_OE	R/W	SPI manual mode output enable	0 : SPI 2I/O or 4I/O input 1 : SPI 2I/O or 4I/O output

16.3.8 P_SPI_Auto_Ctrl (R/W) (0x30E7)

P_SPI_Auto_Ctrl			0x30E7		SPI auto control			
Bit	15	14	13	12	11	10	9	8
Function	ADDR_LEN	ADDR_1-IO	CMD_MBIT	AUTO_ENHM	AUTO_DUMMY_CYCLE[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	AUTO_CMD[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	ADDR_LEN	R/W	Address Length Mode 3-byte (24 bit <= 128M bit) 4-byte (32 bit > 128M bit)	0 : 3-byte address mode 1 : 4-byte address mode
[14]	ADDR_1-IO	R/W	Address 1-IO mode or multi-IO mode	0 : Multi-IO 1 : 1-IO
[13]	CMD_MBIT	R/W	Command 1-IO mode or multi-IO mode	0 : 1-IO 1 : Multi-IO
[12]	AUTO_ENHM	R/W	Auto enhancement mode	0 : Disabled 1 : Enabled
[11:8]	AUTO_DUMMY_CYCLE[3:0]	R/W	Auto Dummy Cycle	0000 = No dummy cycle 0001 = 1 dummy cycle … 1111 = 15 dummy cycle
[7:0]	AUTO_CMD[7:0]	R/W	Auto Command	SPI Command

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Bit	Function	Type	Description	Condition
				Please refer to the data sheet of SPI FLASH.

16.3.9 P_SPI_Enhance_Data (R/W) (0x30E8)

P_SPI_Enhance_Data			0x30E8		SPI enhance data register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	Enhance_Data[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:8]	-	-	Reserved	-
[7:0]	Enhance_Data[7:0]	R/W	Output data on SPI IO pad during transaction in enhance mode. User should refer to vender's specification.	

16.3.10 P_SPI_BANK (R/W) (0x30E9)

P_SPI_BANK			0x30E9		SPI bank control register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-		BANK[5:0]					
Default	0	0	0	0	0	0	1	1

Bit	Function	Type	Description	Condition
[15:6]	-	-	Reserved	-
[5:0]	BANK[5:0]	R/W	SPI Bank Select	000000 = Bank 0 (1M-Word) 000001 = Bank 1 000010 = Bank 2 000011 = Bank 3 (Default)

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Bit	Function	Type	Description	Condition
				… 111111 = Bank 63

The address of SPI Bank 0~2 is 0x8000 ~0x2FFFFF as SPI common bank.
The address of SPI Bank 0~63 map to 0x300000 ~0x3FFFFF.
Every bank has 1M-Word, 6-bit bank register; total of addressing range is 64M-Word.

16.3.11 P_SPI_Auto_Setting (R/W) (0x30EA)

P_SPI_Auto_Settting				0x30EA	SPI auto mode setting register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	HiDrive[1:0]		CONTI	Rx_Delay1T	Rx_Delay[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-			AUTO_SIO23	CS_HI_WAIT[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:14]	HiDrive[1:0]	R/W	SPI1 Interface High Driving	00 : 6 mA 01 : 8 mA 10 : 10 mA 11 : 12 mA
[13]	CONTI	R/W	SPI1 auto mode continuous mode CS will keep low to next discontinuous address	0 : Disabled 1 : Enabled
[12]	Rx_Delay1T	R/W	RX Delay 1 Clock	0 : Disabled 1 : Enabled
[11:8]	Rx_Delay[3:0]	R/W	Rx Delay Time	0000 : No Delay 0001 : Delay 1 Unit 0010 : Delay 2 Unit 0011 : Delay 3 Unit 0100 : Delay 4 Unit 0101 : Delay 5 Unit 0110 : Delay 6 Unit 0111 : Delay 7 Unit 1000 : Delay 8 Unit 1001 : Delay 9 Unit 1010 : Delay 10 Unit 1011 : Delay 11 Unit 1100 : Delay 12 Unit 1101 : Delay 13 Unit 1110 : Delay 14 Unit

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Bit	Function	Type	Description	Condition
				1111 : Delay 15 Unit (1ns < 1 Unit < 2ns @3.0V)
[7:5]	-	-	Reserved	-
[4]	AUTO_SIO23	R/W	Quad Mode 1-IO Command SIO2/SIO3 setting	0 : Floating 1 : Output high
[3:0]	CS_HI_WAIT[3:0]	R/W	CS Deselect Time From Read to next Read at Auto Mode.	0000: Wait 3 SPI Clock 0001: Wait 3 SPI Clock 0010: Wait 4 SPI Clock 0011: Wait 5 SPI Clock 0100: Wait 6 SPI Clock 0101: Wait 7 SPI Clock 0110: Wait 8 SPI Clock 0111: Wait 9 SPI Clock 1000: Wait 10 SPI Clock 1001: Wait 11 SPI Clock 1010: Wait 12 SPI Clock 1011: Wait 13 SPI Clock 1100: Wait 14 SPI Clock 1101: Wait 15 SPI Clock 1110: Wait 16 SPI Clock 1111: Wait 17 SPI Clock

16.3.12 P_SPI_Scramble_Ctrl (R/W) (0x30EB)

P_SPI_Scramble_Ctrl	0x30EB	SPI manual mode scramble control register						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-						ENC_EN	DEC_EN
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:2]	-	-	Reserved	-
[1]	ENC_EN	R/W	SPI1 encryption data enable **Note:** Manual mode omly	0 : Disabled 1 : Enabled
[0]	DEC_EN	R/W	SPI1 decryption data enable **Note:** Manual mode only	0 : Disabled 1 : Enabled

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16.3.13 P_SPI2_Ctrl (R/W) (0x3090)

P_SPI2_Ctrl		0x3090		SPI2 control register				
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	SPI_EN	-		CS_SEL	SPIRST	MIO[1:0]		MOD
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-	DTR	SCK_PHA	SCK_POL	BUSY	SCK_SEL[2:0]		
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	SPI_EN	R/W	SPI2 enabled If SPI is enabled, any further corresponding setting on the selected GPIOs will be no effect.	0 : Disabled 1 : Enabled
[14:13]	-	-	Reserved	-
[12]	CS_SEL	R/W	SPI2 CS Pin Mode Control	0 = SPI_CS controlled by IO 1 = SPI_CS auto controlled
[11]	SPIRST	W	SPI2 Soft Reset If this bit is written by “1”, the state machine of SPI controller and FIFO pointer will return to the original value.	0= No effect 1= Reset SPI Controller
[10:9]	MIO[1:0]	R/W	Multi-IO mode	00 : 1 I/O 01 : 2 I/O 10 : 4 I/O 11 : Reserved
[8]	MOD	R/W	SPI2 Mode Selection register In slave mode only supports SCKPHA = 0, SCKPOL = 0 timing. And SPICS become SPI chip select pin.	0 : Master mode 1 : Slave mode
[7]	-	-	Reserved	-
[6]	DTR	R/W	Double Transfer Rate mode	0 : Normal Transfer Rate 1 : Double Transfer Rate
[5]	SCK_PHA	R/W	SPI2 clock phase This bit should not be changed after **SPI_EN** is 1.	Refer to timing scheme on previous section
[4]	SCK_POL	R/W	SPI2 clock polarity This bit should not be changed after **SPI_EN** is 1.	Refer to timing scheme on previous section
[3]	BUSY	R	SPI2 module busy	0 = not busy 1 = busy
[2:0]	SCK_SEL[2:0]	R/W	Master mode clock selection	000 : SYSCLK / 2

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Bit	Function	Type	Description	Condition
				001 : SYSCLK / 4 010 : SYSCLK / 8 011 : SYSCLK / 16 100 : SYSCLK / 32 101 : SYSCLK / 64 110 : SYSCLK / 128 111 : TMA / 2

16.3.14 P_SPI2_TX_Status (R/W) (0x3091)

P_SPI2_TX_Status			0x3091		SPI2 transmit status register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	TXIF	TXIEN	-					
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	TXFLEV[3:0]				TXFFLAG[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	TXIF	R/W1C	SPI2 Transmit Interrupt flag. This bit is set to “1” by hardware when the transmit FIFO level is lower than the value set by the user. When SMART is set in **P_SPI2_Misc** register, the bit will be cleared as long as the transmit FIFO level is higher than interrupt level, else you should write “1” to this control bit to clear this flag.	Read: 0: Not Occurred 1: Occurred Write: 0: No effect 1: Clear flag
[14]	TXIEN	R/W	SPI2 Transmit Interrupt Enable If this bit is set to “1”, and SPI2 interrupt (when 8-bit TX FIFO level in lower than interrupt level) occurs, hardware will issue an IRQ3 or FIQ to CPU. If this bit is cleared to “0”, this interrupt will be masked. To select between IRQ3 and FIQ, please refer to ***Chapter Interrupt***.	0 : Disabled 1 : Enabled **Note:** If SPI2 TX interrupt is enabled, we suggest programmer enable smart interrupt clear function. (**P_SPI2_Misc[8]**)
[13:8]	-	-	Reserved	-
[7:4]	TXFLEV[3:0]	R/W	Transmit FIFO interrupt level register.	0000 : Data no. in FIFO < 1, 8

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Bit	Function	Type	Description	Condition
			This register is used to indicate how many empty slots are required when issuing interrupt. The smaller value is set, the lower interrupt penalty you have because you can write more data in one interrupt.	write is allowed. 0001 : Data no. in FIFO < 2, 7 write is allowed. 0010 : Data no. in FIFO < 3, 6 write is allowed. 0011 : Data no. in FIFO < 4, 5 write is allowed. 0100 : Data no. in FIFO < 5, 4 write is allowed. 0101 : Data no. in FIFO < 6, 3 write is allowed. 0110 : Data no. in FIFO < 7, 2 write is allowed. 0111 : Data no. in FIFO < 8, 1 write is allowed. 1000~1111 : Not valid
[3:0]	TXFFLAG[3:0]	R	Transmit FIFO Data Level The register is used to indicate how much data is still in the FIFO.	0000 : No data in FIFO or 8 bytes in FIFO. 0001 : 1 byte in FIFO. 0010 : 2 bytes in FIFO. 0011 : 3 bytes in FIFO. 0100 : 4 bytes in FIFO. 0101 : 5 bytes in FIFO. 0110 : 6 bytes in FIFO. 0111 : 7 bytes in FIFO.

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16.3.15 P_SPI2_TX_Data (W) (0x3092)

P_SPI2_TX_Data			0x3092		SPI2 transmit FIFO register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	TXDATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:8]	-	-	Reserved
[7:0]	TXDATA[7:0]	W	Write data to SPI2 Transmit FIFO.

16.3.16 P_SPI2_RX_Status (R/W) (0x3093)

P_SPI2_RX_Status			0x3093		SPI2 receive status register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	RXIF	RXIEN	-				RXFULL	RXFOV
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	RXFLEV[3:0]				RXFFLAG[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15]	RXIF	R/W1C	SPI2 Transmit Interrupt flag. This bit is set to “1” by hardware when the received FIFO level is higher than the value set by the user. When SMART is set in **P_SPI2_Misc** register, the bit will be cleared as long as the received FIFO level is lower than interrupt level, else you should write “1” to this control bit to clear this flag.	Read: 0: Not Occurred 1: Occurred Write: 0: No effect 1: Clear flag
[14]	RXIEN	R/W	SPI2 Receive Interrupt Enable If this bit is set to “1”, and SPI2 interrupt (when 8-bit RX FIFO level in higher than interrupt level) occurs, hardware will issue an IRQ3 or FIQ to CPU. If this bit is cleared to “0”, this interrupt will be masked. To select between IRQ3 and FIQ, please refer to Chapter Interrupt.	0 : Disabled 1 : Enabled **Note:** If SPI2 RX interrupt is enabled, we suggest programmer enable smart interrupt clear function. (**P_SPI2_Misc[8]**)
[13:10]	-	-	Reserved	-
[9]	RXFULL	R	Receive FIFO full register This bit will be set by hardware when the received FIFO is full. And it will be cleared by hardware when the received FIFO is not full.	0 = Not Full 1 = Full
[8]	RXFOV	R/W1C	Received FIFO over run register If the RX FIFO is full, and there is a next data coming in, this flag will be set. The new coming data will overwrite the last input data or be skipped, please refer **P_SPI2_Misc**	Read: 0: Not Occurred 1: Occurred Write: 0: No effect 1: Clear flag

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Bit	Function	Type	Description	Condition
[7:4]	RXFLEV[3:0]	R/W	Receive FIFO interrupt level register. This register is used to indicate number of bytes are stored in received FIFO when issuing interrupt. The larger value is set, the lower interrupt penalty you have since you can read more data in one interrupt.	0000 : Data no. in FIFO >= 1, 1 read is allowed. 0001 : Data no. in FIFO >= 2, 2 read is allowed. 0010 : Data no. in FIFO >= 3, 3 read is allowed. 0011 : Data no. in FIFO >= 4, 4 read is allowed. 0100 : Data no. in FIFO >= 5, 5 read is allowed. 0101 : Data no. in FIFO >= 6, 6 read is allowed. 0110 : Data no. in FIFO >= 7, 7 read is allowed. 0111 : Data no. in FIFO >= 8, 8 read is allowed. 1000~1111 : Not valid
[3:0]	RXFFLAG[3:0]	R	Receive FIFO Data Level The register is used to indicate how much data is still in the FIFO.	0000 : No data in FIFO or 8 bytes in FIFO 0001 : 1 byte in FIFO. 0010 : 2 bytes in FIFO. 0011 : 3 bytes in FIFO. 0100 : 4 bytes in FIFO. 0101 : 5 bytes in FIFO. 0110 : 6 bytes in FIFO. 0111 : 7 bytes in FIFO.

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16.3.17 P_SPI2_RX_Data (R) (0x3094)

P_SPI2_RX_Data			0x3094		SPI2 receive FIFO register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	RXDATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:8]	-	-	Reserved
[7:0]	RXDATA[7:0]	R	Read data from SPI Transmit FIFO

16.3.18 P_SPI2_Misc (R/W) (0x3095)

P_SPI2_Misc			0x3095		SPI2 miscellaneous control register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-						OVER	SMART
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-			BSY	RFF	RNE	TNF	TFE
Default	0	0	0	0	0	0	1	1

Bit	Function	Type	Description	Condition
[15:10]	-	-	-	-
[9]	OVER	R/W	SPI2 FIFO Over Write Mode This register is used to control the data will be overwrite or skipped when TX/RX FIFO is full.	0 : The further write to the full FIFO will be skipped. 1 : The further write to the full FIFO will overwrite the last written data in the FIFO.
[8]	SMART	R/W	SPI2 FIFO SMART Mode Register When this bit is set to “1”, programmer don’t need to clear the transmit/receive interrupt flag when the FIFO status is reach, programmer only needs write-to/read-from transmit/receive FIFO and let the FIFO level lower/higher than the interrupt level, and the interrupt flag will clear automatically.	0 : Normal Interrupt Clear 1 : Smart Interrupt Clear **Note:** In Smart Mode, if IRQ/FIQ is enabled , when interrupt occurs, user must Write/Read FIFO; otherwise, interrupt flag will not be cleared and thus, program will keep entering interrupt subroutine.
[7:5]	-	-	-	-
[4]	BSY	R	SPI2 Controller Busy Flag This bit is used to indicate if the SPI controller is busy or not. For SPI1, When send data to **P_SPI_TX_Data** port, please wait for another one SPI1 clock before probing SPI busy flag. For SPI2, When sending data to **P_SPI2_TX_Data** port, user doesn’t need to wait clock before probing SPI busy flag.	0 : Idle 1 : Busy
[3]	RFF	R	Received FIFO Full Flag This bit is used to indicate if the SPI	0 : Receive FIFO not full 1 : Receive FIFO full, not

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Bit	Function	Type	Description	Condition
			controller is full or not. If the received FIFO is full, that means any data read from SPI bus can't be written into the FIFO, and the RFOV bit will be set in this situation.	empty slot.
[2]	RNE	R	Received FIFO Not Empty Flag This bit is used to indicate if there is any data currently in the received FIFO.	0 : Receive FIFO is empty 1 : Receive FIFO is not empty
[1]	TNF	R	Transmit FIFO Not Full Flag. This bit is used to indicate if there is any empty slots in the transmit FIFO.	0 : Transmit FIFO is full, no more data can be written into it. 1 : Transmit FIFO is not full.
[0]	TFE	R	Transmit FIFO Empty Flag This bit is used to indicate if the transmit FIFO is empty or not.	0 : Transmit FIFO is not empty. 1 : Transmit FIFO is empty.

16.3.19 P_SPI2_Man_Ctrl (R/W) (0x3096)

P_SPI2_Man_Ctrl			0x3096		SPI2 manual control register			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-		TX_DMA_EN	RX_DMA_EN	-	CONTI	MAN_CSB	MAN_OE
Default	0	0	0	0	0	0	1	0

Bit	Function	Type	Description	Condition
[15:6]	-	-	Reserved	-
[5]	TX_DMA_EN	R/W	SPI2 TX DMA enable, TX FIFO will be bypassed when TX DMA is enabled	0 : Disabled 1 : Enabled
[4]	RX_DMA_EN	R/W	SPI2 RX DMA enable	0 : Disabled 1 : Enabled
[3]	-	-	Reserved	-
[2]	CONTI	R/W	SPI2 continuous mode No dummy clock between the TX data.	0 : Disabled 1 : Enabled
[1]	MAN_CSB	R/W	SPI2 Manual mode CSB	0 : CSB = Low 1 : CSB = High
[0]	MAN_OE	R/W	SPI2 Manual mode output enable	0 : SPI 2I/O or 4I/O input 1 : SPI 2I/O or 4I/O output

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17 CTS

17.1. General Description

GPCE500A contains 24 pads Capacitive Touch Sensor (CTS) module for related application. GPCE500A CTS uses constant current for charging/discharging. Two sets of 16- bit timer with preload functions, which can be dedicated for CTS or general timer.

Timer A – number of time Touch Pad will access. After specified number of time is reached, issue an interrupt for Timer B to latch current data to CAP_TMB. (Timer A is slower)
Timer B – If System Clock is selected, overflow may usually occur. User should use a RAM to keep track on number of overflow; or user may lower the clock speed or reduce the number of time accessed by TMA.

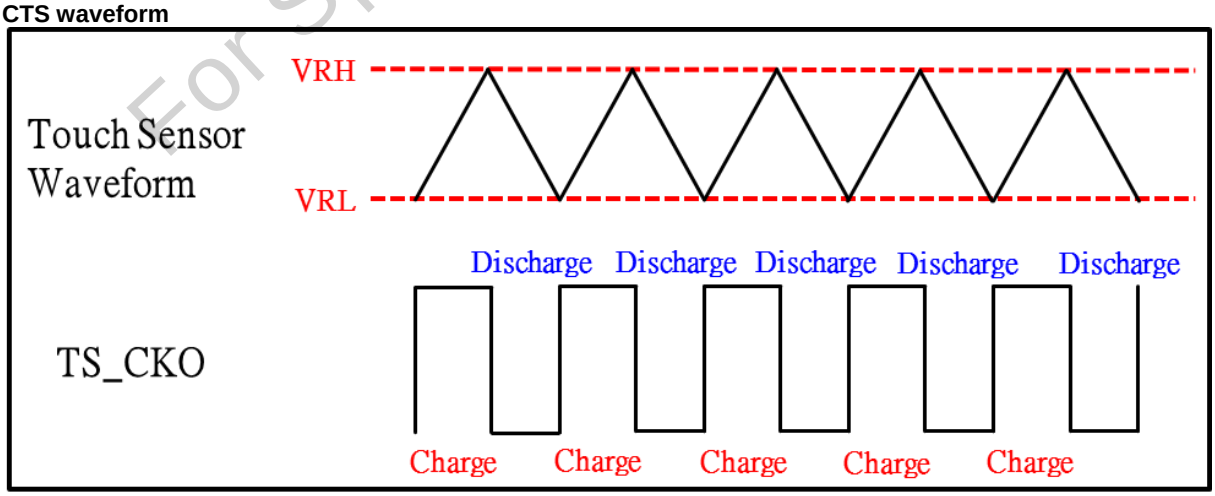
Auto stop mode:

Even in Auto Stop Mode, it can only work after issuing a start command by itself. System will check the first TMA clock and after that, TMB will start counting. When writing CTS_START=1, CTS_TMA/TMB count. When CTS_TMA overflow occurs, interrupt is established. At this moment, CTS_TMB Counter will be recorded on CAP_TMB and reload CTS_TMA/TMB counter again. Wait until the next writing CTS_START=1 to re-detect Touch Sensor.

Manual stop mode:

When writing CTS_START=1, CTS_TMA/TMB count. When CTS_TMA overflow occurs, interrupt is established. At the same time, CTS_TMB Counter will be recorded on CAP_TMB. CTS_TMA/TMB counter will not stop by itself. User need to write CTS_START=0 to stop CTS_TMA/TMB counter. Then, user has to reload CTS_TMA/TMB counter again and precede the next Touch Sensor detection.

CTS waveform



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17.2. Register Map

Name	Address	Description
P_CTS_Ctrl0	0x30D0	CTS control register
P_CTS_Ctrl1	0x30D1	CTS control register1
P_CTS_TMADATA	0x30D2	CTS preload register for timer A
P_CTS_TMACNT	0x30D3	CTS counter for timer A
P_CTS_TMBDATA	0x30D4	CTS preload register for timer B
P_CTS_TMBCNT	0x30D5	CTS counter for timer B
P_CTS_CAPTMB	0x30D6	Capture register for timer B

17.2.1 P_CTS_CTRL0 (R/W) (0x30D0)

P_CTS_Ctrl0	**0x30D0**	CTS control register						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	CHSEL_REF[3:0]				IOSEL_REF[1:0]		-	CTSEN_REF
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	-					AUTO_STP	ISEL	CTS_EN
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:12]	CHSEL_REF[3:0]	R/W	Channel selection for reference self-capacitance	0000 : IOX0 0001 : IOX1 ….. 1111 : IOX15
[11:10]	IOSEL_REF[1:0]	R/W	IO selection for reference self-capacitance	00 : IOA 01 : IOB 1x : No IO selection
[9]	-	-	Reserved	-
[8]	CTSEN_REF	R/W	CTS ref CLK enable	0 : Disabled 1 : Enabled
[7:3]	-	-	Reserved	-
[2]	AUTO_STP	R/W	Enable Auto Stop Mode When this bit is 1, CTS_TMA/TMB counter will be stop automatically when CTS_TMA overflow and CtsStart will be cleared. The new counting will be start after **P_CTS_CTRL1[0]** is written 1. When this bit is 0, CTS_TMA/TMB counter will keep counting after CTS_TMA overflow; users need to write 0 to **P_CTS_CTRL1[0]** to stop	0: Disabled 1: Enabled

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Bit	Function	Type	Description	Condition
			CTS_TMA/TMB.	
[1]	ISEL	R/W	Charge/Discharge current selection	0：25uA 1：50uA
[0]	CTSEN	R/W	Capacitive touch sensor enable	0 : Disabled 1 : Enabled

17.2.2 P_CTS_CTRL1 (R/W) (0x30D1)

P_CTS_CTRL1	0x30D1	CTS control register1						
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	TMB_CLK_SEL[2:0]			TMA_CLK_SEL[2:0]			CTSTMA_EN	CTSTMB_EN
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	CHSEL[3:0]				CTS_IOSEL[1:0]		-	CTS_STR
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[15:13]	TMB_CLK_SEL[2:0]	R/W	CTS Timer B Clock Select	000 : OFF 001 : CTS_REF 010 : SYSCLK 011 : SYSCLK / 2 100 : SYSCLK / 4 101 : SYSCLK / 8 110 : SYSCLK / 16 111 : SYSCLK / 32
[12:10]	TMA_CLK_SEL[2:0]	R/W	CTS Timer A Clock Select	000 : OFF 001 : CTS 010 : SYSCLK 011 : SYSCLK / 2 100 : SYSCLK / 4 101 : SYSCLK / 8 110 : SYSCLK / 16 111 : SYSCLK / 32
[9]	CTSTMA_EN	R/W	CTS TMA is dedicated for touch sensor	0: CTS TMA is dedicated for touch sensor 1: CTS TMA is general timer
[8]	CTSTMB_EN	R/W	CTS TMB is dedicated for touch sensor	0: CTS TMB is dedicated for touch sensor 1: CTS TMB is general timer
[7:4]	CHSEL[3:0]	R/W	Channel selection for self-capacitance	0000: IOX0 0001: IOX1 ….. 1111: IOX15
[3:2]	CTS_IOSEL[1:0]	R/W	IO selection for self-capacitance	00: IOA 01: IOB 1x: No IO selection

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Bit	Function	Type	Description	Condition
[1]	-	-	Reserved	-
[0]	CTS_STR	R/W	Start/Stop CTS Function	0: stops CTS_TMA/TMB counter 1: starts CTS_TMA/TMB counter

17.2.3 P_CTS_TMAData (R/W) (0x30D2)

P_CTS_TMAData			0x30D2		CTS preload register for timer A			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	TMADATA[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	TMADATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	TMADATA[15:0]	R/W	Capacitive touch sensor TMA data register(TMB counter value when TMA overflow)

17.2.4 P_CTS_TMACNTR (R) (0x30D3)

P_CTS_TMACNTR			0x30D3		CTS counter for timer A			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	TMACNTR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Function	TMACNTR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	TMACNTR[15:0]	R	Capacitive touch sensor TMA count register

17.2.5 P_CTS_TMBData (R/W) (0x30D4)

P_CTS_TMBData			0x30D4		CTS preload register for timer B			
Bit	**15**	**14**	**13**	**12**	**11**	**10**	**9**	**8**
Function	TMBDATA[15:8]							
Default	0	0	0	0	0	0	0	0

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Bit	7	6	5	4	3	2	1	0
Function	TMBDATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	TMBDATA[15:0]	R/W	Capacitive touch sensor TMB data register

17.2.6 P_CTS_TMBCNTR (R) (0x30D5)

P_CTS_TMB_CNTR		0x30D5		CTS counter for timer B				
Bit	15	14	13	12	11	10	9	8
Function	TMBCNTR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	TMBCNTR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	TMBCNTR[15:0]	R	Capacitive touch sensor TMB count register

17.2.7 P_CTS_CAPTMB (R/W) (0x30D6)

P_CTS_CAPTMB		0x30D6		Capture register for timer B				
Bit	15	14	13	12	11	10	9	8
Function	TMBCAP[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	TMBCAP[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description
[15:0]	TMBCAP[15:0]	R	TMB counter's value will be captured into CAPTMB while TMA overflow

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